

Indian Battery Industry — Master Research Dossier

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1 INDIAN BATTERY INDUSTRY — MASTER RESEARCH DOSSIER

A Living Institutional Research Notebook

Compiled for: Deepesh Joshi | Indira Securities

Covers: Battery Industry Fundamentals, Raw Materials, Technologies, the Indian Battery Ecosystem, and full company dossiers on **Exide Industries Limited** and **Amara Raja Energy & Mobility Limited**, together with the associated financial models.

Compiled: 24 July 2026

2 Part I — Executive Overview

2.1 1. Purpose of This Dossier

This document is **not** an equity research report, an initiation of coverage note, or a client-facing publication. It is a permanent, continuously-updatable institutional research notebook consolidating everything researched to date on the Indian battery sector — spanning macro industry structure, raw materials and supply chains, battery chemistry and manufacturing, and deep, primary-source-driven company dossiers on the two listed lead-acid incumbents, Exide Industries and Amara Raja Energy & Mobility (ARE&M), both of which are executing a parallel pivot into lithium-ion cell manufacturing.

The objective is that, opened months or years later, this single document should restore full working context on the battery sector and both companies without re-reading dozens of underlying annual reports, concalls, investor presentations, and research chat sessions.

2.2 2. How to Use This Document

- **Parts II–VII** are macro/background material: battery fundamentals, manufacturing processes, raw materials, chemistries, the Indian industry structure, and an ecosystem map of who does what. This material changes slowly and is written to remain broadly valid for several years, with clearly flagged areas (capacity figures, pricing, company-specific pipeline data) that will need periodic refreshes.
- **Parts VIII and IX** are the full, verbatim-preserved company dossiers for Exide Industries and Amara Raja Energy & Mobility respectively, exactly as compiled from primary annual reports, concall transcripts, and investor presentations (Exide) and from NotebookLM-assisted synthesis of primary sources (Amara Raja). Nothing in

the original analysis, conclusions, or tagging has been altered — only re-organized for navigability within this larger document.

- **Part X** builds a comparative lens across the two companies using data already established in Parts VIII and IX.
- **Part XI** documents the financial modelling work completed to date for both companies — the standalone Exide SOTP model and the consolidated ARE&M model — including the reasoning, debates, and unresolved judgment calls behind key assumptions, sourced from the modelling conversations already had.
- **Part XII** is a practical, quarter-by-quarter monitoring checklist combining both companies' dashboards with sector-level triggers.

Throughout the company-specific sections, the same tagging discipline applies:

[FACT] — a verifiable, disclosed fact. **[MGMT COMMENTARY]** — management's own framing, which may be promotional and should be weighed against facts, not treated as fact. **[ANALYST INTERPRETATION]** — a conclusion or hypothesis developed through synthesis; not established truth. **[OPEN QUESTION]** — a genuine gap, unresolved figure, or internal contradiction to chase down.

2.3 3. High-Level Map of the Battery Industry

At the highest level, the industry this dossier tracks can be seen as three overlapping value chains converging on one physical product — the rechargeable battery — and one converging demand story: the electrification of mobility and the grid.

1. **The legacy value chain (lead-acid):** mined/recycled lead → lead-acid cell and battery manufacturing → automotive (OEM + replacement), industrial (UPS, telecom, data centre, solar backup), and specialty applications (submarine, traction, e-rickshaw). This is a mature, high-return, cash-generative business with modest volume growth, now facing a slow but real substitution threat from lithium-ion and (later) sodium-ion in several of its historical strongholds (telecom towers, e-rickshaws, and eventually some UPS/inverter use cases).
2. **The emerging value chain (lithium-ion):** mined lithium/nickel/cobalt/graphite (almost entirely imported into India today) → refined battery-grade chemicals → cathode/anode active materials → cell manufacturing (NMC and LFP) → module/pack assembly with a Battery Management System → end applications spanning electric vehicles, two/three-wheelers, stationary grid storage (BESS), telecom, and data centres. India is at the earliest stage of localizing this chain, with several gigafactories (Reliance, Tata/Agratas, Ola, Exide, Amara Raja, and others) under construction or early ramp as of mid-2026.
3. **The frontier/optionality layer (sodium-ion and beyond):** an emerging, still sub-scale chemistry that substitutes lithium with sodium, offering a structurally cheaper and more India-appropriate raw-material base (no dependence on lithium/cobalt/nickel reserves), currently pursued domestically by Reliance (via its Faradion acquisition) and by KPIT Technologies, among others.

Both Exide and Amara Raja sit at the intersection of value chains (1) and (2): they are the incumbent leaders in chain (1), using its cash generation and distribution moat to fund an entry into chain (2), where they do not yet have an incumbency advantage and are competing against better-capitalized, more diversified conglomerates (Reliance, Tata) as well as pure-play EV-first entrants (Ola Electric).

2.4 4. Relationship Between Macro Trends and Company Performance

The macro trends that matter most for how Exide's and Amara Raja's economics evolve over the next five years are:

- **The pace of India's lithium-ion demand growth** (management and third-party estimates converge on a range of roughly 115–210 GWh/year of demand by the early 2030s, up from a low single-digit GWh base today) — this determines how much runway exists for domestic cell manufacturers to fill before import competition (chiefly from China) squeezes returns.
- **The LFP/NMC chemistry split**, which is shifting decisively toward LFP (an estimated 60–65% share by the early 2030s) for cost, safety, and cycle-life reasons particularly suited to India's climate and to grid-scale BESS — a dynamic both companies have had to navigate in sequencing their own capacity build-outs.
- **The BESS build-out cycle**, driven by the renewable-energy transition and grid-balancing needs, which is a new, large, and largely open addressable market for both companies' lithium-ion ambitions, distinct from (and arguably less competitively crowded than) the EV cell market.
- **Commodity cost cycles** — lead, antimony, tin, silver, and sulfur for the legacy business; lithium carbonate, nickel, cobalt, and graphite for the new business — which drive quarter-to-quarter margin volatility in the core business and capital-intensity assumptions in the new one.
- **Government policy** — GST rate changes on batteries and solar equipment (a direct, dateable demand lever already observed in FY26), PLI-style incentives for Advanced Chemistry Cell (ACC) manufacturing, and China's periodic export-control actions on battery-grade raw materials and lithium-ion manufacturing know-how.

Each of these threads is developed in the sector-level parts below and then applied directly in the company-specific financial model discussions in Part XI.

3 Part II — Battery Industry Fundamentals

Background/Industry Context section — stable, well-established technical material, not company-specific.

3.1 1. What a Battery Is and How It Works

A battery is an electrochemical device that stores energy as chemical potential and releases it as electrical current through a reduction-oxidation (redox) reaction between two electrodes — the **anode** (negative electrode, oxidized during discharge) and the **cathode** (positive electrode, reduced during discharge) — separated by an **electrolyte** that allows ions (but not electrons) to move between them, and a **separator**, a porous membrane that physically keeps the electrodes apart while letting ions pass through. External electrons are forced to travel through the connected circuit rather than through the electrolyte, which is what produces usable electrical current.

A **primary battery** (e.g., a household alkaline cell) can only discharge once; the redox reaction is not practically reversible. A **secondary battery** (e.g., lead-acid or lithium-ion) is rechargeable: applying an external voltage in reverse drives the chemical reaction backward, restoring the original electrode materials. Every battery discussed in this dossier is a secondary/rechargeable battery.

3.2 2. History and Evolution of Battery Technology

- **1800s:** Alessandro Volta's voltaic pile (1800) is the first true battery. The lead-acid battery, invented by Gaston Planté in 1859, is the oldest rechargeable battery technology still in mass commercial production today — a remarkable 165+ years of continuous industrial relevance.
- **Early-to-mid 20th century:** Nickel-cadmium (NiCd) batteries emerge for portable and industrial use; lead-acid becomes the standard automotive starter-battery technology as internal combustion vehicles proliferate.
- **1970s–1990s:** Valve-Regulated Lead-Acid (VRLA) batteries — sealed, maintenance-free variants — are developed and commercialized, enabling lead-acid's expansion into telecom and UPS backup applications. Nickel-metal-hydride (NiMH) batteries emerge as a higher-energy-density alternative to NiCd.
- **1991:** Sony commercializes the first lithium-ion battery (using a lithium cobalt oxide, LCO, cathode), initially for consumer electronics (camcorders, then laptops, then mobile phones).
- **2000s–2010s:** Lithium-ion chemistry diversifies — NMC (nickel-manganese-cobalt) and NCA (nickel-cobalt-aluminium) chemistries emerge for higher energy density (electric vehicles, power tools); LFP (lithium iron phosphate) emerges as a lower-cost, lower-energy-density, but far more thermally stable and longer-cycle-life alternative, initially favoured in China. Tesla's adoption of lithium-ion for mass-market EVs (from 2008 onward) catalyses a global scale-up of lithium-ion manufacturing.
- **2020s:** LFP achieves cost and safety leadership sufficient to win a majority share of the global EV and stationary-storage market from NMC/NCA, particularly in standard-range vehicles and all grid-scale BESS applications. Sodium-ion batteries move from laboratory to early commercial deployment (led by Chinese players such as CATL, and pursued in India via Reliance's Faradion acquisition), positioned as a

lower-cost, geopolitically-independent alternative for cost-sensitive and stationary applications where energy density matters less.

3.3 3. Lead-Acid Batteries

How they work: A lead-acid cell uses a lead dioxide (PbO_2) positive electrode, a sponge (metallic) lead negative electrode, and a sulfuric acid (H_2SO_4) electrolyte. During discharge, both electrodes convert to lead sulfate (PbSO_4), and the electrolyte becomes more dilute (water is produced, decreasing the acid's specific gravity — the basis of the traditional hydrometer state-of-charge test). Charging reverses this reaction.

Advantages: Very low manufacturing cost per unit of storage capacity; mature, well-understood, highly recyclable (lead recovery rates from spent batteries typically exceed 95–99% in a well-run recycling operation, making lead-acid one of the most circular industrial products in existence); high surge-current capability (ideal for engine starting); tolerant of a wide range of operating temperatures and abuse compared to lithium-ion in terms of thermal-runaway risk; simple, well-understood safety profile.

Disadvantages: Low energy density (roughly 30–50 Wh/kg, versus 150–250+ Wh/kg for lithium-ion) — batteries are heavy and bulky per unit of stored energy; shorter cycle life (typically 300–500 deep cycles for flooded/VRLA designs versus 1,000–6,000+ for lithium-ion chemistries); performance and lifespan degrade meaningfully in sustained high heat; contains toxic heavy metal (lead) requiring careful handling and regulated recycling.

Applications: Automotive/two-wheeler starter batteries (SLI — Starting, Lighting, Ignition), industrial standby power (UPS, telecom towers, data centres), traction batteries (fork-lifts, some low-speed EVs), submarine batteries, and — historically — e-rickshaws (a use case now migrating toward lithium-ion).

Typical customers: Automotive OEMs, automotive/two-wheeler aftermarket/replacement channels, telecom tower operators, IT/data-centre operators, solar/inverter system integrators, defence establishments (submarine batteries), and e-rickshaw OEMs/fleet operators (declining share).

Commercial maturity: Fully mature. Growth is now driven primarily by vehicle-parc growth, replacement-cycle economics, and industrial-infrastructure build-out (data centres, telecom towers, UPS) rather than by underlying technology improvement — combined with a structural, slow-moving substitution threat from lithium-ion in select non-automotive use cases.

3.4 4. Lithium-Ion Batteries

How they work: A lithium-ion cell stores energy through the reversible intercalation (insertion) and de-intercalation of lithium ions between layered crystal-lattice structures at the cathode and anode. During discharge, lithium ions move from the anode (typically graphite) through the liquid electrolyte to the cathode (a lithium metal-oxide or lithium metal-phosphate compound, depending on chemistry — see Part V), while electrons travel through the external circuit to do useful work; charging reverses the ion flow. Because no

bulk chemical transformation of the electrode occurs (unlike lead-acid's conversion to lead sulfate), the process is far more reversible over many cycles, which is the fundamental reason lithium-ion cycle life vastly exceeds lead-acid's.

Advantages: High energy density (enabling lighter, more compact batteries for a given stored-energy requirement — critical for vehicles and portable electronics); long cycle life; no “memory effect”; high round-trip efficiency; increasingly modular and scalable from small consumer cells to grid-scale multi-megawatt-hour installations.

Disadvantages: Higher raw-material cost and price volatility (particularly for nickel- and cobalt-containing chemistries); greater sensitivity to overcharge, overheating, and physical damage, requiring a sophisticated Battery Management System (BMS) for safe operation; more complex, capital-intensive manufacturing process (dry-room/clean-room requirements, precision electrode coating, formation and aging steps — see Part III); recycling infrastructure is far less mature globally than lead-acid recycling, though it is improving rapidly.

Applications: Electric vehicles (two, three, and four-wheelers, buses, commercial vehicles), consumer electronics, power tools, grid-scale Battery Energy Storage Systems (BESS), telecom tower backup (replacing lead-acid), data-centre UPS (replacing lead-acid at the premium end), and increasingly, home/residential energy storage paired with rooftop solar.

Typical customers: Automotive OEMs (EV programmes), electric two/three-wheeler manufacturers, utility-scale renewable energy developers and Independent Power Producers (IPPs) procuring BESS, telecom tower companies, data-centre operators, and (in India specifically) government-backed grid-storage tenders issued by bodies such as SECI and NTPC.

Commercial maturity: Globally mature in cell manufacturing (dominated by Chinese and, to a lesser extent, Korean/Japanese producers) but structurally early-stage in India, where domestic cell manufacturing capacity is only now being commissioned (2026–2028 being the critical build-out window across all major Indian projects).

3.5 5. Sodium-Ion Batteries

How they work: Mechanically and electrochemically almost identical to lithium-ion — sodium ions intercalate between a cathode (commonly Prussian Blue analogues or layered transition-metal oxides) and an anode, through a liquid electrolyte — but substitutes sodium for lithium throughout. Because sodium is vastly more abundant and geographically ubiquitous than lithium (it can be sourced from seawater or common salt deposits almost anywhere in the world), this substitution removes the raw-material scarcity and geopolitical-concentration risk inherent to lithium-ion.

Advantages: Substantially lower and more geographically distributed raw-material cost (avoiding lithium, cobalt, and nickel entirely, and permitting cheaper aluminium current collectors on both electrodes instead of copper); excellent cold-weather performance (better capacity retention than LFP at sub-zero temperatures); can be safely discharged to

zero volts for transport and storage, materially simplifying logistics and reducing fire risk in shipping, versus lithium-ion's requirement to ship at a partial (typically ~30%) state of charge.

Disadvantages: Lower energy density than lithium-ion (roughly 140–160 Wh/kg versus 180–220+ Wh/kg for LFP), making it currently less suited to range-sensitive applications such as long-range passenger EVs; a newer, less-proven technology at commercial scale, with a shorter track record on long-term cycle-life and calendar-life data than lithium-ion.

Applications (near-term): Cost-sensitive, weight-insensitive applications — grid-scale stationary storage, low-cost electric two/three-wheelers, and potentially entry-level EVs — where the lower cost and simpler, geopolitically-independent supply chain outweigh the energy-density penalty.

Commercial maturity: Early commercial stage globally (led by Chinese players such as CATL), earlier still in India. Reliance Industries (via its acquisition of UK-based Faradion) and KPIT Technologies are the most visible Indian entrants pursuing indigenous sodium-ion technology, but neither Exide nor Amara Raja has, as of the source documents underlying this dossier, disclosed a committed sodium-ion program — this is a technology to monitor for potential entry by either incumbent.

3.6 6. Flow Batteries and Other Emerging/Stationary-Only Chemistries

Flow batteries (most commonly vanadium redox flow batteries) store energy in liquid electrolytes held in external tanks rather than within the cell itself, allowing power (rate of discharge) and energy (total storage capacity) to be scaled independently — power is set by the size of the electrochemical stack, energy by the size of the tanks. This makes flow batteries attractive for very long-duration (4-hour-plus) stationary storage, where their advantages (near-unlimited cycle life, no degradation from deep discharge, no fire risk from thermal runaway in the same manner as lithium-ion) can outweigh their disadvantages (low energy density, large physical footprint, and materially higher capital cost per kWh than lithium-ion for shorter-duration applications). Flow batteries remain a niche, stationary-only technology globally and are not currently a disclosed focus area for either Exide or Amara Raja, but are worth monitoring as India's grid-storage duration requirements lengthen beyond the 2–4 hour window that lithium-ion BESS currently serves best.

Other emerging chemistries referenced qualitatively in industry commentary include solid-state lithium batteries (replacing the liquid electrolyte with a solid one, promising higher energy density and improved safety, but not yet at commercial scale for automotive/grid applications) and LMFP (lithium manganese iron phosphate — an LFP variant with manganese added to boost energy density by up to ~20% while retaining much of LFP's cost and safety profile), seen as a likely bridge chemistry between LFP and NMC through the early 2030s. # Part III — Battery Manufacturing

3.7 1. Raw Material Value Chain (Overview)

For **lead-acid**, the value chain is short and largely circular: mined lead ore (India has limited domestic reserves and imports refined lead/lead concentrate) or, increasingly, **recycled lead** from spent batteries → smelting/refining → lead oxide, lead alloy grids, and lead paste → cell assembly → finished battery. Because a well-run recycling operation can recover 95%+ of the lead in a spent battery, mature lead-acid manufacturers increasingly integrate backward into recycling (Exide's Chloride Metals subsidiary and Amara Raja's Circular Solutions/ARCS subsidiary are both examples of this), reducing dependence on virgin lead imports and providing a partial natural hedge against lead price volatility.

For **lithium-ion**, the value chain is longer and, for India specifically, currently import-dependent at nearly every upstream stage: mining (lithium, nickel, cobalt, graphite — see Part IV) → refining into battery-grade chemicals (lithium carbonate/hydroxide, nickel/cobalt sulfates) → precursor and cathode-active-material (CAM) / anode-active-material (AAM) synthesis → cell manufacturing (electrode coating, cell assembly, electrolyte filling, formation/aging) → module and pack assembly with a Battery Management System → end-application integration. India's domestic capability today is concentrated at the cell-manufacturing-and-downstream end of this chain; the upstream mining-and-refining stages remain a structural gap the country is only beginning to address (see Part IV).

3.8 2. Mining and Refining

Mining extracts the raw ore or brine containing the target element; refining/processing converts it into a battery-grade chemical form pure enough for use in active-material synthesis. This is capital- and energy-intensive, and for lithium in particular, differs meaningfully by source geology (brine evaporation versus hard-rock/spodumene roasting versus clay-based acid leaching) — detailed in Part IV.

3.9 3. Active Materials

Cathode Active Material (CAM): the compound that gives a lithium-ion cell its chemistry name — e.g., LiFePO_4 for LFP, LiNiMnCoO_2 for NMC. CAM synthesis involves precisely controlled co-precipitation of the constituent metal salts, followed by high-temperature calcination (firing) to form the final crystalline structure, and is generally the single most expensive component of a lithium-ion cell.

Anode Active Material (AAM): overwhelmingly graphite (natural or synthetic) in current commercial cells. Graphite processing (spheroidization and coating for natural graphite, or graphitization at very high temperatures for synthetic graphite) is itself a specialized, capital-intensive, and highly China-concentrated step of the value chain.

3.10 4. Cell Manufacturing

The core cell-manufacturing process for lithium-ion (common to both NMC and LFP, cylindrical, prismatic, and pouch formats) proceeds through:

1. **Electrode slurry mixing:** active material, conductive additive (typically carbon black), and binder are mixed into a slurry with a solvent.
2. **Coating and drying:** the slurry is coated onto thin metal foil current collectors — aluminium for the cathode, copper for the anode — and dried.
3. **Calendaring:** the coated foil is compressed to a precise, uniform thickness and density.
4. **Slitting:** the large coated foil rolls are cut into precise electrode widths.
5. **Cell assembly:** electrodes and separator are wound (cylindrical/some prismatic cells) or stacked (pouch and some prismatic cells) into the cell format, then inserted into the cell casing.
6. **Electrolyte filling and sealing:** performed in a strictly controlled dry room (very low humidity), since lithium-ion electrolytes react violently with moisture.
7. **Formation and aging:** the cell undergoes its first controlled charge/discharge cycles, forming a stable Solid Electrolyte Interphase (SEI) layer on the anode surface that is critical to long-term cycle life and safety, followed by a period of aging/testing to screen out defective cells before shipment.

This process is far more precision- and cleanroom-dependent than lead-acid manufacturing, which is a key reason why lithium-ion gigafactories require multi-year commissioning and qualification timelines (as both Exide's and Amara Raja's dossiers document in detail) even for companies with decades of battery-manufacturing experience in the adjacent lead-acid business.

3.11 5. Module and Pack Assembly

Individual cells are grouped into **modules** (a fixed series/parallel arrangement of cells with basic interconnection and monitoring) and then into a complete **pack** (multiple modules plus the Battery Management System, thermal management, and structural/mechanical housing suited to the end application — an EV chassis-integrated pack, a stationary BESS container, or a telecom-tower cabinet). Pack engineering is where much of the application-specific value-add and customer customization occurs, and is a stage where several Indian companies (including Amara Raja's historical BESS/telecom pack business, and numerous smaller “pack assemblers” referenced in Part VII) have built capability well ahead of achieving domestic cell-manufacturing capability.

3.12 6. Battery Management Systems (BMS)

A BMS is the electronic control system that monitors and manages a lithium-ion pack's voltage, current, and temperature at the cell/module level, balances charge across cells to prevent any individual cell from being over- or under-charged, and triggers protective disconnection in fault conditions. BMS sophistication is a meaningful differentiator in lithium-ion pack quality, safety, and usable cycle life, and is an area where Indian companies with strong power-electronics/software capability (e.g., KPIT Technologies) have built a distinct competitive position even without cell-manufacturing scale.

3.13 7. Recycling and the Circular Economy

Lead-acid recycling is a mature, high-yield (95%+ lead recovery), well-regulated global industry, and — as noted above — both Exide and Amara Raja have integrated recycling operations that materially reduce their virgin-lead import dependence.

Lithium-ion recycling is a much younger and less standardized industry globally, typically using either pyrometallurgical (smelting) or hydrometallurgical (chemical leaching) processes to recover lithium, nickel, cobalt, and copper from spent cells and manufacturing scrap. As Indian lithium-ion manufacturing scales through the later 2020s, a domestic lithium-ion recycling industry will need to develop in parallel — both to manage end-of-life EV/BESS batteries and to recover valuable metals from gigafactory production scrap. Neither Exide nor Amara Raja has, per the source dossiers, announced a dedicated lithium-ion recycling facility as of mid-2026, though Exide's dossier flags this explicitly as an open question to monitor.

4 Part IV — Raw Materials

4.1 1. Lead

Role: The primary active material in lead-acid batteries (as lead dioxide at the cathode and metallic lead at the anode). **Supply chain:** Mined globally (Australia, China, Peru, Mexico, Russia among the largest reserve holders) but, critically for India, an estimated majority of lead-acid manufacturers' input can be sourced from **secondary (recycled) lead** recovered from spent batteries, given lead's near-complete recyclability. **Major producing countries:** China, Australia, Peru, USA, Mexico (primary/mined lead); recycling capacity is far more globally distributed, including a substantial informal/unorganized recycling sector in India that both major manufacturers seek to displace with organized, environmentally compliant recycling (Chloride Metals, Amara Raja Circular Solutions). **India dependence:** Meaningful reliance on imported refined lead and lead concentrate to supplement domestic recycling, though the exact self-sufficiency ratio is a figure both company dossiers flag as inconsistently disclosed (see Exide Part 11, Open Questions). **Price drivers:** London Metal Exchange (LME) lead price, a globally traded commodity; scrap/recycled lead availability and collection-network efficiency; regulatory tightening on informal recycling (a structural tailwind for organized players). **Risks:** Price volatility feeding directly into both companies' gross margins; environmental/regulatory risk around smelting and recycling operations; a slow-moving substitution risk from lithium-ion in several traditional lead-acid strongholds.

4.2 2. Lithium

Role: The core active-material element in all lithium-ion chemistries (LFP, NMC, NCA, LCO, LMO, LTO). **Supply chain:** Extracted from three principal geological sources — continental brines (South America's "Lithium Triangle": Chile, Argentina, Bolivia), hard-rock/pegmatite

spodumene ore (Australia, Canada), and aluminous clay deposits (parts of the USA and, notably for India, parts of Jammu & Kashmir and Chhattisgarh). **Major producing/reserve countries:** Chile and Australia hold the largest global reserves, followed by Argentina and China; production leadership (as distinct from reserves) is currently led by Australia (hard-rock) and Chile (brine). **India dependence:** Almost total import dependence today. India's first major domestic lithium discovery was announced in February 2023 in the Reasi district of Jammu & Kashmir (an inferred resource of 5.9 million tonnes, hosted in bauxitic clay and hard-rock structures), with additional exploratory indications identified in Chhattisgarh (Katghora) and Gujarat (Amba Dungar region, and geothermal brine near Dholera). Because India lacks the brine-flat geology that enables low-cost solar-evaporation extraction used in South America, Indian lithium extraction (where and when it proceeds to commercial mining) will require capital- and chemical-intensive hard-rock mining plus sulfuric-acid leaching and sodium-hydroxide-based purification — a materially more expensive extraction pathway than the South American brine route. Commercial-scale domestic lithium mining is not expected for several years (estimates of 5–7+ years for some deposits to reach production). **Price drivers:** Global EV and grid-storage demand growth, Chinese refining capacity and export-control policy, and the pace at which new brine/hard-rock supply (including from India's own nascent projects) comes online. **Risks:** Extreme price volatility historically (a multi-year swing of several multiples), geographic concentration of both mining and — especially — refining capacity in China, and (for the specific case of Jammu & Kashmir) genuine environmental sensitivity given the region's ecology.

4.3 3. Nickel

Role: A key cathode-active-material component in NMC and NCA chemistries, driving higher energy density (and hence longer EV range) at the cost of thermal stability and raw-material expense relative to LFP. **Supply chain and major producers:** Indonesia is the dominant global producer by a wide margin, followed by Australia, Brazil, and Russia (later two of secondary importance for battery-grade nickel specifically). **India dependence:** Import-dependent; domestic nickel reserves are negligible relative to demand. **Price drivers:** Indonesian export/processing policy (Indonesia has periodically restricted raw ore exports to force domestic downstream processing), stainless-steel demand (nickel's largest end market, competing with batteries for supply), and battery-grade (Class 1) nickel processing capacity specifically, which is more limited than total nickel supply. **Risks:** Concentration risk in a single country (Indonesia); processing-route emissions intensity (particularly for nickel pig iron/HPAL routes) is a growing ESG consideration for battery supply chains globally.

4.4 4. Cobalt

Role: A cathode component in NMC and NCA chemistries, improving cathode structural stability and cycle life; largely or entirely absent in LFP and LMO. **Supply chain and major producers:** The Democratic Republic of Congo (DRC) holds an overwhelming majority of global reserves and mined output (commonly cited at roughly 45–50%+ of reserves and around 70% of mined production), making cobalt the single most geographically

concentrated major battery raw material. **India dependence:** Fully import-dependent. **Price drivers:** DRC political/regulatory stability, informal/artisanal mining supply (a source of significant ESG scrutiny across the global battery industry), and the industry-wide trend of chemistry designs actively reducing or eliminating cobalt content (both to cut cost and to reduce supply-chain and human-rights risk) — a structural demand headwind for cobalt specifically, even as overall lithium-ion demand grows. **Risks:** The highest ESG/reputational risk profile of any major battery raw material, given documented concerns around artisanal mining conditions in the DRC; the primary reason global cell manufacturers (and, by extension, LFP's rising share in India) are moving away from cobalt-containing chemistries wherever performance requirements allow.

4.5 5. Graphite

Role: The anode-active-material in essentially all current commercial lithium-ion cells (both natural and synthetic graphite are used, sometimes blended). **Supply chain and major producers:** China dominates both mining and, especially, processing/graphitization — commonly cited at over 70% of global processed-graphite supply — with Turkey and Brazil holding meaningful natural-graphite reserves that are, as yet, far less processed domestically. **India dependence:** Import-dependent, particularly for battery-grade spheroidized/coated graphite and for synthetic graphite (which requires extremely high-temperature graphitization furnaces). **Price drivers:** Chinese processing capacity and export policy (graphite has been subject to Chinese export-control actions in recent years, given its designation as a dual-use/strategic material in some jurisdictions); the relative cost and performance trade-off between natural and synthetic graphite. **Risks:** Very high single-country concentration risk in processing, arguably the least diversified major link in the entire lithium-ion supply chain globally.

4.6 6. Manganese

Role: A cathode component in NMC (the “M” in NMC) and in LMO/LMFP chemistries; generally cheaper and more geographically abundant than nickel or cobalt, and increasingly used to boost LFP's energy density in the emerging LMFP variant. **Supply chain:** More geographically diversified than lithium, cobalt, or graphite (South Africa, Gabon, Australia, and others hold significant reserves); a lower-severity supply-risk profile than the other cathode metals discussed above. **India dependence:** India has some domestic manganese ore production (used predominantly in steel-making today), but battery-grade manganese sulfate processing capacity is nascent.

4.7 7. Copper and Aluminium

Role: Copper is the anode-side current collector foil in virtually all lithium-ion cells (chosen because it does not electrochemically alloy with lithium at anode potentials, unlike aluminium); aluminium is the cathode-side current collector foil, and — in sodium-ion cells specifically — can replace copper on the anode side too, since sodium does not alloy with aluminium the way lithium does, a meaningful cost advantage for sodium-ion. **Supply chain:** Both are globally traded, well-established commodities with diversified production (Chile/Peru/China for copper; China/Russia/India itself for aluminium, where India is in

fact a significant global producer). **India dependence:** Lower relative risk than the battery-specific metals above, given India’s own aluminium production base and copper’s status as a globally liquid, diversified commodity.

4.8 8. Electrolytes and Separators

Electrolyte: For lithium-ion, typically a lithium salt (commonly lithium hexafluorophosphate, LiPF_6) dissolved in an organic solvent blend; for lead-acid, dilute sulfuric acid; for sodium-ion, an analogous sodium-salt-based liquid electrolyte. **Separator:** A microporous polymer membrane (commonly polyethylene or polypropylene, sometimes ceramic-coated) that physically separates the electrodes while permitting ion flow; separator quality is a meaningful safety differentiator, since separator failure (from puncture, thermal shrinkage, or manufacturing defect) is a leading mechanical cause of internal short-circuits and thermal-runaway events in lithium-ion cells. **India dependence:** Both electrolyte-salt and separator-film manufacturing are nascent domestically; India’s broader chemicals industry (large domestic sulfuric-acid and caustic-soda/chlor-alkali production bases, led by companies such as Tata Chemicals, Grasim Industries, GACL, and DCM Shriram) provides a strong foundation for the acid/alkali inputs needed for lithium extraction and refining specifically, even though electrolyte-salt and separator-film capability remains a separate, less-developed capability gap. # Part V — Battery Technologies (Lithium-Ion Chemistry Deep Dive)

4.9 1. Overview Comparison Table

Chemistry	Cathode Composition	Energy Density	Cycle Life	Thermal Safety	Relative Cost	Primary Applications
LFP (Lithium Iron Phosphate)	LiFePO_4	Moderate (~120–160 Wh/kg cell-level)	Very high (3,000–6,000+ cycles)	Excellent — highest thermal-runaway threshold of the mainstream chemistries	Low (no cobalt/nickel)	Mass-market/standard-range EVs, grid-scale BESS, telecom/stationary backup, commercial 3-wheelers
NMC (Nickel Manganese Cobalt)	LiNiMnCoO_2 (various Ni:Mn:Co ratios, e.g. 6:2:2, 8:1:1)	High (~200–250+ Wh/kg cell-level)	Moderate (1,000–2,000 cycles)	Moderate — higher thermal-runaway risk than LFP	High (uses cobalt and nickel)	Long-range/premium passenger EVs, electric 2-wheelers (energy-dense, space-constrained)

Chemistry	Cathode Composition	Energy Density	Cycle Life	Thermal Safety	Relative Cost	Primary Applications
NCA (Nickel Cobalt Aluminium)	LiNiCoAlO ₂	High (similar to/slightly above NMC)	Moderate	Moderate-to-lower	High	Historically Tesla's primary chemistry for long-range vehicles; less prevalent in India
LMO (Lithium Manganese Oxide)	LiMn ₂ O ₄	Moderate-low	Moderate	Good	Low-moderate	Often blended with NMC for power tools and some EV applications requiring high discharge rates
LCO (Lithium Cobalt Oxide)	LiCoO ₂	High (volumetrically)	Lower	Lower	High (high cobalt content)	Consumer electronics (smartphones, laptops) — the original commercial lithium-ion chemistry; rarely used in automotive/grid scale applications today
LTO (Lithium Titanate)	Li ₄ Ti ₅ O ₁₂ (anode-side chemistry, distinct from the cathode-based naming above)	Low	Extremely high (10,000+ cycles)	Excellent	High	Fast-charging applications, some transit-bus and grid-frequency-regulation use cases where cycle life and charge speed matter more than energy density

4.10 2. LFP (Lithium Iron Phosphate) — Detail

Energy density/safety/cost/life-cycle: LFP sacrifices roughly 20–35% energy density versus NMC in exchange for a substantially higher thermal-runaway threshold (commonly cited around 270°C versus roughly 210°C for standard NMC), 2–4x longer cycle life, and 20–30% lower cost per kWh, since it eliminates the two most expensive and geopolitically sensitive cathode inputs — nickel and cobalt — entirely.

Applications: The default chemistry choice for standard-range passenger EVs globally (led by China’s BYD and CATL), for all forms of grid-scale BESS (where weight/volume is not a binding constraint but 10+ year cycle life is essential), for telecom and stationary backup, and — in India specifically — for electric three-wheelers and other cost-sensitive commercial mobility applications.

Future outlook: LFP is projected to hold roughly 60–65% of India’s lithium-ion demand by the early 2030s, and both Exide (via its Bengaluru gigafactory Phase 1 prismatic line, targeted principally at telecom/BESS/3-wheeler use cases, and the Hyundai/Kia passenger-EV supply agreement) and Amara Raja (which has flagged LFP as the likely eventual chemistry for its BESS-facility capacity, pending final confirmation) are building meaningful LFP exposure alongside their initial NMC/cylindrical-cell lines.

4.11 3. NMC (Nickel Manganese Cobalt) — Detail

Energy density/safety/cost/life-cycle: NMC’s higher energy density is what enables longer driving range in a given battery-pack volume/weight — the primary reason premium and long-range EVs still favour it over LFP — but at the cost of shorter cycle life, higher thermal-runaway risk, and materially higher and more volatile raw-material cost given its nickel and cobalt content.

Applications: Long-range/premium passenger EVs (particularly in markets or vehicle segments less sensitive to sticker price), and high-speed electric two-wheelers, where cabin/chassis space is severely constrained and maximizing range per litre of battery volume matters more than absolute cost.

Future outlook: Projected to hold a shrinking but still-meaningful roughly 25–30% share of India’s lithium-ion demand by the early 2030s, retaining a “premium stronghold” in long-range passenger vehicles and 2-wheelers even as LFP wins on volume. Both Exide’s and Amara Raja’s Phase 1 gigafactory lines began with (or split evenly with) NMC cylindrical-cell capacity, reflecting the earlier commercial readiness of cylindrical NMC cell formats and technology-partner expertise (SVOLT for Exide, Gotion for Amara Raja) at the time each project was launched.

4.12 4. NCA, LMO, LCO, LTO — Brief Notes

These four chemistries are of secondary relevance to the two companies covered in this dossier as of mid-2026 (neither Exide nor Amara Raja has disclosed NCA, LMO, LCO, or LTO as a primary planned chemistry), but are included for completeness given their roles elsewhere in the global lithium-ion landscape: NCA remains most associated with certain

long-range EV programmes historically pioneered by Tesla; LMO is typically blended with NMC rather than used standalone, valued for high discharge-rate capability; LCO remains the dominant chemistry for consumer electronics given its high volumetric energy density in small form factors, but is essentially absent from automotive/grid applications; LTO trades energy density for exceptional cycle life and fast-charge capability, finding niche use in some transit and grid-frequency-regulation applications where those attributes are paid a premium.

4.13 5. LMFP — The Emerging Bridge Chemistry

Adding manganese to an LFP cathode (creating LMFP, lithium manganese iron phosphate) is expected to lift energy density by up to roughly 20% while largely preserving LFP's cost and safety advantages, positioning it as a credible bridge between the LFP and NMC camps by 2030. This is a chemistry to monitor for potential future adoption by either Exide or Amara Raja as their initial LFP/NMC lines mature and as global technology transfer partners (SVOLT, Gotion, and others) bring LMFP capability to market. # Part VI — The Indian Battery Industry

4.14 1. Market Structure

India's battery industry today is best understood as two markets at very different stages of maturity, layered on top of each other:

- **Lead-acid** is a mature, consolidated, largely domestic-manufacturing duopoly. Exide Industries and Amara Raja Energy & Mobility together dominate the organized lead-acid market across automotive (OEM and replacement) and industrial applications, with smaller organized players and a meaningful unorganized/informal sector (particularly in recycling) alongside them.
- **Lithium-ion** is an early-stage, capital-intensive, multi-player race to localize a value chain that today is almost entirely import-dependent (cells, and much of the upstream chemical/material supply chain, are sourced overseas — overwhelmingly from China). As of the underlying research for this dossier (mid-2026), India's actual *operational* domestic cell-manufacturing capacity remains under 1 GWh—low single-digit GWh, against a cumulative *announced* pipeline exceeding 180–220+ GWh across all projects — a gap between ambition and delivered capacity that is the central execution risk across the entire domestic industry, not specific to any one company.

4.15 2. Growth Drivers

- **Automotive parc growth and replacement-cycle demand** underpin steady, mid-single-to-low-double-digit lead-acid volume growth.
- **Electrification of mobility** (two, three, and four-wheelers) is the largest single demand driver for new lithium-ion capacity, alongside a parallel, slower substitution of lead-acid within existing internal-combustion vehicles' 12V auxiliary-battery role (see Part II).

- **Renewable energy build-out and grid-balancing needs** drive the BESS opportunity — India's growing solar and wind capacity requires proportionate storage to manage intermittency, creating large utility-scale tender pipelines (SECI, NTPC, state utilities, and private developers such as Adani Green, Reliance New Energy, JSW Energy, and Tata Power Renewable).
- **Data centre and telecom infrastructure growth** support continued industrial lead-acid demand (data centres still favour lead-acid-based UPS in much of the installed base, even as some premium/hyperscale facilities begin trialing lithium-ion) while simultaneously being one of the fastest lithium-ion-conversion fronts within the industrial segment (telecom towers have already substantially converted from lead-acid to lithium-ion at the private tower-company level).
- **Government policy**, discussed in detail below.

4.16 3. Government Policy

- **GST reform:** A significant, dateable demand catalyst — battery GST was cut from 28% to 18%, and solar-equipment GST from 12% to 5%, directly lowering end-consumer prices and pulling forward demand (both Exide's and Amara Raja's FY26 results explicitly attribute part of their H2 FY26 demand strength to this GST 2.0 pass-through).
- **Production-Linked Incentive (PLI) schemes for Advanced Chemistry Cells (ACC):** The principal central-government mechanism incentivizing domestic lithium-ion cell manufacturing capacity, allocating capacity to winning bidders (Reliance and Ola Electric are cited as PLI-scheme allocation recipients in the source material) and shaping the competitive geography of the gigafactory build-out.
- **Import-duty structure:** Import duties on finished cells/batteries (cited at up to ~18% in some contexts) are a meaningful cost disadvantage for importing finished lithium-ion product versus domestic manufacturing, once domestic capacity is qualified and price-competitive.
- **Critical minerals strategy:** The government-backed joint venture Khanij Bidesh India Limited (KABIL — jointly owned by NALCO, Hindustan Copper, and MECL) is tasked with securing overseas critical-mineral mining rights and joint ventures, reflecting the strategic priority placed on reducing import dependence for lithium, cobalt, nickel, and related inputs.
- **China policy interaction:** Chinese export-control actions on lithium-ion-related raw materials, precursor chemicals, and manufacturing know-how are a recurring, real risk factor for Indian gigafactory projects that rely on Chinese technology-transfer partnerships (both Exide's SVOLT partnership and Amara Raja's Gotion partnership are exposed to this dynamic) — both companies' dossiers flag specific dates (e.g., a China export-restriction delay to November 2026) as items to track.

4.17 4. Import Dependence and the Domestic Manufacturing Ecosystem

As detailed in Part IV, India's lithium-ion supply chain is import-dependent at nearly every upstream stage — raw lithium, nickel, cobalt, and graphite; refined battery-grade chemicals; and, to a large extent, cathode/anode active materials and cell-manufacturing

equipment itself (most Indian gigafactory projects rely on technology transfer and/or joint ventures with established Chinese or other international players: SVOLT for Exide, Gotion for Amara Raja, and various technology partnerships underpinning the Reliance, Tata/Agratas, and Ola projects). The strategic logic across the industry is to first localize cell manufacturing and downstream pack/module assembly — capturing the largest single value-add step and reducing landed-cost/import-duty exposure — while building out upstream mining, refining, and materials capability over a longer horizon.

4.18 5. The Domestic Gigafactory Pipeline (Illustrative Snapshot, Mid-2026)

The following reflects the state of publicly discussed projects as of mid-2026 research; given the pace of announcements and revisions in this sector, figures should be treated as directional and re-verified against primary sources (company filings, investor presentations) before being relied upon in isolation.

Project Initiator	Technology/J V Partner	Target Capacity	Chemistry Focus	Approx. Project Cost	Indicative Commercialization Timeline
Reliance Industries (Reliance New Energy)	In-house R&D (including the Faradion sodium-ion acquisition)	~120 GWh ultimate (40 GWh Phase 1)	Heavily LFP-focused (plus sodium-ion)	Part of a much larger, tens-of-thousands-of-crore energy-complex investment	Ramp-up through H2 2026 and beyond
Tata Group (Agratas Energy Storage Solutions)	In-house/global R&D	~20 GWh Phase 1	Balanced NMC/LFP split	~₹13,000 crore	2027
Ola Electric	In-house R&D	~20 GWh ultimate (6 GWh Phase 1)	Primarily NMC (cylindrical, e.g. 4680-format)	~₹8,000+ crore	Operational and expanding through late 2026
Exide Energy Solutions (Exide Industries)	SVOLT Energy (China)	12 GWh ultimate (6 GWh Phase 1, split evenly 3 GWh NMC	50:50 Phase 1 split; revenue mix skews LFP-first (telecom/BESS/3-wheeler qualify faster than	~₹6,000 crore (cumulative equity infusion ₹4,802cr by end-FY26)	Customer samples from May 2026 (cylindrical) / June–July 2026 (prismatic); commercial revenue from

Project Initiator	Technology/J V Partner	Target Capacity	Chemistry Focus	Approx. Project Cost	Indicative Commercialization Timeline
		cylindrical / 3 GWh LFP prismatic)	automotive)		~Q3 FY27
Amara Raja (Giga Corridor, via ARACT)	Gotion High-Tech (via the GIB EnergyX structure)	16 GWh ultimate	Starts NMC (Giga-1, cylindrical), scaling toward LFP for the separate BESS facility	~₹9,500 crore total commitment	Giga-1 Phase 1 (2 GWh cylindrical) targeted for H1 CY2027 start of production; BESS facility targeted for 5 GWh by Q4 FY27/end-CY2026; Customer Qualification Plant commissioned in FY26

Chemistry split: LFP is estimated to command roughly 60–65% of India’s planned lithium-ion capacity, favoured for grid-scale BESS and mass-market/commercial EVs; NMC roughly 25–30%, retained for premium passenger EVs and high-speed two-wheelers; the balance split across other chemistries including early sodium-ion capacity.

Year-wise operational (not merely announced) capacity is expected to scale from a low single-digit GWh base in 2026 to a realistic 25–40 GWh range by 2027–2028 as the first wave of gigafactories reach commercial production, with total domestic demand by 2030–2031 projected in the 115–210 GWh/year range depending on the pace of EV adoption and grid-storage build-out — implying that announced *capacity* (180–220+ GWh) may substantially exceed near-term *demand*, a dynamic worth monitoring for potential overcapacity/pricing-pressure risk in the back half of this decade.

4.19 6. BESS (Battery Energy Storage Systems) — A Distinct, Fast-Growing Sub-Market

BESS deployment in India spans two layers: (a) **manufacturers** who physically build battery packs, containerized modules, or cells for storage applications, and (b) **project developers/utilities** who deploy storage at grid scale under government or private tenders. As of mid-2026 research, more than 15 active manufacturers and over a dozen major utility-scale developers were identified as participating in India’s BESS build-out, led on the developer side by Adani Green Energy (which has commissioned one of India’s largest single BESS installations, at its Khavda Renewable Energy Park in Gujarat), Reliance

New Energy, JSW Energy, NTPC Green Energy, and Tata Power Renewable Energy — each pursuing multi-gigawatt-hour-scale pipelines through 2027–2030.

For Exide and Amara Raja specifically, BESS represents a large, still relatively open (i.e., less competitively saturated than the EV-cell market) addressable market for their new lithium-ion capacity, distinct from the automotive-OEM qualification cycle, and one where Amara Raja in particular has built an early lead in distributed lithium-storage deployment (telecom-site BESS) well ahead of its own cell-manufacturing capacity coming online.

4.20 7. Future Trends to Monitor

- The pace at which *announced* gigafactory capacity converts into *operational, commercially-qualified* capacity (the single largest execution-risk variable across the sector).
- The LFP/NMC/LMFP chemistry mix as it actually plays out versus current projections.
- Whether India’s own lithium mining projects (Jammu & Kashmir, Chhattisgarh) progress from exploration/inferred-resource status toward commercial extraction, which — on a multi-year horizon — would be the single most significant structural change to India’s import-dependence profile.
- The emergence (or non-emergence) of sodium-ion as a credible third chemistry pathway for cost-sensitive Indian applications.
- Continued GST/policy support (or withdrawal) for both lead-acid and lithium-ion end markets.
- Chinese export-control policy on raw materials, precursor chemicals, and technology transfer, given the current dependence of nearly every major Indian gigafactory project on a Chinese (or in some cases global) technology partner. # Part VII — Ecosystem Mapping

This section maps the players across the Indian battery value chain, based on industry research conducted to date. Company inclusion/exclusion and capacity figures should be treated as a snapshot (mid-2026) requiring periodic refresh rather than a static, exhaustive registry.

4.21 1. Lithium-Ion Cell Manufacturers (Gigafactory Scale)

Company	Manufacturing Capability	Declared/Target Capacity
Exide Energy Solutions (Exide Industries)	Cell manufacturing (NMC cylindrical + LFP prismatic), module assembly, utility-scale stationary storage	6 GWh initial phase, expanding to 12 GWh
Amara Raja Energy & Mobility (via ARACT)	Lithium-ion cell engineering, LFP integration, cell prototyping	16 GWh customer-validation pipeline under development
Reliance Industries (Reliance New Energy)	Cell manufacturing, in-house R&D, sodium-ion (via Faradion)	40 GWh Phase 1, up to ~120 GWh ultimate

Company	Manufacturing Capability	Declared/Target Capacity
Tata Group (Agratas Energy Storage Solutions)	Cell manufacturing (NMC + LFP)	20 GWh Phase 1
Ola Electric (Ola Cell Technologies)	Domestic cell manufacturing (NMC cylindrical), EV and BESS-oriented	~20 GWh under the PLI scheme
Waaree ESS	Multi-MWh container integration, structural integration for solar-plus-storage	~20 GWh long-term roadmap
DC&T Global	Turnkey containerized BESS manufacturing	~3 GWh operational
IB Solar	LFP-focused safety-standard manufacturing	~1.8 GW

4.22 2. Pack, Module, and Specialized Assembly Manufacturers

Company	Focus
Log9 Materials	India's first fully indigenous commercialized lithium-ion cells; rapid-charging commercial-fleet focus
TDK Corporation	Large battery-pack facility (Sohna, Haryana) serving consumer electronics and energy storage
Tata AutoComp Gotion	EV battery pack supply to automakers (joint venture)
Cygni Energy	Automated EV battery-pack facility, Telangana
Bharat Heavy Electricals Ltd. (BHEL)	Specialized industrial pack assembly, Bengaluru
GP Eco Solutions (iNVERGY)	Commercial/industrial BESS blocks, 400 kWh–5 MWh range
Luminous Power Technologies, Servotech, Delta Electronics India, Replus Engitech, Statcon Energias, Battrix	Distributed/modular pack assembly for residential, commercial, and light-EV applications

4.23 3. Chemical, Component, and Materials Suppliers

Company	Role in the Battery Value Chain
Neogen Chemicals	Electrolyte salts and related battery chemicals
Gujarat Fluorochemicals	PVDF binders and other battery materials
Himadri Speciality Chemical	Cathode and anode material expansion
Aether Industries	Specialty chemicals relevant to battery

Company	Role in the Battery Value Chain
	manufacturing
Tata Chemicals, Grasim Industries, Gujarat Alkalies and Chemicals (GACL), DCM Shriram	Sulfuric acid and caustic soda/soda ash — key inputs for lithium ore leaching and purification
Coromandel International, Paradeep Phosphates, Hindalco Industries, Vedanta (Sterlite/HZL), Khaitan Chemicals & Fertilizers	Sulfuric acid production (largely byproduct-based, from fertiliser and metal-smelting operations)

4.24 4. Lead Recyclers (Circular Economy — Lead-Acid Side)

Company	Role
Chloride Metals Limited (wholly owned subsidiary of Exide Industries)	Three lead-recycling plants; backward integration for Exide's lead-acid value chain
Amara Raja Circular Solutions Private Limited (ARCS)	Lead-recycling refinery (Cheyyar), backward integration for Amara Raja's LAB value chain; battery-breaking operations targeted from Q2 FY27

4.25 5. OEM Customers and Offtake Relationships

Segment	Representative Customers/Relationships
Automotive OEM (lead-acid)	Passenger-vehicle and two-wheeler manufacturers across the Indian market
Automotive OEM (lithium-ion, emerging)	Hyundai Motor Group / Kia Corporation (Exide — separate, incremental supply agreement to the Phase 1 gigafactory); Ather Energy, Piaggio (Amara Raja — MoU-stage as of the underlying research, not yet confirmed as firm binding offtake)
Electric-vehicle assembly (broader ecosystem)	Tata Motors, Mahindra & Mahindra, Ola Electric, TVS Motor — positioned as beneficiaries of localized cell supply reducing import-duty and logistics cost

4.26 6. Battery Integrators, BESS Developers, and Utility-Scale Buyers

Company	Role
Adani Green Energy (AGEL)	India's largest single active grid-scale BESS site footprint (Khavda Renewable Energy Park, Gujarat); ~15 GWh deployment target by March 2027, scaling toward 50 GWh over five years
Reliance New Energy	Jamnagar Green Industrial Nerve Centre; large PLI-linked ACC allocation; multiple solar-plus-storage PPAs (including with SECI)
JSW Energy (JSW Neo Energy)	~4.2 GWh active energy-storage pipeline; ₹58+ billion approved for dedicated BESS plant construction; 40 GWh global target by 2030
NTPC Green Energy Limited (NGEL)	BESS integration across thermal-plant zones; ~2,334 MWh awarded/under construction
Tata Power	Co-located BESS within a 26+ GW clean-energy pipeline; landmark

Company	Role
Renewable Energy (TPREL)	100 MW Mumbai load-centre BESS project

4.27 7. Key Global Technology/Supply Partners

Partner	Relationship
SVOLT Energy (China)	Technology-transfer partner for Exide Energy Solutions' Bengaluru gigafactory
Gotion High-Tech (via GIB EnergyX)	Technology-transfer partner for Amara Raja's Giga Corridor
Johnson Controls International (historical)	21-year joint-venture technology partner for Amara Raja's lead-acid business (1995/1997–2019; VRLA and PowerFrame® Grid Technology)
Leclanché SA (historical)	Original joint-venture partner for Exide's Nexcharge lithium-ion pack/module business (2018–2022, before Exide bought out the stake)

4.28 8. The Rajan Raheja Group — Exide's Ownership Context

Exide Industries is the flagship listed asset of the **Rajan Raheja Group**, a diversified Indian conglomerate whose other significant holdings span:

- **Automotive & Industrial Manufacturing:** Exide Industries Ltd. (flagship); Supreme Petrochem Ltd. (polystyrene/polymer materials joint venture); Standard Batteries Ltd.; Innovassynth Technologies (India) Ltd. (custom pharmaceutical/chemistry research and manufacturing).
- **Building Materials & Construction:** Prism Johnson Ltd. (formerly Prism Cement); H&R Johnson (India) (ceramic tiles, bath products); RMC Readymix (India) Pvt. Ltd.
- **Information Technology & Media:** Sonata Software Ltd. (IT consulting/software services); Hathway Cable & Datacom Ltd. (co-owned with Reliance Industries); Asianet Satellite Communications; Outlook Publishing (India) Pvt. Ltd.
- **Retail & Hospitality:** Globus Stores Pvt. Ltd.; Health & Glow; EIH Associated Hotels Ltd. (hotel assets alongside the Oberoi group).
- **Financial Services:** Raheja QBE General Insurance Co. Ltd. (joint venture with Australia's QBE Insurance).

[ANALYST INTERPRETATION] This diversified group structure is relevant primarily as governance/related-party context (Exide's dossier separately tracks related-party transactions and board composition in Section 6.3) rather than as a source of direct operational synergy with the battery business — none of the group's other holdings appear, from the research conducted to date, to feed materially into Exide's battery-industry economics. # Part VIII — Exide Industries Limited

The following is the complete Exide Industries Internal Research Dossier, integrated in full and unabridged. Only heading levels have been adjusted (each demoted by one level) so the

dossier nests correctly within this master document's table of contents; no content, conclusion, tag, or figure has been altered, shortened, strengthened, or weakened from the original.

EXIDE INDUSTRIES LIMITED

Internal Research Dossier

(Living Document — Institutional-Quality Research Notebook)

Prepared by: Deepesh Joshi | Indira Securities (Junior Equity Research Analyst)

Document version: v1.0 | Compiled: 14 July 2026

NSE: EXIDEIND | BSE: 500086 | CSE: 15060/10015060 | CIN: L31402WB1947PLC014919

Source documents integrated in this version:

- NotebookLM research conversation session — "Exide Industries Limited" (22 sources: Annual Reports FY20–FY26, Credit Ratings, Earnings Calls, Equity Research, Investor Presentations), sessions dated 2–13 July 2026
- Q2 FY26 Earnings Conference Call transcript, 17 November 2025 (management: Avik Roy – MD & CEO, Manoj Kumar Agarwal – Director Finance & CFO, Pravin Saraf – then Executive Director)
- Q3 FY26 Earnings Conference Call transcript, 3 February 2026 (Pravin Saraf now MD & CEO, Exide Energy Solutions Ltd.)
- Q4 & FY26 Earnings Conference Call transcript, 6 May 2026
- Investor Presentation, November 2025 (covering Q2/H1 FY26 results)
- Investor Presentation, Q4 & FY26 Results, May 2026

How to Use This Dossier

This is not an equity research report, initiation note, or client presentation. It is the master internal working file for everything learned about Exide Industries across annual reports, concalls, investor presentations, and independent NotebookLM-assisted research. It is meant to let you (or another analyst) rebuild full context on the company after months away, without re-reading every source document.

Throughout the document, statements are tagged so you can tell fact from interpretation:

[FACT] A verifiable, disclosed fact (financial figure, stated policy, board composition, etc.)

[MGMT COMMENTARY] *Management's own framing/commentary, which may be promotional and should be weighed against facts, not treated as fact itself*

[ANALYST INTERPRETATION] A conclusion, pattern, or read-between-the-lines interpretation developed either by the NotebookLM research sessions or by synthesis across sources — treat as a hypothesis, not established truth

[OPEN QUESTION] A genuine gap in current knowledge — something management has not disclosed, something sources conflict on, or something to chase in the next call

Update this document after every quarterly result, annual report release, or major corporate action. When new information contradicts an old conclusion, don't delete the old conclusion silently — note that the view changed and why.

4.29 1. Company Overview

4.29.1 1.1 Corporate Identity & Ownership

[FACT] Exide Industries Limited is India's largest lead-acid storage battery manufacturer, headquartered at Exide House, 59E Chowringhee Road, Kolkata 700020. CIN: L31402WB1947PLC014919. Listed on NSE (EXIDEIND), BSE (500086) and the Calcutta Stock Exchange (15060/10015060) since 1979.

[FACT] Exide is part of the Rajan Raheja Group, which acquired control of the company in 1993. The primary promoter entity is Chloride Eastern Limited (UK), indirectly controlled by individuals including Mr. S. B. Raheja. As of 31 March 2026, promoters held 46% of the equity (45.99% as reported mid-2026), Domestic Institutional Investors ~18–19%, Foreign Institutional Investors ~10–11%, and the remainder (~25%) with other public shareholders. This shareholding mix has been essentially stable through FY26.

[FACT] The company has been profitable in every year since inception and has a long, continuous dividend payout history — averaging roughly a 16% payout ratio in recent years.

[FACT] Credit ratings (ICRA): Long-term AAA/Stable, Short-term A1+. Standalone balance sheet has been debt-free for over a decade. Statutory auditor: B S R & Co. LLP.

4.29.2 1.2 History & Key Milestones

Year	Milestone
1946	Commenced manufacturing operations
1973	Pioneered India's first polypropylene battery
1976	Established in-house R&D centre in Kolkata (now ~50 years old, 100+ R&D staff, 4 NABL-accredited labs)
1979	Listed on stock exchanges
1983	Launched India's first indigenous OEM battery (for Maruti)
1992	Delivered first submarine batteries to the Indian Navy
1993	Acquired by the Rajan Raheja Group
1998	Acquired Chloride Metals (lead recycling business) and Standard Batteries
2001	Pioneered India's first tubular inverter battery
2011–12	Ventured into the Home UPS business
2018	Foray into Li-ion packs & modules via the "Nexcharge" joint venture with

Year	Milestone
	Leclanché SA (Switzerland); Prantij, Gujarat facility established (1.5 GWh pack/module assembly)
2022	Bought out Leclanché's JV stake (Nexcharge became wholly owned); incorporated Exide Energy Solutions Limited (EESL) as the dedicated Li-ion cell manufacturing subsidiary; original roadmap for a 12 GWh cell plant announced with an 8–10 year phased build-out
Mar 2024	Amalgamated the erstwhile Exide Energy Private Limited (Nexcharge pack/module business) into EESL
FY25	Transitioned organisation model from Strategic-Business-Unit (SBU) structure to the functional "One Exide" model (Trade/B2C, Institutional/B2B, International clusters)
2025–26	Bengaluru lithium-ion cell gigafactory (Phase 1, 6 GWh) nears commissioning; cylindrical (NMC) customer sample deliveries begin May 2026

Source: Q4 FY26 Investor Presentation, "Our Journey: Key Milestones"; NotebookLM chat session; Q2/Q3/Q4 FY26 concall transcripts.

4.29.3 1.3 Corporate Structure & Subsidiaries

- Exide Industries Limited (standalone) — core lead-acid battery manufacturing, India's largest, 11 domestic manufacturing plants across product categories.
- Chloride Metals Limited (CML) — wholly owned subsidiary; operates 3 lead-recycling plants; backward integration into the lead-acid value chain. Management states CML meets a majority of the group's lead requirement through recycling (see reconciliation note below on the exact %).
- Exide Energy Solutions Limited (EESL) — wholly owned subsidiary housing the lithium-ion business: (a) the Prantij, Gujarat pack/module assembly plant (1.5 GWh, operating, ex-Nexcharge) and (b) the new Bengaluru cell-manufacturing gigafactory (Phase 1: 6 GWh of a planned 12 GWh). Pravin Saraf is MD & CEO of EESL (appointed effective the Q3 FY26 quarter; joined the Exide board as Executive Director 1 September 2025).

[ANALYST INTERPRETATION] Group manufacturing footprint is quoted inconsistently across documents as "11 plants" (core business only, Nov-2025 PPT) vs. "13 plants" (Q4 FY26 PPT, explicitly stated as 11 core + 2 EESL plants). Treat 13 as the current all-in figure; the discrepancy is simply whether EESL plants are included, not a change in the underlying facilities.

4.29.4 1.4 Board of Directors (as of Q4 FY26, May 2026)

Name	Role	Background
Sridhar Gorthi	Chairman & Independent Director	LLB, National Law School of India University, Bengaluru. Leading M&A/JV/PE legal expert.
R B Raheja	Vice-Chairman & Non-	B.Com. 50+ years of business experience;

Name	Role	Background
	Executive, Non-Independent Director	promoter-family representative; active in educational/charitable trusts.
Avik Roy	Managing Director & CEO	BE (Electrical), Jadavpur University; EMBA, Asian Institute of Management, Manila. 34+ years' industrial leadership incl. senior roles at Siemens (India, Germany, Bangladesh).
Manoj Kumar Agarwal	Director – Finance & CFO (joined board 1 May 2025)	Commerce graduate, Goenka College of Commerce; qualified CA and CS. 30+ years in financial strategy, treasury, M&A, overseas subsidiaries.
Surin Kapadia	Independent Director	FCA; partner, G.M. Kapadia & Co. 25+ years in taxation, exchange-control law, valuation.
Radhika Govind Rajan	Independent Director	MBA, IIM Ahmedabad; MSc Physics, IIT Bombay. 40+ years in global financial markets (incl. New York); cross-border finance & PE specialist.
Jaidit Singh Brar	Independent Director	24 years at McKinsey; industrial growth, restructuring and energy-sector strategy across Asia.
Rajeev Khandelwal	Executive Director (joined board 1 September 2025)	BE + MBA (Marketing). 30+ years as a P&L leader in FMCG, building consumer brands across India/ASEAN/Africa. Leads the B2C trade business.
Pravin Saraf	Executive Director; also MD & CEO, Exide Energy Solutions Ltd. (joined board 1 September 2025)	BE (Mechanical) + MBA (Materials Management). 35+ years global manufacturing/supply-chain/digital-transformation experience; ex-Bosch.

[FACT] Governance structure: Chairman and CEO roles are separated. As of June 2026 the board composition is 89% men / 11% women, with 56% of directors classified independent. All major committees (Audit; Nomination & Remuneration) are chaired by Independent Directors.

[ANALYST INTERPRETATION] Two new executive appointments (Khandelwal for B2C trade, Saraf for EESL) both landed in September 2025 — coinciding almost exactly with the "One Exide" operating-model transition and the ramp-up phase of the lithium-ion project. Read this as management deliberately building out dedicated leadership bandwidth for (a) the core trade business and (b) the new-energy bet, rather than running both through the existing MD/CEO structure. Also note: Pravin Saraf's move from Executive Director (Exide Industries, generic operations/automation remit in FY26 Q2 call) to dedicated MD & CEO of EESL by Q3 signals EESL's growing strategic weight and the need for a focused P&L owner.

4.29.5 1.5 Scale Statistics (Quick Reference)

Metric	Detail
Years of legacy	79 years (as of Q4 FY26 PPT); ~78 years cited by Avik Roy on the Q3 FY26 call
Countries served	60+ (Nov-2025 PPT) / 70+ (Q4 FY26 PPT) — exports footprint is expanding into new geographies even as export revenue share has fallen
Manufacturing plants	11 core lead-acid + 2 EESL (Bengaluru + Prantij) = 13 total
Lead recycling plants	3 (via Chloride Metals Ltd.) — only Indian battery manufacturer with 3 large captive recycling plants
Global technical collaborations	5 (Nov-2025 PPT) / 3 "strategic" collaborations (Q4 FY26 PPT) — terminology/count varies by document; core partnerships include SVOLT (Li-ion technology) and East Penn Manufacturing/"Deka" technology (data-centre front-terminal battery design)
Network touchpoints	1,00,000+ dealers/distributors (Nov-2025 PPT) growing to 1,20,000+ network touchpoints (Q4 FY26 PPT), incl. 1,200+ spare-parts/accessories outlets and 60+ warehouses
OEM relationships	75+ OEMs served across the automotive industry
Product range	2.5 Ah to 20,200 Ah — one of the widest ranges of any battery maker globally
Market capitalisation	₹33,214 cr (30 Sep 2025) → ₹30,647 cr (30 Apr 2026) — a ~7.7% decline over ~7 months, tracking margin/growth softness and possibly EESL execution-risk repricing

4.30 2. Business Model

4.30.1 2.1 The "One Exide" Operating Model

[FACT] In FY25, Exide transitioned from a Strategic Business Unit (SBU)-led organisational model to a functional "One Exide" operating model, organised around three clusters: Trade (B2C), Institutional (B2B), and International.

[MGMT COMMENTARY] *Management states this was "intended to recalibrate our go-to-market approach across business verticals" and that during FY26 the model "enabled the Company to be more agile and customer-focused, while bringing synergies across the organisation."*

[ANALYST INTERPRETATION] The SBU→functional shift is one of the most important "soft" facts in the whole research base. Read literally, it is an efficiency/agility initiative. Read between the lines (developed through the NotebookLM cross-year synthesis), the previous SBU structure likely created internal silos that slowed the company's response to the lithium-ion disruption hitting Telecom and E-Rickshaw lead-acid volumes. The re-org

can be seen as an organisational admission that a battery is no longer a standalone product sale but an entry point into a broader customer energy relationship — which is also consistent with the parallel build-out of digital/data infrastructure (see section 2.4).

4.30.2 2.2 Revenue Segments — Structure

Exide's revenue can be sliced two ways: by channel (Trade/B2C vs. Institutional/B2B vs. International) and by end-application vertical. The vertical view is more useful for modelling:

Vertical	Channel	Notes
Automotive OEM	Institutional (B2B)	2W/3W/4W new-vehicle battery supply. Roughly half of total automotive revenue (auto OEM+aftermarket ~50% of total company revenue per FY26 commentary); replacement:OEM mix within automotive is roughly 70:30.
Automotive Replacement / Aftermarket	Trade (B2C)	2W/4W battery replacement — Exide's highest-margin, most defensible business; ~10–14% YoY growth sustained through FY26.
Industrial UPS (IUPS) incl. Data Centers	Institutional (B2B)	Part of the ~28–32% "Industrial" (non-automotive) revenue share; Data Centers is a fast-growing sub-vertical inside IUPS (see section 3).
Railways / Traction / Motive Power	Institutional (B2B)	Benefiting from near-100% railway electrification (targeted by FY26) and e-commerce/warehouse material-handling electrification.
Telecom	Institutional (B2B)	Structurally declining lead-acid vertical as towers migrate to lithium-ion. Fell from ~10% of revenue at its 5G-rollout peak to ~1% of revenue by Q3 FY26 (management describes it as "largely bottomed out").
Solar (Trade + Institutional)	Both	Off-grid battery+panel+inverter combos (Trade, GST-supported) and grid-scale utility solar (Institutional, driven by the 500 GW renewable target). Crossed ₹1,000 cr revenue in FY26; target ₹1,500 cr within 2–3 years.
Home UPS / Inverter Batteries	Trade (B2C)	Highly seasonal (peaks Mar–Jul, ~3x off-peak volumes); weather-sensitive (monsoon timing materially swings quarterly results).
Power & Projects / Industrial Infra	Institutional (B2B)	Lumpy, tender-driven; tied to large thermal-power and utility infrastructure capex cycles.
Exports	International	Historically ~8% of standalone turnover; fell to ~5.3% in FY26 due to geopolitics (West Asia conflict), Western European recession (Germany/UK/Italy), and US/EU tariff uncertainty.

Vertical	Channel	Notes
New Energy (Li-ion cells, packs, modules)	EESL (separate P&L)	Currently mostly pack/module assembly (Prantij) using imported cells (~₹100 cr/9 months run-rate per Q3 FY26 call); Bengaluru cell manufacturing (Phase 1, 6 GWh) approaching commercial revenue.

4.30.3 2.3 Distribution, Manufacturing & Value Chain

[FACT] Distribution: 1,20,000+ network touchpoints (Q4 FY26), including 1,200+ dedicated spare-parts/accessories outlets, 60+ warehouses, and 1,700+ "Exide Care" service outlets. Rural push via "Exide Battery Champions" village-level outreach, with presence in 18,000+ villages and 500+ rural meets conducted in FY26.

[FACT] "Batmobile" doorstep service — described by management as the industry's only such platform — handles 1.08 lakh+ customer requests per year across 300+ Indian cities, with an average response time of ~30 minutes and 300+ dedicated technicians. Customers can request service via a WhatsApp number (70440 00000).

[FACT] Manufacturing: 11 core lead-acid plants strategically located pan-India, integrated with 3 captive lead-recycling plants (Chloride Metals Ltd.) that provide raw-material recovery and cost/quality control. Recent capex has focused on automation — "Punched Grid" technology for 2-wheeler batteries (motorcycle capacity fully converted to Punched Grid by end of calendar 2025 per Q2 FY26 call) and "CONCAST" (continuous casting) for 4-wheeler/industrial/CV batteries — plus robotic palletising/depalletising, robotic sticker print & apply, and robotic vision systems for quality checks.

[MGMT COMMENTARY] *Management frames the automation programme as delivering three benefits simultaneously: cost savings, quality improvement (and consequently lower warranty costs), and "deskilling"/safety improvements on the shop floor.*

[FACT] Value chain / raw materials: Lead and lead alloys represent >70% of the bill of materials (BOM) by value; together with sulfuric acid and plastics (polypropylene), these three inputs cover ~95–96% of the BOM. Alloying elements include tin, silver, and antimony.

[ANALYST INTERPRETATION] There is a data conflict in the underlying research on CML's lead self-sufficiency ratio: one NotebookLM answer states CML meets ~79% of lead requirements via recycling; a separate answer in the same research thread states ~50–53%. This is flagged as an open item (see section 11) — the true figure should be pulled directly from the FY26 Annual Report's raw-material disclosure rather than relied upon from either NotebookLM answer.

[FACT] Pricing mechanics: ~20–45% of lead is imported or priced at import parity even when domestically sourced/recycled, because Indian lead trades at import-parity pricing. This means INR depreciation can fully offset falling LME lead prices — as happened through FY26 (LME softened, but INR depreciation of ~10% erased the benefit).

4.30.4 2.4 Digital & Customer-Facing Initiatives

- Dealer digitisation: "Exide Hub" app, dealer microsites ("Exide Care"), Google Search presence for 2,000+ dealers generating unique leads monthly, automated WhatsApp service-reminder alerts.
- Sales-force & channel-partner tools: AI-enabled sales analyzer and invoice processing, visit planning and churn alerts, and the "Exide Humsafar" loyalty app targeted at mechanics and electricians (scan-and-earn points programmes).
- E-commerce: Exide products listed on Amazon and Flipkart with an average 4+ user rating.
- 60 lakh+ paperless warranty registrations completed to date (per NotebookLM synthesis) — an explicit data-capture initiative.

[ANALYST INTERPRETATION] This digital layer (loyalty apps, warranty capture, AI sales tools, WhatsApp service alerts) is one of the clearer emerging-moat candidates in the research base. The strategic logic, as inferred by NotebookLM's cross-document synthesis, is that Exide is trying to build a direct data relationship with end consumers and channel partners that a regional/unorganised lead-acid competitor cannot replicate — turning a one-time battery purchase into a tracked, repeatable relationship (useful for both lead-acid replacement-cycle marketing and, longer-term, potential Li-ion/EV ecosystem cross-sell).

4.31 3. Segment Analysis

4.31.1 3.1 Automotive — OEM

[FACT] Auto OEM delivered back-to-back quarters of +20–30% YoY growth through FY26 (Q3 FY26: +25% YoY, highest-ever quarterly revenue; Q4 FY26: second consecutive quarter of +25% YoY growth, another highest-ever quarter). Driven by the GST 2.0 rate cut (28%→18% on batteries, effective 22 September 2025) and broader auto-sector demand recovery.

[FACT] New OEM wins/expansions: 100% supplier for the Tata Sierra petrol relaunch and the domestic Kia Seltos; 100% auxiliary-battery share on the Maruti e-Vitara; auxiliary battery supply to VinFast VF6/VF7 EVs.

[MGMT COMMENTARY] *Management repeatedly frames auto OEM growth as a leading indicator for the aftermarket/replacement business roughly two years forward — i.e., today's OEM fitment becomes tomorrow's replacement-cycle demand.*

4.31.2 3.2 Automotive — Replacement / Aftermarket

[FACT] 2W and 4W replacement demand grew consistently in the low-double-digit range through FY26 (Q2: ~10–11%; H1: ~11–12%; full-year commentary reiterates "mid-teens" for 2W/4W replacement in some quarters). This is Exide's highest-margin business.

[MGMT COMMENTARY] *Management was "a little worried" through H1 FY26 that a weak OEM cycle two years prior would eventually soften aftermarket demand, but stated by Q3*

FY26 that the OEM recovery had removed this concern, given the ~2-year replacement-cycle lag.

4.31.3 3.3 Industrial (Non-Automotive) — Overview

[FACT] The Industrial Division (ex-automotive) represented roughly 28–32% of standalone turnover across FY22–FY26, comprising Industrial UPS (incl. Data Centers), Railways/Traction/Motive Power, Telecom, and Power & Projects.

[MGMT COMMENTARY] *Management characterises B2B industrial demand as trailing the broader capex cycle — i.e., Exide's industrial order book reflects the tail end of infrastructure capex commitments made 1–2 years earlier.*

[FACT] Key demand drivers cited: near-complete railway electrification (~100% targeted by FY26), a new railway maintenance policy that replaces batteries on a fixed overhaul cadence rather than waiting for end-of-life (a structural tailwind for battery replacement volumes), 10–15% material-handling-equipment (forklift/motive power) electrification tied to e-commerce/quick-commerce warehouse growth, and rising data-centre capex.

4.31.4 3.4 Data Centers

[FACT] Data Centers sit within the Industrial UPS (IUPS) vertical, itself the largest sub-segment of the Industrial Division. Standalone quarterly Data Center revenue was quantified at only ~₹75–100 crore per quarter as of Q2 FY26 — a small base but with a strong pipeline of RFQs/tenders.

[FACT] Product: the "EHP" (Extended High Power) series and the Front Terminal Planished Grid (FTPG) range, developed via the East Penn Manufacturing ("Deka") technology partnership. Front-terminal design reduces installation footprint 20–40% vs. standard batteries — Exide states it is currently the only Indian manufacturer offering a full 340W–750W front-terminal portfolio for this segment.

[FACT] India's data-centre market is projected to grow at a ~20–21% CAGR medium-term, reaching ~3,400 MW of capacity by 2030.

[MGMT COMMENTARY] *Management notes the segment has a long ~3–4 quarter gestation period from "zero date" (initial RFQ) to order confirmation due to multi-consultant approval processes, and explicitly states not all data-centre demand is addressable by lead-acid — enterprise-tier facilities favour lead-acid for reliability/cost, while hyperscale/colocation facilities with severe footprint constraints increasingly favour lithium-ion (BESS), which is why EESL's LFP prismatic cell line is being developed partly with stationary/BESS use cases in mind.*

[ANALYST INTERPRETATION] Data Centers is being positioned by management as evidence that Exide is an "Infrastructure/Digital India" play, not merely an auto-ancillary stock — likely an attempt to earn a higher valuation multiple less tied to auto-sector cyclicality. The current revenue base (~₹300–400 cr/year estimated) is still too small to move the needle materially, so this remains a multi-year re-rating thesis rather than a near-term earnings driver.

4.31.5 3.5 Telecom

[FACT] Telecom lead-acid revenue has collapsed from ~10% of standalone turnover at the peak of the India 5G tower rollout to ~1% of revenue by Q3 FY26, as tower operators (Jio, Indus Towers, BSNL, etc.) migrated to lithium-ion. Revenue contraction in this vertical alone was 25–30% in FY25.

[MGMT COMMENTARY] *Management states this decline was fully anticipated and "factored into our business plan," and highlights that Exide was the first company to supply lithium-ion packs to Indus Towers, positioning the company to capture this volume migration through its own Li-ion business rather than losing it entirely to competitors.*

[ANALYST INTERPRETATION] Telecom is the clearest, most concrete real-world case study of lead-acid-to-lithium cannibalization actually happening inside Exide's own numbers — useful as a template for thinking about how fast E-Rickshaw (next) and eventually other verticals could migrate.

4.31.6 3.6 E-Rickshaw / Last-Mile

[FACT] Similar to Telecom, the e-rickshaw market has seen a faster-than-anticipated shift to lithium-ion, causing degrowth in Exide's lead-acid e-rickshaw battery volumes. This segment, together with Telecom, is grouped by management within the ~3% "E-Rickshaw + Telecom" component of the disclosed "8% drag."

[FACT] EESL's LFP prismatic cell line (Line 3/4, part of the 6 GWh Phase 1) is explicitly targeted at 3-wheeler and e-rickshaw applications, alongside BESS/stationary and Telecom, precisely because this product does not require the lengthy OEM homologation cycle that passenger-vehicle cylindrical cells require — making it likely to be Exide's first Li-ion cell revenue stream even though the cylindrical (2W) line begins production first.

4.31.7 3.7 Solar

[FACT] Solar crossed the ₹1,000 crore standalone revenue mark for the first time in FY26 (up from ~₹800 crore the prior year), with management targeting ₹1,500 crore within 2–3 years. As of the Q2 FY26 call, solar was ~4–5% of total revenue.

[FACT] Two distinct sub-markets: (1) residential/commercial rooftop solar (Exide's core focus — off-grid systems use batteries; on-grid systems use panel+inverter only), supported by GST reduction on solar combo packs from 12% to 5% and the 'PM Surya Ghar' rooftop electrification scheme (targeting 1 crore households); and (2) grid-scale utility solar, driven independently by India's 500 GW renewable-energy target by 2030 and MNRE's 40 GW rooftop-solar target by 2026.

[MGMT COMMENTARY] *Management explicitly states that lithium-ion's core advantages (higher energy density/smaller footprint, fast charging) are largely irrelevant for solar backup applications, since footprint is dictated by the solar panels rather than the battery, and fast charging is not a priority for a backup-power use case. Consequently, lead-acid is expected to remain the primary chemistry for solar for the foreseeable future, and EESL's Li-ion product roadmap is not prioritising the solar use case.*

4.31.8 3.8 Home UPS / Inverter Batteries

[FACT] Home UPS/inverter batteries typically represent ~20–25% of full-year revenue and are highly seasonal — volumes in the March–July season run at roughly 3x the off-peak (Oct–Feb) level. Q1 is typically Exide's highest-revenue quarter for this reason.

[FACT] An extended monsoon in 2025 delayed the seasonal inverter-demand ramp into H2 FY26; combined with GST-related channel destocking in Q2 FY26, this business saw a rare negative growth quarter (Q2 FY26 solar/inverter mix reportedly went from +35% in Q1 to roughly -5% in Q2), before rebounding strongly in Q3/Q4.

- New launches: "EL Ultra" (premium segment) and "PowerBox" (economy segment) inverter batteries; Solar Grid-Tie inverters under the "Exide Aditya" brand; the "Invazen" range launched exclusively for the White Goods retail channel.

4.31.9 3.9 Exports

[FACT] Exports were historically ~8% of standalone turnover but fell to 5.3% of standalone turnover in FY26 (from 8.1% in FY25), driven by the West Asia geopolitical conflict, a Western European recession (Germany, UK, Italy — impacting industrial traction battery sales), and US/EU tariff uncertainty that delayed the ramp-up of premium products developed for those markets.

[MGMT COMMENTARY] *Management has guided for an export recovery in multiple consecutive quarters ("Q4 onwards we will report an uptick" — Q2 FY26 call; "Exports will have volatility for the first half but will recover" — Q4 FY26 call) without the recovery materialising at the pace originally guided, though management points to a new European distribution partner and improving tariff clarity (referencing a US tariff announcement around the February 2026 timeframe) as positive catalysts. As of the Q4 FY26 call, exports were still guided to remain soft through at least H1 FY27.*

[ANALYST INTERPRETATION] Given the repeated "recovery next quarter" guidance on Exports that has not yet been borne out, this is a useful management-credibility tracker — the pattern of guidance should be checked against actual results each quarter and flagged if it becomes chronic over-optimism rather than a one-off macro delay.

4.32 4. New Energy Business — Exide Energy Solutions Ltd. (EESL)

4.32.1 4.1 History & Corporate Structure

[FACT] Exide entered lithium-ion in 2018 via the "Nexcharge" joint venture with Leclanché SA (Switzerland). In 2022, Exide bought out Leclanché's stake, making the business a wholly owned subsidiary, and simultaneously announced its own 12 GWh cell-manufacturing gigafactory project in Bengaluru with an original 8–10 year phased build-out roadmap. In March 2024, the Nexcharge pack/module business (formerly Exide Energy Private Limited) was amalgamated into the newly formed Exide Energy Solutions Limited (EESL), consolidating all Li-ion activities under one subsidiary.

4.32.2 4.2 Two Distinct Facilities

Facility	Status	Detail
Prantij, Gujarat	Existing / operational	1.5 GWh pack & module assembly (not cell manufacturing). Fully automated, robotised lines handling cylindrical, prismatic and pouch cells. Products: battery packs, modules, racks, in-house-developed Battery Management Systems (BMS). Customers: global commercial-vehicle manufacturers, large Indian E2W producers, telecom tower operators (incl. first-mover supply to Indus Towers), buses/CVs, stationary ESS. Order book ~₹700 cr (as of FY23); pack/module revenue run-rate ~₹100 cr per 9 months (Q3 FY26 call), i.e. roughly ₹130–150 cr annualised — currently built on imported cells.
Bengaluru, Karnataka	Under construction / commissioning	The new cell-manufacturing "gigafactory." Total planned capacity 12 GWh; Phase 1 = 6 GWh, split evenly 3 GWh cylindrical NMC (2-wheeler focus) + 3 GWh prismatic LFP (3-wheeler/e-rickshaw, stationary/BESS, Telecom, and some 4-wheeler/bus applications). Land and utilities were purchased/built for the full 12 GWh; only Phase-1-scale sheds and machines have been installed, meaning Phase 2 (next 6 GWh) will cost meaningfully less than Phase 1.

4.32.3 4.3 Investment / Capex Trajectory

Point in time	Cumulative investment
As of Oct 2025 (H1 FY26 call)	₹3,947 cr cumulative equity infused into EESL
As of Jan 2026 (Q3 FY26 call)	₹4,252 cr cumulative equity infused
As of end-FY26 / May 2026 (Q4 FY26 call)	₹4,802 cr cumulative equity infused; ₹1,500 cr invested in FY26 alone (₹600 cr in Q4 FY26 alone)
FY27 plan (board-approved, Jan 2026)	₹1,400 cr further equity infusion — a mix of capex and opex/working-capital funding
Total Phase 1 project cost	Escalated from an original ₹5,200 cr to ₹5,600 cr "due to operational reasons" — a ~7.7% cost creep before the first commercial cell has shipped

[FACT] Financing: EESL has achieved financial closure for Phase 1 including a ₹2,000 crore undrawn term-loan facility at the subsidiary level. The standalone Exide Industries balance sheet remains debt-free; borrowing (where it exists) sits only at EESL.

[ANALYST INTERPRETATION] Management's "hybrid funding model" — core lead-acid cash flows fund EESL's equity requirement while EESL itself can draw term debt — lets

Exide preserve its zero-debt standalone identity while still leveraging up the new-energy bet at the subsidiary level. This is a capital-structure choice worth watching closely: if Phase 1 underperforms, the debt sits at EESL, but continued equity infusions from the parent (as planned for FY27) still represent an opportunity cost to standalone shareholders even without technical parent-level debt.

4.32.4 4.4 Timeline & Milestones

[FACT] Commissioning roadmap disclosed in the Q4 FY26 investor presentation as a six-stage process: (1) Facility & utilities → (2) Equipment trials → (3) Process validation → (4) Customer samples → (5) Homologation → (6) Commercial supply. As of May 2026, both cylindrical and prismatic lines are described as "leading the validation" stage — i.e., roughly stage 3–4 of 6.

[FACT] As of the Q4 FY26 presentation: 95%+ of utilities were nearing completion across all 4 lines; all equipment was on-site for the current phase; ~45 Master Purchase Agreements (MPAs) had been signed for raw-material supply-chain readiness (up from 30+ as of the Nov 2025 presentation); 450+ shopfloor associates had been onboarded and were supporting trials/commissioning (up from 300+ in Nov 2025).

[FACT] Line-by-line status (May 2026): Line 1 (cylindrical NMC) — commissioning near completion, customer sample deliveries beginning around May 2026. Line 2 — installation completed, commissioning ongoing. Line 3 (prismatic LFP) — final installation and commissioning ongoing. Line 4 — installation nearing completion.

[FACT] Warehouse infrastructure (electrode warehouse, finished-goods & assembly warehouse, hazardous-chemicals warehouse) was under simultaneous construction as of Nov 2025.

Timeline evolution across the last three quarterly calls

Call	Guidance given
Q2 FY26 (Nov 2025)	"Company expects to start production towards the end of FY26." Equipment installation/commissioning nearing completion; ~100% of utility systems nearing commissioning.
Q3 FY26 (Feb 2026)	Cylindrical line: internal validation samples made, internal validation in progress; next step is sending samples to customers. Prismatic Line 3: near completion of internal acceptance criteria, sample manufacturing to start "next month" (i.e., ~March 2026). Line 4 installation targeted for completion by April 2026.
Q4 FY26 (May 2026)	Cylindrical line: customer sample delivery expected to begin "this month or next" (i.e., May–June 2026); process validation completed. Prismatic: internal validations for cylindrical complete; prismatic customer samples targeted for June–July 2026. Cylindrical customer homologation typically takes 2–3 months from sample delivery. Commercial dispatch date still explicitly undisclosed — management repeatedly calls this "a material disclosure" and declines to commit to a date even when directly pressed by analysts.

[ANALYST INTERPRETATION] The original 2022 roadmap implied a ~27–30 month operationalisation window (referenced explicitly by an analyst on the Q3 FY26 call). Measured from a 2022 project announcement, that would have implied production by roughly mid-2024 to early 2025. The actual trajectory — utilities still finishing in late 2025, customer samples only starting mid-2026, and commercial dispatch date still undisclosed as of May 2026 — represents a materially slower ramp than the original public framing, even though management has never explicitly acknowledged the slippage in those terms. This is one of the most important "soft" red flags in the entire research base and should be tracked quarter by quarter until a firm commercial-revenue date is finally given.

4.32.5 4.5 Chemistry, Technology Partner & Multiple Use-Cases

[FACT] Two chemistries: NMC (Nickel Manganese Cobalt) in cylindrical form factor, targeted at 2-wheelers; LFP (Lithium Iron Phosphate) in prismatic form factor, targeted at 3-wheelers/e-rickshaws, BESS/stationary storage, and Telecom, with some overlap into 4-wheeler/bus/stationary use given the flexible prismatic form factor. A third chemistry, LTO (Lithium-Titanium-Oxide), is mentioned as part of the broader technology roadmap without a specific product line yet disclosed.

[FACT] Technology partner: SVOLT (a global Li-ion cell technology player) is providing end-to-end technical support — process know-how, machinery selection, and crucially, access to SVOLT's own upstream raw-material supply chain, which underpins the Master Purchase Agreements (MPAs) signed with suppliers.

[MGMT COMMENTARY] *Management states that ahead of China's August 2025 export-restriction notification on rare-earth technology, equipment, and raw materials for the Li-ion sector, Exide had already secured its required machinery and technical know-how — i.e., the first two of three restricted categories were locked in before the restrictions were announced. The raw-material restriction was subsequently delayed by China to November 2026, giving roughly an extra year of buffer, but management notes explicitly that "even the whole world doesn't know" what happens after that date.*

4.32.6 4.6 Yield, Utilisation & Margin Framework

[MGMT COMMENTARY] *Yield is repeatedly described by management (Pravin Saraf, Q4 FY26 call) as "the most critical factor governing cost of cell manufacturing." The stated best-case/target yield is 90%. Utilisation targets are framed in stages: 60% → 70% → 80%+, with the key strategic threshold being 80–85% utilisation combined with 90% yield — this combined benchmark is repeatedly cited as (a) the trigger for evaluating Phase 2 investment and (b) the point at which management will finally provide a steady-state margin outlook.*

[MGMT COMMENTARY] *On pricing: management states cell pricing negotiations are a "bilateral" mix of import-parity and cost-plus approaches, varying customer by customer, and that the ultimate target is to reach landed-import-cell-price parity through a combination of (i) plant utilisation/yield improvements, (ii) raw-material localisation over time, and (iii) an indirect tailwind from China's own VAT-structure changes on exported cells (a ~3% VAT increase already implemented, rising to a cumulative ~9% by January 2027), which makes imported Chinese cells relatively more expensive and narrows the gap Exide needs to close.*

[MGMT COMMENTARY] *On margins: management's most consistent framing (repeated across Q2, Q3, and Q4 FY26 calls) is that steady-state EESL margins should land "somewhere between" lead-acid OEM margins (thin) and lead-acid aftermarket margins (rich) — because unlike lead-acid, Li-ion is a largely B2B/OEM business (limited trade/aftermarket component, mainly confined to 2W/3W applications) but the OEM relationship here is akin to supplying "the engine of the vehicle" rather than a peripheral SLI battery, which management argues commands better economics than a typical lead-acid OEM part. Management has consistently declined to give a specific margin number, deferring this to "once we hit 85% utilisation and 90% yield."*

[FACT] A key derisking point repeatedly emphasised by management: unlike lead-acid trade-channel commodity risk (which is only partially and slowly passed through), the core Li-ion raw material (lithium/cell inputs) is expected to be substantially indexed/pass-through in OEM contracts, similar to how lead is indexed in lead-acid OEM contracts — reducing (though not eliminating) commodity-margin volatility in the new business.

4.32.7 4.7 The Hyundai/Kia Relationship — A Distinct, Separate Capex Cycle

[FACT] Exide/EESL has a co-investment arrangement with Hyundai/Kia that is explicitly separate from, and incremental to, the 6 GWh Phase 1 capacity (confirmed unambiguously on the Q4 FY26 call: "That's a little different. This will not have an impact on the Hyundai [capacity]."). Management has consistently declined to disclose the investment size, capacity, or timeline for this arrangement, calling it "still not in public" and "a material disclosure."

[ANALYST INTERPRETATION] This is arguably the single most important red-team finding surfaced by the NotebookLM research (see section 8). Investors who had assumed the already-committed ₹5,600 cr Phase 1 investment included guaranteed Hyundai/Kia offtake were mistaken — Hyundai/Kia requires its own additional capacity and, by extension, its own additional capital commitment, layered on top of a Phase 1 that has not yet proven its own commercial viability. This raises both funding-sequencing risk (a second capex cycle before the first has generated a rupee of profit) and execution-bandwidth risk for EESL's still-young organisation.

4.32.8 4.8 India Lithium-Ion Industry Context

[FACT] Per industry estimates cited in the Q4 FY26 investor presentation: India's current Li-ion demand is 20+ GWh, largely met through imports (commercial-scale domestic cell manufacturing is still nascent). Demand is projected to reach 140–150 GWh by 2030 (EV + stationary applications combined), with EVs expected to represent 60–70% of that demand and overall market growth of 65%+ (particularly 2W and PV) through 2030.

[FACT] Expected 2030 EV penetration by end-market (vs. 2026 actual): 3-wheelers ~36% → 70%+; 2-wheelers ~9–10% → 30%+; passenger vehicles/commercial vehicles ~3–4% → 15%+.

[MGMT COMMENTARY] *Management frames the localisation opportunity as a "chicken and egg" problem: government policy support (PLI-type incentives, favourable import-duty*

structuring) for domestic cell manufacturing is unlikely until India reaches a critical mass of ~20–25 GWh of local capacity (roughly matching total current EV-sector demand), at which point 2–3 large domestic players reaching that combined scale should catalyse policy intervention.

[FACT] Cited policy tailwind example: the Draft EV Policy 2026–30 (Government of NCT of Delhi) includes EV purchase incentives, fleet-electrification mandates (government/school/delivery/aggregator fleets), EV-charging-ready building codes, and broader local-ecosystem support — illustrative of the type of state-level policy push expected to accelerate demand across E2W, E3W, passenger EVs, e-buses, and e-goods delivery fleets.

[FACT] "True" Make-in-India manufacturing, per management, requires domestic sourcing of anode and cathode materials, not merely importing electrodes and assembling/packaging them domestically — a distinction management uses to differentiate genuine backward-integrated cell manufacturing (which Exide claims to be pursuing) from more superficial "assembly-only" localisation.

4.32.9 4.9 Circularity / Lithium-ion Recycling

[FACT] Exide has not yet announced a dedicated lithium-ion battery recycling facility. It has, however, stated an intent to build "scalable circular material systems" for the new-energy platform, and is already registered under the Battery Waste Management Rules 2022 and E-Waste Management Rules 2023, with compliance steps under way (engaging authorised e-waste recyclers, establishing end-of-life collection mechanisms).

[FACT] Extended Producer Responsibility (EPR) compliance costs began showing up as accruals from Q2 FY26 onward, inflating "other expenses"; management confirmed on the Q2 FY26 call that this is now a recurring cost, not a one-time provision, and stated an aspiration to eventually pass this through to customers as a pricing premium for "100% recycled" positioning, drawing an analogy to how other commodity industries have achieved this over time.

[ANALYST INTERPRETATION] Given Exide's stated "molecule to megawatt" vision for EESL and its existing lead-recycling moat via Chloride Metals, the most likely medium-term path (per NotebookLM synthesis) is that Exide eventually replicates its lead-recycling backward integration in the lithium space once domestic Li-ion volumes reach critical mass — but there is no committed project, capex figure, or timeline for this today. Treat as a multi-year optionality, not a near-term catalyst.

4.33 5. Financial Analysis

4.33.1 5.1 Standalone Quarterly & Annual Performance (FY25–FY26)

Metric	Q4 FY25	Q2 FY25	Q1 FY26	Q2 FY26	H1 FY25	H1 FY26	FY25	FY26
Revenue (₹ cr)	4,159	4,267	—	4,178	8,580	8,688	16,588	17,269

Metric	Q4 FY25	Q2 FY25	Q1 FY26	Q2 FY26	H1 FY25	H1 FY26	FY25	FY26
YoY growth	—	—	5.0% (Q1)	-2.1%	—	+1.3%	—	+4.1%
EBITDA (₹ cr)	467	484	—	395	978	943	1,893	1,943
EBITDA margin	11.2%	11.3%	—	9.4%	11.4%	10.9%	11.4%	11.3%
PBT (₹ cr)	343	399	—	298	773	727	1,441	1,500*
PBT margin	8.2%	9.4%	—	7.1%	9.0%	8.4%	8.7%	8.7%
PAT (₹ cr)	255	298	—	221	577	541	1,077	1,111
PAT margin	6.1%	7.0%	—	5.3%	6.7%	6.2%	6.5%	6.4%

*Source: Q2 FY26 and Q4 FY26 Investor Presentations. *FY26 PBT is before an exceptional item of ₹9 cr. Q1 FY26 standalone figures were not separately disclosed in the source presentations beyond the derived +5% YoY growth commentary; treat H1 FY26 as the reliable aggregate.*

[FACT] Full-year FY26 (standalone): Revenue ₹17,269 cr (+4.1% YoY); EBITDA ₹1,943 cr (11.3% margin, +2.6% YoY); PBT ₹1,500 cr before exceptional item (8.7% margin, +4.1% YoY); PAT ₹1,111 cr (6.4% margin, +3.2% YoY).

[FACT] Q4 FY26 alone: Revenue ₹4,551 cr (+9.4% YoY, a record quarterly revenue); EBITDA ₹530 cr (11.7% margin, +13.7% YoY, ~50 bps YoY margin expansion); PBT ₹420 cr (9.2% margin, +22.6% YoY); PAT ₹312 cr (6.9% margin, +22.7% YoY).

[ANALYST INTERPRETATION] Q2 FY26 was the weak point of the year — a genuine, GST-driven channel-destocking air-pocket (-2.1% Q2 revenue growth, -18% EBITDA, -26% PAT) — followed by a strong V-shaped recovery through Q3/Q4. The full-year growth number (4.1%) understates how much stronger the H2 exit rate was (Q4 alone grew revenue 9.4% YoY with EBITDA up 13.7%), so quarter-by-quarter momentum, not the annual average, is the better signal of underlying business health heading into FY27.

4.33.2 5.2 Consolidated vs. Standalone Growth — An Important Distinction

[FACT] Consolidated revenue growth (which includes EESL and other subsidiaries) was reported by NotebookLM's cross-annual-report analysis as only 2.79% in FY25 and 4.39% in FY26 — figures that are close to, but not identical with, the standalone growth rates quoted directly by management on the earnings calls (standalone FY26 revenue growth of 4.1% per the Q4 FY26 call).

[ANALYST INTERPRETATION] This is worth reconciling carefully: standalone and consolidated growth rates being this close to each other (rather than consolidated running meaningfully lower, given EESL is pre-revenue/early-revenue and diluting the blended growth rate) suggests either (a) EESL/CML consolidation effects are currently small in absolute rupee terms relative to the standalone base, or (b) some intercompany eliminations are offsetting each other. This is flagged as an item to verify directly against

the FY26 consolidated financial statements rather than relying solely on NotebookLM's synthesis.

4.33.3 5.3 Margin Drivers

[FACT] Gross margin moved from 31.6% (Q3 FY26) to 30.1% (Q4 FY26), a ~150 bps sequential decline, driven by a ~₹150 crore net negative commodity-cost impact in Q4 alone (non-lead inputs: antimony, sulfuric acid, LPG, plastics).

[FACT] Despite the gross-margin decline, EBITDA margin was held flat sequentially at ~11.7% through cost control on three specific levers explicitly cited by the CFO: factory cost as % of sales, employee cost, and — notably — a reduction in warranty cost as % of sales (attributed to the manufacturing-automation programme improving product quality).

Commodity price trends (as disclosed across the three FY26 calls)

Commodity	Trend
Antimony	\$11,000/ton (base) → peaked near \$60,000–\$66,000/ton (~6x) around Q2 FY26 → softened to ~\$33,000–35,000/ton by Q3 FY26 (~3x YoY, still elevated). Driver: Chinese export bans/restrictions.
Sulfur (feedstock for sulfuric acid — note: management is precise that this is sulfur, not sulfuric acid itself)	₹15/kg (one year prior) → ₹58/kg (March 2026 exit) → ₹74–75/kg (April 2026 exit) — roughly a 5x increase YoY. Driver: sulfur is a byproduct of petrochemical refining; reduced refinery utilisation (linked to crude shortages from the West Asia conflict) reduced sulfur supply even as demand held.
Tin / Silver / Copper	Tin +12% and Copper +13% sequentially (Q3 FY26); Silver +50% YoY (Q3 FY26). These are alloying elements used mainly in automotive-battery grids.
LME Lead	Softer year-on-year through most of FY26 in USD terms, but the benefit was more than offset by INR depreciation (~10% weaker over the relevant period) because Indian lead — even recycled/domestic lead — trades at import-parity pricing. LME lead spiked back up to ~\$1,950/ton by April 2026 exit, reintroducing volatility even in USD terms.
INR/USD	Moved from ~₹80/USD to ~₹90/USD across the period referenced in the Q2 FY26 call, materially pressuring input costs even where global commodity indices were flat or falling.

Price-hike cadence (FY26)

- January 2026: ~2% price increase taken (per Q3 FY26 call).

- Q3 FY26 (Oct–Dec 2025): No price correction taken — management consciously absorbed cost inflation to pass the full benefit of the GST 2.0 rate cut through to end consumers.
- 20 March, 1 April, and a further tranche around 20 April 2026: three-stage price increases cumulating to roughly 5–6% (varying by product line), explicitly to recover accelerating non-lead commodity inflation.
- Further increase announced/taken around 1 April and again in early April 2026, totalling ~3% in that specific round per the Q4 FY26 call; management indicated more increases were likely in May/June 2026 given the trajectory of April-exit commodity prices.

[MGMT COMMENTARY] *Management states OEM price-correction negotiations typically take about one quarter to conclude ("the lead time is typically... a quarter"), versus trade/aftermarket price hikes which can be implemented immediately by company decision.*

4.33.4 5.4 Returns — ROCE, ROE, Asset Turnover

[FACT] Two different ROCE figures appear across the source presentations and must not be conflated:

- Nov 2025 Investor Presentation: "ROCE (Core business)" 17.6% (Mar '25) → 16.3% (Sep '25).
- Q4 FY26 Investor Presentation: "ROCE*" 26.0% (FY25) → 28.8% (FY26), explicitly footnoted as "calculated for the core business excluding investments in HDFC Life and EESL project."

[OPEN QUESTION] The two ROCE series (16–17% band vs. 26–29% band) are clearly built on different capital bases or annualisation conventions, even though both are labelled "core business." This must be reconciled directly against the underlying calculation (likely: whether HDFC Life investment and EESL equity are excluded from the capital-employed denominator, and whether the numerator is annualised from a half-year run-rate) before using either figure in a valuation model. Until reconciled, treat the Q4 FY26 PPT's 28.8%/26.0% figures — which come with an explicit exclusion footnote — as more analytically useful and probably closer to a true "core lead-acid business" ROCE.

[FACT] Asset turnover ratio (core business, ex-HDFC Life/EESL): improved from 3.1x (FY25) to 3.4x (FY26).

[FACT] Working-capital usage (as % of revenue, core business): improved from 11.3% (FY25) to 7.3% (FY26) per the Q4 FY26 presentation; a separate H1 FY26 snapshot in the Nov 2025 presentation showed inventory days falling from 110 (Mar '25) to 93 (Sep '25) and working-capital usage falling from 11.8% to 8.4% over the same window.

[FACT] ROE: per NotebookLM's cross-annual-report analysis, consolidated ROE spiked to 53.5% in FY22 (driven by the one-time ₹4,587 crore exceptional gain on the sale of the life-insurance business to HDFC Life) before normalising to roughly 7.6% by FY26, reflecting

both a larger post-sale equity base and the large, not-yet-revenue-generating EESL capital deployment (~₹4,802 cr cumulative equity as of FY26-end) sitting on the balance sheet.

[ANALYST INTERPRETATION] The core-business ROCE trend (26.0%→28.8%) and the consolidated ROE trend (normalising toward ~7.6%) tell two very different stories that must both be held in mind simultaneously: the underlying lead-acid business is getting structurally more efficient and higher-returning, while the consolidated entity's headline returns are being diluted by a large, currently non-earning capital base at EESL. This is the crux of the entire investment debate (see section 9).

4.33.5 5.5 Cash Flow, Working Capital & Inventory Management

[FACT] Net Cash from Operating Activities (per NotebookLM's cross-annual-report synthesis): ₹2,231 cr in FY26 vs. ₹1,298 cr in FY25 — a sharp improvement attributed to materially better working-capital management (net capital turnover ratio improved to 11.36x from 8.22x).

[FACT] A deliberate, management-acknowledged working-capital action occurred in Q2 FY26: facing GST-related channel destocking, management consciously cut production in the second half of August and all of September 2025, prioritising cash generation over volume/market share. This generated an incremental ₹500 crore+ of cash through H1 FY26 via reduced inventory, at the direct cost of Q2 profitability (production cuts reduced fixed-cost absorption, contributing to the -18% EBITDA quarter).

[MGMT COMMENTARY] *CFO Manoj Agarwal describes this as a deliberate cash-first strategic call during periods of demand uncertainty, contrasted with a subsequent conscious pivot back to "running the factory at full capacity" in Q3 once GST-driven demand momentum became clear — i.e., management explicitly toggles between a cash-preservation posture and a growth/volume posture depending on near-term demand visibility.*

4.33.6 5.6 Capital Allocation & Capex

[FACT] Stated core-business capex policy: capex is targeted at roughly the same level as annual depreciation (~₹400–500 crore per year in recent years), split across manufacturing-technology upgrades (competitiveness/quality), factory automation (productivity), and, since ~2022, the large incremental commitment to EESL/lithium-ion (see section 4.3 for detail).

[MGMT COMMENTARY] *Management (Avik Roy, Q3 FY26 call) frames the capital-allocation philosophy in three sequential "waves": (1) invest in manufacturing technology for cost competitiveness and quality — largely completed; (2) invest in factory automation for productivity — ongoing; (3) invest for the future via the lithium-ion project — the current large-scale priority.*

[FACT] Standalone depreciation was ₹589 crore in FY26 (per NotebookLM synthesis). A material step-up in consolidated depreciation is expected once EESL's Capital-Work-in-Progress (CWIP — quoted at ₹3,937.83 crore in one NotebookLM analysis) is moved into the gross fixed-asset block upon commissioning, likely from H2 FY27 onward.

4.33.7 5.7 Balance-Sheet Strength & the HDFC Life Stake

[FACT] Exide's standalone balance sheet has been debt-free for over a decade. The company also holds a significant equity stake in HDFC Life Insurance (received as consideration when Exide sold its life-insurance business to HDFC Life), valued at roughly ₹5,000–5,140 crore as of the FY26 period discussed.

[MGMT COMMENTARY] *Management repeatedly describes this stake as providing "ample financial flexibility" — effectively a synthetic liquidity reserve that can, if needed, be monetised to fund the lithium-ion capex programme without taking on parent-level debt, thereby preserving the zero-debt standalone identity as "a point of pride."*

[FACT] The stake is subject to mark-to-market (MTM) volatility, routed through Other Comprehensive Income (OCI) rather than the P&L. Disclosed MTM movements: a ~₹850 crore reduction in the value of non-current investments through H1 FY26 (per the Q2 FY26 call, confirmed by the CFO as MTM-driven, with no shares actually sold), and a further ~₹714 crore negative OCI impact cited for FY26 as a whole in the NotebookLM synthesis. As of all FY26 disclosures reviewed, management states no HDFC Life shares have actually been sold.

[ANALYST INTERPRETATION] Relying on a volatile, listed minority equity stake as an implicit backstop for a fixed-cost, multi-year industrial capex programme is a genuine treasury-structure risk, not merely a bear-case talking point — if HDFC Life's share price were to fall meaningfully at the same time EESL needed incremental funding, Exide would either have to monetise the stake at a depressed value or take on debt (breaking the zero-debt standalone framing) to keep EESL funded on schedule.

4.34 6. Management Assessment

4.34.1 6.1 Communication Style & Consistency

[ANALYST INTERPRETATION] Across the three FY26 earnings calls reviewed, Avik Roy (MD & CEO) is a consistent, detail-forward communicator who reliably repeats the same core framework quarter to quarter (the "92% of the business growing double-digit vs. 8% drag" framing appears, essentially unchanged, in Q2, Q3, and Q4 FY26 calls). This consistency is a positive credibility signal — the story is not being reshaped quarter to quarter to fit whatever result was delivered.

[ANALYST INTERPRETATION] At the same time, management is notably and consistently unwilling to give forward numerical guidance on several specific, recurring topics: (a) next-year EBITDA margin guidance (explicitly declined in Q2, Q3 FY26 calls, with only a loose "12–13% corridor" offered once), (b) EESL steady-state margins (deferred every quarter to "once we hit 85% utilisation/90% yield"), (c) exact commercial-dispatch date for Bengaluru cells (called "a material disclosure" every time it is asked), and (d) Hyundai/Kia capacity, investment size and timeline (explicitly declined as "still not in public"). This pattern of specific, recurring non-disclosure — rather than general reticence — is worth tracking as its own signal.

4.34.2 6.2 Capital Allocation Track Record

- Positive: maintained a zero-debt standalone balance sheet through a multi-thousand-crore new-business capex programme, funding it via internal accruals plus targeted subsidiary-level debt at EESL only.
- Positive: sustained a ~16% dividend payout ratio even while funding the EESL capex programme, indicating no funding-driven pressure (yet) to cut shareholder returns.
- Mixed: repeated cost overruns/timeline slippage on the single largest capital project in company history (Phase 1 cost creep of ~7.7%; commercial-dispatch date still undisclosed well past the originally implied 27–30 month operationalisation window).
- Mixed: the Hyundai/Kia "separate, incremental" capacity requirement was only clarified/confirmed as distinct from Phase 1 in the May 2026 (Q4 FY26) call — a fact that, in hindsight, investors modelling Hyundai/Kia offtake as already covered by the committed ₹5,600 cr Phase 1 spend would have gotten wrong for some time.

4.34.3 6.3 Governance

[FACT] Chairman/CEO roles separated; board is majority independent-director-influenced in key committees (Audit; Nomination & Remuneration both chaired by Independent Directors); board gender composition as of June 2026 is 89% men/11% women with 56% independent directors.

[FACT] Recent board refresh: Manoj Kumar Agarwal (CFO) added 1 May 2025; Rajeev Khandelwal (B2C trade) and Pravin Saraf (EESL) both added 1 September 2025 — a deliberate strengthening of executive bandwidth coinciding with the "One Exide" transition and the EESL ramp-up.

[FACT] Auditor: B S R & Co. LLP (a Big-Four-network firm), a positive audit-quality signal.

[FACT] Governance/leadership churn was explicitly addressed on the Q3 FY26 call regarding senior-level exits within EESL: management stated this reflects a young industry with limited experienced local talent (heavy poaching/migration across the nascent Indian Li-ion sector, in both directions), and that successor appointments were pre-planned (announced the same day as departures in at least one instance) rather than reactive.

4.34.4 6.4 Sustainability/ESG Trajectory (Governance-Adjacent)

[FACT] S&P Global Corporate Sustainability Assessment (CSA) score improved to 65/100 (scored 12 November 2025 per the Nov-2025 PPT reference; a later Q4 FY26 PPT references a 65/100 score dated 11 February 2026 — likely the same underlying assessment cycle referenced twice), roughly a ~50% improvement vs. the 2024 score, and roughly 2x the disclosed industry average of 33/100, placing Exide in the "Industry Mover" category and the top-5-percentile globally among ~9,200 assessed companies (16 auto-component companies made the global CSA Yearbook; Exide is one of only 2 Indian companies to do so).

[FACT] EcoVadis Sustainability Rating: Bronze medal, top 35% globally, score 67/100 (75th percentile), assessed against ~1,75,000 companies. Pillar-level rankings: Environment (top 14%), Labour & Human Rights (top 10%), Sustainable Procurement (top 21%), Ethics (top 29%). Top 14% specifically within the "Manufacture of Batteries & Accumulators" industry peer set.

[FACT] CSR is organised around five branded pillars: Exide Aarogya (health & wellness), Exide Akshar (education), Exide Kausal (skill upgradation/livelihood), Exide Saksham (empower & elevate), and Exide Paryavaran (environment & climate action). ~80% of CSR-project beneficiaries come from vulnerable communities, with project outreach across 10 states and 24 districts. Recent recognitions include the ICSI CSR Excellence Award (8th edition), the BCC&I Social Leadership Award (3rd edition), and the British Safety Council Award 2025 (ESG & EHS standards).

4.35 7. Competitive Advantages / Moat Assessment

4.35.1 7.1 Existing Moats (Core Lead-Acid Business)

- Backward-integrated lead recycling via Chloride Metals Ltd. (3 plants) — provides a structural cost floor and input-supply resilience that most competitors, who rely on volatile imported/third-party recycled lead, cannot match. Durability: high — recycling infrastructure of this scale takes years and significant capital to replicate, and regulatory tightening on lead handling (see risk section) actually raises the barrier to entry further for new competitors, even as it is also a risk for Exide itself.
- Distribution scale and OEM relationships — 1,20,000+ network touchpoints, 75+ OEM relationships, decades of brand trust ("70+ years of unmatched trust" per the Batmobile service statistics). Durability: high in the near-to-medium term; this is the type of asset that erodes slowly even under competitive pressure, given switching costs for both dealers and OEM design-in relationships.
- Manufacturing scale and automation lead (Punched Grid, CONCAST, robotics) — delivering cost and quality (lower warranty cost) advantages that are being actively reinvested rather than harvested. Durability: medium-high; requires continuous capital reinvestment to sustain, but currently widening rather than narrowing based on FY26 disclosures.
- R&D depth (50-year-old Kolkata R&D centre, 100+ R&D staff, 4 NABL-accredited labs, 5 global technical collaborations incl. East Penn/"Deka" and SVOLT) — enabling fast, differentiated product launches (front-terminal data-centre batteries, AGM for premium OEMs, SUV-specific batteries). Durability: medium-high, contingent on continued investment.

4.35.2 7.2 Emerging Moats

- Digital/data layer — dealer digitisation, loyalty apps for mechanics/electricians, AI-enabled sales tools, 60 lakh+ paperless warranty registrations, WhatsApp-based service alerts and Batmobile doorstep service. Durability: uncertain but potentially high if it succeeds in building a durable, switching-cost-generating direct consumer

relationship that regional/unorganised competitors structurally cannot replicate given their smaller scale.

- First-mover position in indigenous, at-scale Li-ion cell manufacturing in India — an 18–24 month experiential learning-curve head start over any future domestic entrant (per management's own framing), plus early access to SVOLT's supply chain via signed MPAs. Durability: uncertain — genuinely contingent on execution (yield/utilisation ramp) actually succeeding, and on India's broader domestic supply chain (anode/cathode materials) developing fast enough to matter.
- "Last man standing" dynamic in lead-acid — management notes that global lead-acid manufacturers are retreating from the category worldwide, while India's ICE-vehicle replacement-cycle demand is expected to persist far longer than in Western markets, potentially allowing Exide to gain share simply by outlasting competitors who exit lead-acid prematurely, all while using that cash-generative core to fund the Li-ion transition. Durability: this is a demand-timing bet, not a structural moat per se — it depends on India's ICE replacement cycle actually persisting as long as management expects.

4.35.3 7.3 Weaknesses / Threats to the Moat

- Regulatory fragility of the lead-recycling moat — lead is a highly toxic material subject to increasingly strict pollution-control and hazardous-waste-management norms; a major environmental non-compliance event, or a step-change tightening of Hazardous Waste Management Rules, could turn the recycling advantage into a significant reputational and financial liability.
- Chemistry/technology-obsolescence risk in the new-energy bet — EESL is committing ₹5,600 cr+ to current-generation NMC/LFP chemistries via the SVOLT partnership; a breakthrough in solid-state or another emerging chemistry over the next 3–5 years could strand this asset before it completes even its first depreciation cycle.
- Cannibalisation risk understated by the "92% growing / 8% drag" framing — the segments in the "8% drag" (Telecom, E-Rickshaw, Exports) include some of the fastest-growing frontiers of the prior decade; if this pattern spreads to other high-growth lead-acid verticals faster than Li-ion can be commercialised and monetised by Exide itself, the core cash engine could erode faster than currently modelled.

4.36 8. Key Business Drivers (Value-Relevant Variables)

4.36.1 8.1 Structural Drivers

- India's underlying vehicle parc growth and low passenger-vehicle penetration (~38 vehicles per 1,000 people cited by NotebookLM synthesis) — a long structural runway for both OEM and replacement demand.
- Data-centre capacity build-out (~20–21% CAGR, targeting 3,400 MW by 2030) and railway electrification (~100% target by FY26) — structural, policy-anchored infrastructure demand largely insulated from auto-sector cyclicity.

- India's renewable-energy build-out (500 GW target by 2030) and rooftop-solar policy support (PM Surya Ghar, MNRE 40 GW rooftop target) — structural tailwind for the Solar vertical.
- Lead-acid-to-lithium-ion chemistry migration across Telecom, E-Rickshaw, and eventually parts of the automotive/industrial space — a structural headwind for legacy lead-acid volumes that Exide is attempting to convert into an opportunity via EESL.

4.36.2 8.2 Cyclical Drivers

- Automotive OEM production cycles (SIAM data) — a ~2-year leading indicator for the higher-margin aftermarket/replacement business.
- Monsoon timing/intensity — directly swings the Home UPS/Inverter seasonal ramp (an extended 2025 monsoon materially dented Q2 FY26 results).
- LME lead, antimony, sulfur, tin, silver, copper prices, and USD/INR — quarter-to-quarter gross-margin swing factors, only partially and with a lag (~1 quarter for OEM contracts) passed through to customers.
- GST/tax-policy changes (e.g., the September 2025 GST 2.0 rate cut) — can cause short-term channel destocking/restocking distortions (as seen in Q2 FY26) even when the medium-term demand effect is positive.
- Geopolitical conditions (West Asia conflict, Western European macro conditions, US/EU tariff policy) — directly driving the Exports vertical's volatility.

4.36.3 8.3 Management-Controlled Drivers

- Pace and cost discipline of EESL's Bengaluru gigafactory ramp (yield%, utilisation%, cost overruns, timeline adherence) — arguably the single largest company-specific, management-controlled variable determining medium-term equity value.
- Pricing discipline / speed of commodity pass-through in the trade/aftermarket channel.
- Capital-allocation sequencing between core-business capex, EESL equity infusions, dividend payout, and (optionally) monetisation of the HDFC Life stake.
- Digital/data-capture initiative execution — determines whether the "emerging digital moat" thesis actually compounds into a durable advantage or remains a cosmetic feature set.

4.36.4 8.4 External / Uncontrollable Drivers

- Chinese policy on rare-earth/technology/raw-material export restrictions for the Li-ion sector (raw-material restriction delayed to November 2026 — a hard date to track).

- Indian government policy support (or lack thereof) for domestic Li-ion cell manufacturing, contingent on India reaching the ~20–25 GWh critical-mass threshold across all domestic players combined.
- Global lithium and NMC/LFP raw-material price volatility, driven by evolving EV demand/supply dynamics in China and elsewhere.

4.37 9. Investment Thesis

4.37.1 9.1 Bull Case

- The standalone lead-acid business is a genuinely improving, highly cash-generative "cash engine": core-business ROCE improved from 26.0% (FY25) to 28.8% (FY26), asset turnover improved from 3.1x to 3.4x, and standalone operating cash flow rose from ₹1,298 cr (FY25) to ₹2,231 cr (FY26) — all while the standalone balance sheet remains debt-free.
- That cash engine is funding a first-mover position in indigenous, at-scale Li-ion cell manufacturing without diluting shareholders or breaking the zero-debt standalone identity (debt, where it exists, sits only at EESL).
- A "last man standing" dynamic in lead-acid, where global competitors are retreating from the category and India's long ICE-replacement runway could allow Exide to capture disproportionate share simply by continuing to invest while others exit.
- Multiple megatrend-exposed growth verticals beyond auto-ancillary cyclicality — Data Centers (~20% CAGR), Solar (crossed ₹1,000 cr, targeting ₹1,500 cr), and railway/infrastructure electrification — potentially supporting a valuation re-rating away from a pure-auto-parts multiple.
- An emerging digital/data moat (loyalty apps, 60 lakh+ paperless warranty registrations, AI sales tools) that smaller/unorganised competitors structurally cannot replicate.
- The HDFC Life stake (~₹5,000+ cr) provides a funding cushion/synthetic liquidity reserve for the capex programme, reducing near-term financing-risk even if EESL's own cash generation is delayed.
- Strong historical compounding track record: 25-year revenue/EBITDA/PAT CAGRs of 12.3%/11.2%/14.0% respectively, and a 25-year share-price CAGR of 19.0% — comfortably ahead of Nifty 50 (12.6%), BSE Sensex (12.7%), and Nifty 500 (13.5%) over the same period.

4.37.2 9.2 Bear Case

- The "92% growing / 8% drag" framing may understate genuine cannibalisation risk: the "drag" segments (Telecom, E-Rickshaw, Exports) were among the highest-growth frontiers of the prior decade, and consolidated revenue growth of only 2.79% (FY25) and 4.39% (FY26) suggests Li-ion migration may be eating the core

business's most profitable growth segments faster than the core can organically replace them.

- The Hyundai/Kia relationship requires a separate, incremental capacity/capital commitment beyond the already-committed ₹5,600 cr Phase 1 spend — a second capex cycle layered on top of a first one that has not yet proven commercial viability, raising funding-sequencing and execution-bandwidth risk.
- EESL/Phase 1 valuation risk: models that value the Li-ion business at roughly 1x invested capital (~₹5,000–6,000 cr) may be too generous for an asset expected to face 10–12% start-up yield rejection rates and likely negative-to-thin EBITDA for its first 1–2 years of commercial operation.
- The HDFC Life stake is a volatile, mark-to-market asset (a ~₹714–850 cr negative OCI hit disclosed across FY26/H1 FY26) — using it as an implicit funding backstop for a fixed-cost industrial capex programme is a genuine treasury-structure mismatch, not merely a theoretical risk.
- Commercialisation timeline for the Bengaluru gigafactory has already slipped materially versus the originally implied 27–30 month operationalisation window (2022 announcement), and management still declines to give a firm commercial-dispatch date as of May 2026 — every additional month of delay makes the eventual domestic cell price less competitive against falling global (particularly Chinese) LFP/NMC cell prices.
- Phase 1 project cost has already crept up ~7.7% (₹5,200 cr → ₹5,600 cr) before the first commercial cell has shipped, a pattern that, if it continues, would further pressure the project's ultimate return on capital.
- Technology-obsolescence risk: a committed, multi-thousand-crore bet on current-generation NMC/LFP chemistry could be stranded by a breakthrough in solid-state or another emerging chemistry within the asset's depreciation life.
- ESG/regulatory fragility: Exide's lead-recycling moat depends on continued favourable (or at least stable) hazardous-waste/pollution regulation; a single major non-compliance event or regulatory tightening could turn this advantage into a significant liability.

4.37.3 9.3 Variant Perception / What the Market May Be Missing

[ANALYST INTERPRETATION] The market narrative on Exide likely oscillates between two extremes — treating it either as a legacy lead-acid compounder facing terminal EV disruption, or as a speculative Li-ion "story stock" riding India's EV narrative. The more interesting, less obviously priced-in framing (developed via the NotebookLM cross-document synthesis) is that Exide is running two genuinely distinct businesses on two genuinely distinct risk/return profiles simultaneously: a highly efficient, high-ROCE, self-funding legacy cash machine, and a high-risk, high-optionality industrial start-up housed inside the same corporate shell. Whether the current market price appropriately separates

and weights these two components — rather than blending them into a single, muddled auto-ancillary multiple — is the core valuation question this dossier should keep returning to as more data (particularly EESL utilisation/yield disclosures and a firm commercial-dispatch date) becomes available.

4.37.4 9.4 Key Catalysts to Watch

- A firm, credible commercial-dispatch date for Bengaluru cells (would materially reduce execution-risk discount).
- First disclosed EESL utilisation% and yield% figures once production begins (the key threshold management has flagged repeatedly: 80–85% utilisation + 90% yield).
- Disclosure of Hyundai/Kia capacity, investment size, and timeline.
- Export recovery actually materialising after multiple quarters of "next quarter" guidance.
- Government policy movement on domestic cell-manufacturing incentives once the ~20–25 GWh local-capacity threshold approaches.
- Any monetisation (partial or full) of the HDFC Life stake, which would clarify management's actual funding intentions for EESL Phase 2 and beyond.

4.38 10. Financial Model Notes

Per the research brief that generated much of this analysis, these notes deliberately describe how to think about forecasting each line item rather than providing point forecasts, since numeric assumptions should be built and stress-tested separately in the working Excel model, not hard-coded into this narrative dossier.

Line item	Forecasting approach & key considerations
Revenue	Disaggregate into Trade (B2C) / Institutional (B2B) / International, then layer in incremental EESL revenue starting FY27 (contingent on the commercial-dispatch date, still unconfirmed). Key drivers: 2W/4W SIAM production data (2-year lead indicator for aftermarket), data-centre capex, PM Surya Ghar solar scheme uptake. Sensitivity: high sensitivity to 2W/4W OEM production cycles. Bull case: 11–13% core CAGR + rapid Li-ion offtake incl. Hyundai/Kia. Bear case: structural lead-acid decline in Telecom/E-Rickshaw/Exports exceeds the Li-ion ramp-up.
Volume & Realisation	Model volume growth by segment; link realisation to LME lead price and premiumisation mix (AGM/EFB). Risk: sudden INR depreciation can offset falling LME lead prices given 20–45% import/import-parity exposure.
Gross Margin	BOM is ~95–96% lead/acid/plastic. Use a lagged LME pass-through model for the ~30% institutional/OEM business (implemented with

Line item	Forecasting approach & key considerations
	roughly a 1-quarter lag per management commentary); CML's recycled-lead ratio (see open question in section 11) provides a structural cost floor. Bull case: antimony/other alloy prices normalising toward pre-spike levels; higher CML utilisation.
EBITDA Margin	Standalone has been resilient in the ~11.3–11.7% band despite commodity spikes, protected by automation/warranty-cost improvements. Consolidated margins likely to dip in FY27–28 due to EESL start-up yield losses (10–12% rejection rates assumed during ramp). Key monitoring threshold: 85% utilisation / 90% yield at EESL as the point where margin visibility improves.
Depreciation & Interest	Model a step-change in depreciation once EESL's CWIP (~₹3,900+ cr) moves to gross block upon commissioning (likely H2 FY27+). Standalone interest is near-zero (debt-free); model EESL-level interest against the ₹2,000 cr undrawn term-loan facility.
Capex	Core lead-acid business: model at roughly the level of depreciation (~₹500 cr/yr, per stated policy). EESL: factor in completion of the ₹5,600 cr Phase 1 project, FY27 guided incremental equity infusion of ₹1,400 cr, a still-undisclosed Hyundai/Kia incremental capex requirement, and a potential Phase 2 (next 6 GWh) decision contingent on Phase 1 hitting 80–85% utilisation/90% yield (roughly FY27–28/29 decision window).
Working Capital	Inventory turnover ~3.9x (FY26); receivables turnover ~11.3x, supported by a high cash-and-carry replacement-sales mix (~65–70%) which keeps receivable risk low. Model an increase in working-capital intensity in FY27 to support the EESL commercial-launch ramp.
Other Income / Treasury	Model returns on the HDFC Life stake (~₹5,000+ cr) and the liquid mutual-fund treasury pool at a ~5–7% yield assumption; flag MTM volatility on the HDFC Life stake (routed through OCI, not P&L) as a balance-sheet — not earnings — risk factor, though large moves could still influence sentiment/financing flexibility.
Operating & Free Cash Flow	Standalone OCF has been strong (₹2,231 cr FY26). Consolidated FCF will likely remain negative-to-neutral through FY27 given the heavy EESL investment cycle; look for an FCF inflection once EESL Phase 1 clears the 85% utilisation threshold. Bull-case funding path: EESL self-funds future expansion partly via monetising the HDFC Life stake rather than incremental parent equity or subsidiary debt.

4.38.1 10.1 Leading Indicators to Track for the Model

- SIAM vehicle-production data — high correlation with the OEM revenue line and a ~2-year leading indicator for the higher-margin aftermarket line.
- LME lead spot price and INR/USD — primary drivers of realisation and COGS (must be modelled jointly given the import-parity pricing dynamic).

- EESL CWIP-to-fixed-asset conversion date — the trigger for the depreciation step-change and for genuine commercial Li-ion revenue recognition.
- EESL yield-maturity curve (utilisation% and yield% once disclosed) — the primary driver of eventual consolidated EBITDA recovery/dilution.

4.39 11. Open Questions & Items to Revisit

[OPEN QUESTION] Exact CML lead-recycling self-sufficiency ratio — conflicting figures seen in different NotebookLM research answers (~79% vs. ~50–53%). Needs to be pinned down against the primary Annual Report disclosure.

[OPEN QUESTION] Reconciliation of the two divergent "core business ROCE" series shown across the Nov-2025 PPT (16.3%/17.6%) and the Q4 FY26 PPT (28.8%/26.0%) — almost certainly different capital-base/annualisation methodologies, but the exact definitional difference has not yet been confirmed.

[OPEN QUESTION] Firm commercial-dispatch date for Bengaluru Li-ion cells — never disclosed by management despite being directly asked in at least three consecutive quarterly calls.

[OPEN QUESTION] Steady-state EESL margin profile — management has consistently deferred this to "once we hit 85% utilisation and 90% yield," with no numeric guidance given as of the Q4 FY26 call.

[OPEN QUESTION] Hyundai/Kia partnership: capacity allocation, investment size, and timeline — explicitly undisclosed ("material disclosure") as of Q4 FY26.

[OPEN QUESTION] Phase 2 (next 6 GWh) capex quantum and firm timing — contingent on Phase 1 hitting its utilisation/yield benchmarks, expected in the FY27–28/29 window but not yet committed.

[OPEN QUESTION] Whether/when a dedicated lithium-ion recycling facility will be announced, and what capital commitment it would require.

[OPEN QUESTION] Precise ongoing run-rate cost of Battery Waste Management/EPR compliance as a % of revenue — management has confirmed it is now a recurring (not one-time) cost but has declined to quantify it.

[OPEN QUESTION] Standalone-vs-consolidated growth-rate reconciliation (FY25/FY26 consolidated growth of 2.79%/4.39% quoted by NotebookLM vs. standalone growth of ~4.1% quoted directly by management) — verify against the consolidated financial statements directly.

[OPEN QUESTION] Data Center segment — only ever disclosed qualitatively/partially (~₹75–100 cr/quarter as of Q2 FY26); no clean full-year standalone disclosure has been given. Track whether management ever breaks this out as a standalone reporting line.

[OPEN QUESTION] Credibility check on Export-recovery guidance — management has guided a recovery "next quarter" across multiple consecutive quarters without it fully materialising; worth tracking whether this becomes a chronic pattern of over-optimism.

4.40 12. Monitoring Dashboard — Quarterly Checklist

Use this checklist every quarter alongside the results release and earnings call. Items are grouped by theme.

4.40.1 12.1 Core Business Health

- Segment trend/outlook grid (Auto OEM, 2W/4W Replacement, Industrial Infra, Telecom, Solar, Home UPS/Inverters, Exports) — reproduced verbatim in every investor presentation; compare quarter to quarter for direction changes.
- Standalone revenue growth (YoY and QoQ), domestic-ex-Telecom growth specifically (management's preferred "clean" growth metric).
- Gross margin and EBITDA margin, sequential and YoY, plus the specific commodity/price-hike commentary given each quarter.
- Price-hike announcements: date, quantum, and which channel (trade/OEM/both).
- Commodity trackers: LME lead, antimony, sulfur, tin, silver, copper, and INR/USD.

4.40.2 12.2 New Energy (EESL) Execution

- Cumulative equity infusion into EESL (track the running total against the ₹5,600 cr Phase 1 budget and any Phase 2/Hyundai-Kia disclosures).
- Line-by-line commissioning status (L1–L4), number of MPAs signed, shopfloor-associate headcount.
- Progress along the 6-stage roadmap: Facility & utilities → Equipment trials → Process validation → Customer samples → Homologation → Commercial supply.
- Any disclosed utilisation% or yield% figures once production begins.
- Any update on commercial-dispatch date, Hyundai/Kia specifics, or Phase 2 timing.

4.40.3 12.3 Balance Sheet & Capital Allocation

- Standalone debt status (should remain zero); EESL-level debt drawdown against the ₹2,000 cr facility.
- HDFC Life stake fair value and any OCI/MTM movement disclosed; any indication of partial monetisation.
- Working-capital metrics: inventory days, receivable/payable turnover, and the "core ROCE" figure (confirm which methodology is being used, per the open question in section 11).
- Dividend declaration and payout ratio trend.

4.40.4 12.4 External Environment

- China's Li-ion raw-material export-restriction status (delayed to November 2026 — a hard date to revisit).
- Any Indian government policy movement on domestic cell-manufacturing incentives (PLI-type schemes, import-duty changes).
- Export-market conditions: West Asia conflict status, Western European macro data, US/EU tariff developments.
- GST or other tax-policy changes affecting batteries/solar.

4.40.5 12.5 Governance & Sustainability

- Any board changes; committee composition; related-party transactions.
- S&P Global CSA score and EcoVadis rating (next assessment cycles).
- Any environmental/regulatory compliance events at lead-recycling or manufacturing facilities.

4.41 13. Source Log & Documentation Standards

Document	Date	Primary content contributed to this dossier
NotebookLM research session, "Exide Industries Limited"	22 sources (Annual Reports, Credit Ratings, Earnings Calls, Equity Research, Investor Presentations); chat sessions dated 2 & 13 July 2026	Cross-year financial-trend analysis (FY20–FY26), management-language evolution analysis, raw-material deep dive, red-team/skeptical-analyst exercise, 5-year model-thinking framework, Prantij vs. Bengaluru comparison, data-centre segment detail
Q2 FY26 Earnings Call transcript	17 November 2025	H1 FY26 results discussion; GST 2.0 destocking impact; lithium-ion Line 1/Line 3 progress; EESL cumulative investment ₹3,947 cr; Rajeev Khandelwal & Pravin Saraf induction to the board
Q3 FY26 Earnings Call transcript	3 February 2026	Q3/9M FY26 results; commodity-cost detail (antimony/tin/silver/sulfur); EESL cumulative investment ₹4,252 cr; Pravin Saraf confirmed as MD & CEO, EESL; Hyundai/Kia capacity questions first raised
Q4 & FY26 Earnings Call transcript	6 May 2026	Full FY26 results; EESL cumulative investment ₹4,802 cr, FY27 ₹1,400 cr guidance; Hyundai/Kia confirmed as separate/incremental to Phase 1; cylindrical customer samples beginning

Document	Date	Primary content contributed to this dossier
Investor Presentation, November 2025	17 November 2025	Q2/H1 FY26 financial highlights; strategy framework (Revenue Growth/Customer Centricity/Digitization/Process Automation); long-term growth-driver slides; EESL facility/line-by-line construction photos; sustainability CSA score
Investor Presentation, Q4 & FY26 Results	5–6 May 2026	Full FY26 financials; expanded "Exide at a glance" statistics; new product launches (EL Ultra, PowerBox, Exide DARE); Gigafactory 6-stage milestone tracker; industry-context slide (India Li-ion demand 20→140-150 GWh by 2030); governance & ESG recognitions

4.41.1 13.1 Notes on Documentation Discipline

- Where two source documents disclose materially different figures for a metric described with the same label (e.g., the two "core ROCE" series, or CML's lead self-sufficiency ratio), both figures are recorded here rather than silently reconciled or discarded, with the discrepancy flagged as an open question (section 11).
- Management commentary is recorded with attribution to the specific earnings call/quarter where possible, since management's framing has evolved over time (e.g., the increasing specificity around Hyundai/Kia being a separate capex cycle, only fully clarified in the Q4 FY26 call).
- This document should be updated after every quarterly result release, annual report publication, or major corporate action (capex approval, board change, regulatory development). When updating, add new information to the relevant existing section rather than appending a new undifferentiated block, and explicitly flag where a new data point changes a previously stated conclusion.

— *End of Dossier v1.0* — # Part IX — Amara Raja Energy & Mobility Limited

The following is the complete Amara Raja Energy & Mobility Internal Research Dossier, integrated in full and unabridged, using the same approach applied to the Exide dossier in Part VIII. Only heading levels have been adjusted so the dossier nests correctly within this master document's table of contents; no content, conclusion, tag, or figure has been altered from the original.

AMARA RAJA ENERGY & MOBILITY LIMITED

Internal Research Dossier

(Living Document — Institutional-Quality Research Notebook)

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Document version: v1.0 | Compiled: 22 July 2026

Formerly Amara Raja Batteries Limited; renamed 2023

Source documents integrated in this version:

- Gemini NotebookLM research session, "Amara Raja Energy & Mobility" (16 sources: Annual Reports FY19–FY26, Earnings Call transcripts Q1–Q4 FY26, Investor Presentations Q1–Q4 FY26); chat sessions dated 21–22 July 2026

4.42 How to Use This Dossier

This is the master internal working file for Amara Raja Energy & Mobility (ARE&M) research, built entirely from a Gemini NotebookLM research session covering 16 source documents (annual reports FY19–FY26, and Q1–Q4 FY26 concalls/investor presentations). It is a companion document to the existing Exide Industries dossier, and should eventually support direct peer comparison between the two lead-acid incumbents as both pivot into lithium-ion.

This version is built solely from the NotebookLM chat log — i.e., a second-hand synthesis layer, not the primary annual reports/transcripts themselves. Treat every number here as provisional until cross-checked against the primary source once those PDFs are uploaded directly. Several internal inconsistencies surfaced within the NotebookLM session itself (flagged throughout as Open Questions) and should be the first things resolved once primary documents are available.

Tagging convention (same as the Exide dossier):

[FACT] A verifiable, disclosed fact (financial figure, stated policy, corporate action, etc.)

[MGMT COMMENTARY] *Management's own framing/commentary — weigh against facts, don't treat as fact itself*

[ANALYST INTERPRETATION] A conclusion or pattern developed by the NotebookLM research persona or by synthesis across the chat — a hypothesis, not established truth

[OPEN QUESTION] A genuine gap, unresolved figure, or internal contradiction in the current research base

4.43 1. Company Overview

4.43.1 1.1 Identity & Corporate Evolution

[FACT] Amara Raja Energy & Mobility Limited (ARE&M), formerly Amara Raja Batteries Limited, is one of two dominant players (alongside Exide Industries) in India's organised lead-acid battery market — management itself describes the market structure as a duopoly. The company renamed itself from "Amara Raja Batteries" to "Amara Raja Energy & Mobility" in 2023 to reflect a strategic pivot beyond lead-acid into lithium-ion cells, battery packs, EV charging infrastructure, and stationary energy storage (BESS).

[FACT] Founded over four decades ago by Dr. Ramachandra N Galla. Current Chairman & Managing Director is Jayadev Galla. The company's stated guiding philosophy/mantra is "Gotta be a better way."

[MGMT COMMENTARY] *Jayadev Galla on the 2023 rebrand: "Our rebranding... represents more than just a new name; it symbolises our broader vision and dedication... This differentiation allows us to continue delivering growth in our core business while taking dedicated and proactive steps to make ready solutions for emerging applications."*

[ANALYST INTERPRETATION] The stated business model is a deliberate "Dual-Pillar Strategy": maximise the value/cash generation of the legacy Lead-Acid Battery (LAB) business while using that cash to fund the capital-intensive New Energy Business (NEB) build-out. The NotebookLM research repeatedly frames this as: the LAB business is a "synthetic bond" funding a "New Energy call option." This framing (echoing a similar structure identified independently in the Exide dossier) is a useful mental model to carry through the rest of this document — but note that ARE&M's NEB bet is structured very differently from Exide's (separate subsidiary ecosystem with four distinct entities, vs. Exide's single EESL subsidiary — see section 3).

4.43.2 1.2 History & Key Milestones

Period	Milestone
~1985	Amara Raja Group founded; Amara Raja Power Systems Ltd. (later ARPS) was the first company in the group
1990s	Pioneered VRLA (Valve Regulated Lead-Acid) battery technology in India for telecom applications
1995	Diversified into automotive batteries; entered a 21-year Joint Venture with Johnson Controls International (JCI), which took a 26% equity stake
1997	Technical Assistance Agreement with JCI for technology absorption
2000s	Disrupted the Indian automotive aftermarket with "factory-charged, ready-to-fit" AMARON® batteries, shifting the industry away from the high-maintenance "dry-charged" battery model
Sep 2018	Licensed JCI's proprietary PowerFrame® Grid Technology for advanced flooded lead-acid batteries (stamped-grid plate-making know-how)
Nov 2018 – Apr 2019	JCI announced the sale of its global power-solutions business to Brookfield Business Partners; the Amara Raja–JCI Shareholders' Agreement, Investment Agreement, Technical Assistance Agreement and Technology Collaboration Agreement were all terminated effective 1 April 2019, ending the 21-year JV
FY20	JCI's 26% stake was split: the Galla family acquired 2% (raising Galla-family promoter holding from ~26% to 28.06%); the remaining 24% was acquired by an institutional investor, Panther ARBL Holdings LP (later renamed Clarios ARBL Holdings LP). JCI was formally de-classified as a promoter, causing the headline "promoter holding" to fall from ~52.06% to ~28.06%
Jun 2022	Amara Raja Circular Solutions Private Limited (ARCS) incorporated — lead-

Period	Milestone
	recycling subsidiary
Nov 2022	Amara Raja Advanced Cell Technologies Private Limited (ARACT) incorporated — houses the New Energy Business / Giga Corridor
2023	Corporate rebrand to "Amara Raja Energy & Mobility"; groundbreaking for the Amara Raja Giga Corridor in Divitipally, Telangana
1 Jun 2023	"Slump sale" of the New Energy Business into ARACT made effective
29 Sep 2023	Amara Raja Power Systems Limited (ARPS) — historically the group's original 1984-founded company — was acquired from the promoter group and became a wholly-owned subsidiary of ARE&M, bringing chargers/power-electronics in-house
FY25	Construction start of the Giga-1 cell factory; lead-recycling refinery operations (ARCS, Cheyyar) commence
Mid-2025	New tubular battery plant at the Amara Raja Growth Corridor (ARGC) begins commercial production (1.5–1.8 million units p.a.), following reinstatement after a fire (partially insurance-funded)
FY26	e+ Energy Labs (Hyderabad R&D hub) and the Customer Qualification Plant (CQP, multi-chemistry pilot cell line) ramp up; NEB revenue recovers with a 61.4% YoY surge

4.43.3 1.3 The Johnson Controls Joint Venture — Detail

[FACT] The Johnson Controls (JCI) relationship was a formal 21-year Joint Venture (1995/1997–2019), not merely a supply or licensing arrangement. JCI held a 26% equity stake via Johnson Controls (Mauritius) Private Limited and had board representation through nominee directors.

[FACT] Scope of the JV: technology transfer that enabled ARE&M's entry into VRLA batteries for telecom, foundational technology behind the AMARON® brand's OEM and aftermarket leadership, and (from September 2018) a license for JCI's PowerFrame® Grid Technology for advanced flooded lead-acid batteries. ARE&M engineers regularly visited JCI facilities for technology absorption and equipment design suited to Indian conditions.

[FACT] Termination was involuntary from ARE&M's perspective in the sense that it was triggered by JCI's own corporate action — the global sale of JCI's power-solutions business to Brookfield Business Partners LP — rather than by any dispute or ARE&M decision. All four underlying agreements (Shareholders' Agreement, Investment Agreement, Technical Assistance Agreement, Technology Collaboration Agreement) were terminated together effective 1 April 2019.

[ANALYST INTERPRETATION] Despite losing its long-standing technology partner, ARE&M had already absorbed the critical underlying technologies (VRLA, PowerFrame Grid, AMARON manufacturing know-how) over the prior two decades and continued operating and licensing these processes post-exit. The apparent ~24-percentage-point

collapse in "promoter holding" (52.06%→28.06%) is a reclassification artefact of JCI's exit, not a reduction in the founding family's actual economic stake — the Galla family's own holding in fact increased slightly (26%→28.06%) during this transition. Any historical shareholding-pattern chart that shows a cliff-drop in FY20 promoter holding should be read with this context, not interpreted as an actual sell-down or loss of control.

4.43.4 1.4 Corporate Structure & Subsidiaries

Subsidiary	Incorporated / Status	Role & Financials
Amara Raja Advanced Cell Technologies Pvt. Ltd. (ARACT)	Nov 2022	Wholly-owned. Houses the New Energy Business: Li-ion cell manufacturing (Giga Corridor, 16 GWh target) and battery pack assembly (5 GWh target). FY26 revenue ₹786.51 cr; FY26 net loss ₹73.23 cr (FY25 net loss was ₹12.14 cr — losses have widened materially as capex/R&D scaled).
Amara Raja Circular Solutions Pvt. Ltd. (ARCS)	Jun 2022	Wholly-owned. Operates the lead-acid battery recycling plant at Cheyyar, Tamil Nadu; scaling from an initial 50,000 MT refinery capacity toward an eventual 150,000 MTPA, targeting 25–30% of the group's captive lead requirement. FY26 gross revenue ~₹1,456.36 cr, almost entirely captive/internal (parent purchased ₹1,424.19 cr of goods from ARCS in FY26, eliminated on consolidation).
Amara Raja Power Systems Ltd. (ARPS)	Est. 1984; became ARE&M subsidiary 29 Sep 2023	Historically the first company in the Amara Raja Group; re-acquired from the promoter group in 2023 to bring chargers and power-electronics in-house. Focus: industrial/EV chargers, integrated power systems for railways, BESS. FY24 revenue ₹105.38 cr / PAT ₹9.10 cr; by FY26, despite an order book of 5,000+ charger units, ARPS posted a net loss of ₹20.69 cr — still in an investment/ramp-up phase.
Amara Raja Batteries Middle East FZE (ARBME)	Jul 2018, Sharjah, UAE	Wholly-owned international marketing arm serving the Middle East and Africa. FY26 revenue ₹15.84 cr, fully external (no internal elimination).

[FACT] Joint Ventures / Associates: The historical JCI JV ended in FY19-20 (see 1.3). As of the FY26 filings reviewed, ARE&M explicitly states it has no existing joint ventures or associates.

4.44 2. Business Model & Segment Structure

4.44.1 2.1 Reporting Structure — Standalone vs. Consolidated

[FACT] Standalone: ARE&M reports as a single statutory operating segment under Ind AS 108 ("manufacturing and marketing of storage batteries"). Operationally, however, management and the research base consistently bifurcate standalone revenue into an Automotive vertical (~67–71% of revenue) and an Industrial vertical (~25–29%).

[FACT] Consolidated: Effective 1 April 2023, the group reports two distinct segments under Ind AS 108: (i) Lead-Acid Batteries & Allied Products, and (ii) "Others" / New Energy Business — broadly corresponding to the standalone entity plus ARBME on one side, and ARACT + ARPS on the other.

[ANALYST INTERPRETATION] An important nuance for modelling: some lithium pack revenue still flows through the standalone entity as "trading revenue" even though the bulk of NEB activity now sits in ARACT — meaning the standalone/consolidated split is not a perfectly clean LAB-vs-NEB line. This is visible in the sharp rise in standalone "Trading Activity" revenue share (from ~4% in FY23 to 15–16% in FY25/FY26), which includes lubricants, traded Home UPS systems, some tubular batteries, and likely some lithium-product resale.

4.44.2 2.2 Automotive Vertical

[FACT] Brands: Amaron® (premium), PowerZone™ (value/rural-focused), ELITO® (international markets).

[FACT] Products: 4W and 2W batteries including AGM and EFB technologies (for start-stop/hybrid applications), plus 3W, commercial-vehicle, and farm-equipment batteries.

[FACT] Internal split: Replacement/Aftermarket ~70% of the Automotive vertical vs. OEM ~30% (a working assumption validated by the NotebookLM research, consistent with management's own framing of the aftermarket as "3x" the OEM opportunity over a vehicle's life given 2–3 replacement cycles per vehicle).

[MGMT COMMENTARY] *Management frames the OEM segment as a leading indicator: OEM volume growth today converts into aftermarket/replacement revenue roughly 3 years later, once those vehicles first need a battery replacement.*

[FACT] Adjacent/allied revenue: Home UPS/inverter batteries, and a newer Lubricants business ("Amaron Hi-Life Lubes") that reportedly crossed ₹100 crore in FY25 and was running at a ~₹50 crore/quarter rate in FY26 — riding on the existing 100,000+ retail-touchpoint distribution network.

[OPEN QUESTION] Whether the Lubricants business is margin-accretive in its own right, or primarily a volume-filler leveraging existing distribution infrastructure, has not been established from the research base and needs direct verification.

4.44.3 2.3 Industrial Vertical

[FACT] Brands: PowerStack®, Amaron Quanta®, Amaron Volt®, Amaron Sleek®, Amaron Brute™. Products: VRLA batteries spanning 7 Ah to 6,000 Ah.

[FACT] Key sectors: Telecom (market leadership position, but lead-acid revenue share has structurally collapsed to under 5% of total revenue as the industry migrates to lithium-ion — cumulative lithium telecom installations have crossed 1 GWh); UPS and Data Centers (ARE&M states it holds the largest installed capacity in the Indian Data Center battery segment, with disclosed exposure to the Adani Group as a client); Railways (AC/lighting batteries, integrated power systems); Power & Utilities, Oil & Gas, Solar integrators; Motive Power (aviation cargo equipment, forklift-fleet rental, heavy material handling).

[MGMT COMMENTARY] *Management explicitly frames Telecom+UPS as historically ~20% of total revenue combined, with the Telecom lead-acid component now shrunk to <5% — i.e., the segment has structurally de-risked from a Telecom-concentration standpoint even as absolute Telecom LAB revenue has declined ~30–35%, because Data Center/UPS growth (15%+) has more than offset it.*

[ANALYST INTERPRETATION] This is functionally the same lead-acid-to-lithium cannibalization story documented independently in the Exide dossier for the Telecom vertical — useful for cross-checking industry-wide migration speed assumptions between the two companies rather than treating either company's disclosure in isolation.

4.44.4 2.4 New Energy Business (NEB)

[FACT] Housed principally in ARACT (cells/packs) and ARPS (chargers/power systems/BESS). Products: lithium-ion battery packs (NMC and LFP chemistries), Battery Management Systems (BMS), and — pending Giga Corridor commissioning — lithium-ion cells.

[FACT] The Giga Corridor: a 262-acre site in Divitipally, Telangana, covering a planned 16 GWh Li-ion cell manufacturing facility and a 5 GWh battery-pack assembly plant, with a total committed outlay of ₹9,500 crore over 5–6 years. Phase 1 is a 2 GWh cylindrical-cell line, with commercial start-of-production (SOP) targeted for H1-CY2027.

[FACT] Customer Qualification Plant (CQP): a commercial-scale pilot line supporting multiple chemistries (NMC/LFP) and multiple form factors, intended to validate samples with OEMs ahead of full Giga-1 commissioning. Full-scale CQP operation is targeted for Q2-FY27 — management and the research base both flag this as the single most important near-term "go/no-go" leading indicator for Phase 1 offtake, since successful 4W-OEM sample qualification here is the prerequisite for justifying further capital deployment.

[FACT] e+ Energy Labs: an advanced R&D/innovation hub in Hyderabad, with 300+ professionals hired, housing dedicated cell- and pack-level test/validation labs.

[FACT] BESS (Battery Energy Storage Systems): a 10 GWh integration-facility target, with an initial 5 GWh facility targeted for SOP by end-CY2026/Q4-FY27, carrying a specific disclosed capex outlay of ₹280 crore. Management claims BESS will have a very high **asset-**

turnover profile of 9–10x — materially higher than the legacy lead-acid business — which, if realised, could be the first NEB sub-vertical to reach EBITDA neutrality (potentially FY28 per the research synthesis).

[FACT] Chargers (via ARPS): fully operational and commercially scaled, not merely an R&D project. ARE&M states it was the first Indian company to localise IP67-rated off-board EV chargers (materially better water resistance than the industry-standard IP20), and has achieved 50%+ local content on DC fast chargers under India's Phased Manufacturing Programme (PMP). Product range spans 2 kW/3 kW off-board chargers (Piaggio, Mahindra), 3.5 kW/7.4 kW AC Type-2 portable chargers (ARAI type-tested) for 4-wheelers, 30–120 kW DC fast chargers supplied to Charge Point Operators (e.g., Yellow Stone EV, GO EC), and 180–240 kW chargers under development for buses/heavy trucks. Order book had crossed 5,000 units as of Q2 FY26.

[MGMT COMMENTARY] *On the strategic rationale for the charger business: "Our EV charger business has yielded promising results. We have secured a strong order pipeline, which has helped us capture a strong share of the growing charging infrastructure market." Management also frames the combined "Battery + Charger" bundled offering as a way to increase share-of-wallet and customer stickiness with EV OEMs.*

[ANALYST INTERPRETATION] The charger business is a genuinely under-appreciated proof point in the research base: it is commercially live, revenue-generating, and technologically differentiated (IP67, PMP-compliant localisation) — unlike the cell-manufacturing bet, which remains pre-revenue. However, ARPS itself posted a net loss (₹20.69 cr, FY26) despite a healthy order book, indicating the segment has not yet reached profitable scale even where it is commercially operational.

4.44.5 2.5 Exports & International Business

[FACT] Exports reach 70+ countries, historically concentrated in the Indian Ocean Rim, South-East Asia, Middle East, and Africa, with recent expansion into the USA (the world's largest single automotive-battery market) and further planned expansion into Europe and Latin America.

[FACT] Management explicitly positions the export strategy as "technology-agnostic" — i.e., not confined to lead-acid. ARE&M is already shipping lithium packs and energy-storage solutions internationally (including a disclosed major Energy Storage battery-system order from Africa) and is using an "expanded LiB portfolio" to grow international UPS/distribution business, partly to offset overseas Telecom lead-acid volume losses as international customers also migrate to lithium.

[MGMT COMMENTARY] *Management characterised FY26 as a "tough year" for exports specifically due to tariff issues in North America and geopolitical tensions in the Middle East, with no clear resolution guided for H1 FY27.*

[OPEN QUESTION] There is an unreconciled discrepancy in the research base between a standalone export figure of ~₹1,404 crore and a consolidated export figure of ₹1,684.37 crore (12.2% share) for FY26 — both quoted at different points in the same NotebookLM

session. This likely reflects the standalone-vs-consolidated distinction (i.e., ARBME and other subsidiary export sales adding to the standalone base) but has not been explicitly reconciled and should be checked directly against the FY26 Annual Report segment note.

4.45 3. Financial Analysis

4.45.1 3.1 Consolidated Revenue — Three-Year Trend

Segment (₹ Cr)	FY24	FY25	FY26	YoY growth
Lead-Acid Batteries & Allied (LAB)	11,186.17	12,345.13	13,005.29*	+10.36% (FY24→25); ~+5.3–7.5%** (FY25→26)
New Energy Business (NEB) / Others	522.27	501.19	808.71	-4.04% (FY24→25); +61.4% (FY25→26)
Consolidated Total	11,708.44	12,846.32	13,814.00	+9.72% (FY24→25); +7.5-8.6%** (FY25→26)

*Source: Gemini NotebookLM synthesis of Annual Reports FY24–FY26 and Q1–Q4 FY26 concalls/PPTs. *The FY26 LAB figure and its growth rate were quoted inconsistently across different points in the same research session (LAB growth vs. FY25 base is described elsewhere as ~7.5% blended, which would imply a slightly different FY26 LAB absolute figure than ₹13,005.29 cr) — flagged as an open item in section 9. **Both a ~7.5% and an ~8.6% total consolidated growth figure for FY26 appear at different points in the source chat.*

4.45.2 3.2 Standalone vs. Consolidated Revenue Bridge (FY26)

[FACT] FY26 standalone revenue: ₹13,548.86 crore. FY26 consolidated revenue: ₹13,814.00 crore. Delta: ₹265.14 crore, representing the net external (non-eliminated) revenue contributed by the subsidiary structure.

Subsidiary	Gross FY26 Revenue (₹ Cr)	Internal Sales to Parent (₹ Cr)	Implied External Contribution (₹ Cr)	Share of the ₹265 Cr delta
ARACT (New Energy)	786.51	~640.33	~146.18	~55% of the delta
ARPS (Power Systems / Chargers)	149.32	~38.34	~110.98	~42% of the delta
ARBME (Middle East)	15.84	0.00	15.84	~6% of the delta
ARCS (Recycling)	~1,456.36	1,424.19	~32.17	Mostly eliminated (captive)

[ANALYST INTERPRETATION] The small size of the net consolidation delta (₹265 cr) against gross subsidiary turnover exceeding ₹2,400 crore combined is itself a meaningful data point: it confirms that ARCS (recycling) and, to a significant extent, ARACT

(cells/packs) function largely as captive internal suppliers to the parent today, rather than external revenue generators. This is a genuine vertical-integration moat (capturing manufacturing margin, insulating the parent from LME lead volatility and Chinese cell-price swings) — but it also means headline consolidated growth understates how much of the NEB/recycling capacity build-out is not yet monetised externally.

4.45.3 3.3 Segment Profitability — Where the Losses Sit

[FACT] NEB segment: FY26 consolidated revenue ₹808.71 cr against a segment loss of ₹135.15 cr — an implied EBIT margin of approximately -16.7% (i.e., total segment costs ran at ~117% of segment revenue).

[FACT] ARACT specifically: FY25 net loss ₹12.14 cr (on a much smaller revenue base) widening to a FY26 net loss of ₹73.23 cr (on revenue of ₹786.51 cr) — losses have grown roughly 6x year-on-year even as revenue grew ~61%, reflecting the scaling of R&D and pre-commissioning capex-related overhead.

[FACT] ARPS specifically: FY26 net loss ₹20.69 cr, despite an order book exceeding 5,000 charger units — indicating the charger business, while commercially live, has not yet reached break-even scale.

[MGMT COMMENTARY] *Management's own framing of the loss driver: "The consolidated level margin dilution is due to additional expenses that we are incurring on our New Energy business product development and ramping up some of the production facilities." On the future profitability path: management projects an EBITDA margin of 10–11% for NEB once the business reaches a scale of 8–10 GWh.*

[ANALYST INTERPRETATION] Three structural reasons underlie the current losses, per the research synthesis: (1) under-absorption of fixed costs — infrastructure and headcount (300+ engineers at e+ Energy Labs alone) have been built for an eventual 16 GWh scale but are currently supporting only a small pack-assembly/CQP-testing revenue base; (2) R&D intensity — NMC 21700 cell development and LFP validation work is largely expensed rather than capitalised, unlike the lead-acid business's more "value maximisation"-oriented capex; (3) import dependency — until Giga-1 cell production begins, ARE&M is effectively a pack assembler buying Chinese cells, making it a price-taker in a market currently characterised by Chinese oversupply/aggressive pricing (cells reportedly near ~\$50/kWh).

[OPEN QUESTION] How much of the ₹135 crore FY26 NEB segment loss is non-cash depreciation/amortisation versus genuine cash R&D burn has not been disclosed or estimated in the research base — this materially affects how the loss should be read from a cash-flow-runway perspective.

[OPEN QUESTION] The exact cash break-even revenue level for the standalone pack-assembly business (i.e., excluding Giga-factory pre-operative overhead) has not been established.

4.45.4 3.4 Margins, COGS & Cost Structure

[FACT] Material cost as a % of Sale of Products (standalone): 68.5% (FY25) → 69.7% (FY26), reflecting higher alloy and sulfuric-acid input costs.

[FACT] Primary raw materials: lead, tin, and antimony alloys, plus sulfuric acid. Antimony ore prices surged roughly 5x during FY25 — a commodity-shock example directly comparable to the antimony spike independently documented in the Exide dossier for the same period, suggesting an industry-wide (not company-specific) input-cost shock.

[FACT] Consolidated EBITDA margin: ~10.8% in FY26, down from a historical FY15–FY19 band of 14–16%, and below management's own long-standing 14% long-term aspiration.

[MGMT COMMENTARY] *On the margin gap: "Considering the higher lead base... close to ₹2,10,000 [per ton]... we will have to first reach the first milestone of 13% [EBITDA]... thereafter move back to our original 14%."*

[FACT] Lead prices have structurally re-based from ~₹1.5 lakh/ton to ~₹2.1 lakh/ton over the past three years.

[FACT] Recycling (ARCS) margin accretion: management states the captive recycling plant currently provides a 0.5–0.6% accretion to consolidated EBITDA margin by supplying internally sourced lead, with an aspiration to scale this toward 1.0–1.5% as the Cheyyar plant moves from refining-only operations to full "battery breaking" (expected to start around Q2 FY27).

[FACT] Li-ion cell cost target: management explicitly states ARE&M cannot be cost-competitive with Chinese cell imports today, and is targeting "China plus 15–20%" as the achievable near-term cost gap, aiming to close this over roughly the next 5 years. Cell manufacturing capital cost was originally pegged at \$55–60 million per GWh, though management notes this has since fallen 20–25% due to global equipment-price corrections.

[FACT] Government-related cost support: the research base found no specific government-mandated "production cost" for the industry. It does reference broader industry-level ACC (Advanced Chemistry Cell) PLI subsidies exceeding ₹50,000 crore (an India-wide scheme, not ARE&M-specific) and a 5% basic customs duty concession on cells intended to support domestic localisation.

[FACT] Extended Producer Responsibility (EPR) liability: a ₹35 crore provision was booked in Q2 FY26, with management guiding to a recurring impact of roughly ₹1 crore/month going forward — directly analogous to the EPR cost dynamic independently documented in the Exide dossier.

4.45.5 3.5 Returns, Debt & Capital Structure

[FACT] ROCE: 19.2% (FY24) declined to 16.2% (FY25), attributed to the large capital deployment into long-gestation New Energy assets that are not yet generating commensurate revenue.

[FACT] Debt profile: ARE&M has been debt-free for over a decade at the group level, historically carrying only interest-free sales-tax deferment loans obtained under state industrial policies. Gearing ratio remains in the 0.00x–0.03x range even as of FY26.

[FACT] On 14 August 2025, the Board approved an enhancement of working-capital limits by ₹500 crore and commercial-paper issuance capacity of up to ₹500 crore (₹1,000 crore combined enhanced flexibility). Current borrowings rose to ₹271.46 crore in FY26 (working-capital related).

[MGMT COMMENTARY] *CFO commentary on funding philosophy: "The business continues to benefit from a conservative capital structure... internal funding sources also provides flexibility in sequencing future investments."*

[ANALYST INTERPRETATION] The August 2025 CP/working-capital enhancement is a meaningful signal even though gearing remains near-zero today: it suggests management is building optionality for short-term bridge funding — likely to smooth working-capital swings from geopolitical/supply-chain disruption and to maintain a safety buffer given the sheer scale of the Giga Corridor programme — without yet committing to structural long-term debt. Whether the company eventually needs to shift to long-term project debt for Giga Corridor Phases 2 and 3 (the remaining ~₹8,000 crore of the ₹9,500 crore commitment) if ROCE remains suppressed in the low-double-digit range is an open strategic question (see section 9).

4.46 4. Capital Expenditure & Capacity Roadmap

4.46.1 4.1 Historical Capex (FY20–FY26)

FY	Capex (₹ Cr)	Key focus
FY20	272.66	Debottlenecking VRLA lines; poly-container line for global markets
FY21	948.45	Capacity augmentation; board approval for the lead-recycling plant
FY22	444.57	Capacity augmentation despite inflationary/pandemic headwinds
FY23	1,176.00	Giga Corridor groundbreaking (Telangana); solar-plant commissioning
FY24	478.00	Rebranding phase; first high-speed pack-assembly facility established
FY25	1,200.00	Giga-1 factory construction start; recycling-refinery (ARCS) operations commence
FY26	1,341.00	e+ Energy Labs and Customer Qualification Plant (CQP) ramp-up

[ANALYST INTERPRETATION] Capex has scaled roughly 6x from FY20 (₹273 cr) to FY26 (~₹1,341 cr) — explicitly described in the research base as the most capital-intensive phase in the company's 40-year history. The step-change from FY22 (₹445 cr) to FY23 (₹1,176 cr) cleanly marks the point where the Giga Corridor moved from strategic announcement to physical construction.

4.46.2 4.2 The ₹9,500 Crore Giga Corridor Commitment

[FACT] Total outlay: ₹9,500 crore over 5–6 years, covering a 16 GWh Li-ion cell facility and a 5 GWh battery-pack assembly plant at a 262-acre site in Divitipally, Telangana.

[FACT] Cumulative equity infused into ARACT as of end-FY26: ~₹1,500 crore — i.e., roughly 16% of the total ₹9,500 crore commitment has been deployed so far, with the vast majority of the capital commitment still ahead.

[FACT] Phase 1: 2 GWh cylindrical-cell line; commercial start-of-production (SOP) targeted for H1-CY2027.

[FACT] FY27 guided consolidated capex: ₹1,500–1,700 crore, of which ₹1,100–1,200 crore is earmarked for the New Energy Business and roughly ₹300–500 crore for the core LAB business and recycling infrastructure combined. A specific ₹280 crore outlay is guided for the 5 GWh BESS integration facility.

4.46.3 4.3 Capacity Roadmap Table

Segment	Current Capacity (FY26)	Target Capacity	Key Milestone
Li-ion Cells	0 GWh (CQP testing only)	16 GWh (FY30/31 target)	2 GWh Phase-1 SOP targeted H1-CY2027
Li-ion Packs	~2.5 GWh (consolidated)	5 GWh	Scaling via new mobility OEM relationships
BESS	0 GWh (under construction)	10 GWh	5 GWh SOP targeted end-CY2026 / Q4-FY27
Automotive LAB	~66 million units	Incremental (throughput-led)	Capacity gains via process/"throughput" improvements rather than new greenfield lines
Lead Recycling (ARCS)	50,000 MT (refinery only)	150,000 MTPA	"Battery breaking" operations targeted to start Q2-FY27, moving beyond refining-only

[ANALYST INTERPRETATION] Note the important distinction management draws between the legacy LAB business (where recent capacity gains — e.g., 4W capacity from 59 Mn to 66 Mn units — have come from "sweating" existing assets/throughput optimisation rather than new plants) and the New Energy business (which is genuinely greenfield, long-gestation capex). This mirrors the same "two-speed" capital-intensity pattern documented in the Exide dossier's core-vs-EESL capex framing.

4.46.4 4.4 Segment-Wise Capex Guidance Summary

Segment	FY27 Guidance	Long-term / Project Total	Key Projects
New Energy	₹1,100–	₹9,500 cr over 5–6	16 GWh Giga cell factory, CQP,

Segment	FY27 Guidance	Long-term / Project Total	Key Projects
Business (NEB)	1,200 cr	years	e+ Energy Labs
Lead-Acid Battery (LAB) & Recycling	~₹300–500 cr	₹1,001.44 cr committed to SIPCOT (Tamil Nadu) for the recycling plant; ~₹488 cr spent as of FY25	Maintenance capex, throughput/process improvements, 1.8 Mn-unit tubular-battery plant
BESS	Included within NEB above	₹280 cr specific plant outlay	5 GWh integrated solution plant for grid/C&I applications
Consolidated Total	₹1,500–1,700 cr	₹9,500 cr+	Highest investment phase in company history

4.47 5. Management Guidance Track Record — Hit / Miss Scorecard

This section is one of the most analytically valuable outputs of the NotebookLM research: an explicit, cross-year tracking of management guidance against actual results. Building and maintaining this scorecard every quarter is one of the highest-value habits for judging management credibility over time.

Guidance Area	Original Target	Actual Result	Verdict
NEB Revenue	₹1,000 crore by FY25 (guided in the FY24 Annual Report)	₹501.2 crore (FY25)	MAJOR MISS (~50% shortfall)
NEB Revenue	"Rescaled" momentum guidance for FY26	₹808.7 crore (FY26)	MISS vs. original ₹1,000 cr target, though +61.4% YoY recovery
EBITDA Margin	14% long-term aspiration	10.8% consolidated (FY26)	MISS
LAB Revenue Growth	8–10% guided	~7.5% (FY26 actual, per one synthesis point)	NEAR HIT (slightly below range)
Capex (FY26)	₹1,400–1,500 crore guided	~₹1,341 crore reported	Under-Spend vs. guidance
Telecom Market Share	Maintain >50% market share through the lithium transition	Protected >50%	HIT
Consolidated Topline CAGR (5-	15–17% CAGR	~12.7% actual (FY24, vs.	MISS on the high end; broadly in line with the

Guidance Area	Original Target	Actual Result	Verdict
yr, from FY21 guidance)		restated FY23 base)	separately-guided 10–12% automotive-industry-specific growth range

4.47.1 5.1 The NEB Guidance Pattern — A Closer Look

[MGMT COMMENTARY] *FY24 Annual Report language: the battery-pack and charging-solution business was "projected to reach close to ₹1,000 crore by next financial year [FY25]." Actual FY25 result: ₹501.19 crore — roughly half the guided figure, and management attributed the miss to "lower OEM offtake" in the EV battery-pack segment.*

[MGMT COMMENTARY] *By FY27 guidance, management has notably shifted from an absolute-rupee target to a revenue-share target: "At least move to a 7% to 8% [NEB revenue share] kind of number."*

[ANALYST INTERPRETATION] This shift — from a specific absolute rupee figure (₹1,000 cr) that was missed by ~50%, to a relative revenue-share target (7–8%) — is itself worth flagging as a pattern: relative/share-based targets are structurally easier to hit or explain away (since they move with the denominator too) and harder for outside analysts to hold management precisely accountable against. The research base's own recommendation is to apply a 15–20% "execution haircut" to any NEB forecast derived from management guidance, given the FY25 track record.

[ANALYST INTERPRETATION] On the implied math for the FY27 target: to reach an 8% NEB revenue share on a consolidated base of roughly ₹15,500 crore (a plausible FY27 consolidated total under base-case LAB growth assumptions), NEB revenue would need to reach approximately ₹1,240 crore — implying required YoY NEB growth of roughly 50–60%, similar in magnitude to the 61.4% surge actually delivered in FY26. Whether this repeats depends heavily on: (i) continued telecom-lithium migration (installations already past 1 GWh), (ii) the BESS facility actually reaching SOP by Q4 FY27 as guided (even one quarter of grid-scale BESS offtake could add ₹100–150 cr), and (iii) the charger order book (5,000+ units) converting to delivered revenue faster than a conservative ~15% growth assumption would imply.

4.47.2 5.2 What This Scorecard Says About Management Style

[ANALYST INTERPRETATION] The pattern that emerges is consistent with what was independently observed in the Exide dossier for a different management team: strong, credible execution on the legacy core business (LAB growth, market-share defence in Telecom) paired with a track record of over-optimism on new-energy ramp timelines and absolute targets. Unlike Exide — where management has been notably tight-lipped and has repeatedly declined to give specific forward numbers on its EESL business — ARE&M's management has been more willing to give specific numeric targets (₹1,000 cr NEB revenue, 14% EBITDA margin, specific GWh timelines), which has in turn made it easier to document specific misses. This is a useful contrast for peer-comparison purposes: ARE&M's management is more "talkative"/specific but has a visible miss record; Exide's

management is more guarded/vague but has therefore not yet been caught missing a specific number (because so few specific numbers have been given).

4.48 6. FY27 Management Guidance (Current)

Metric	FY27 Guidance
LAB Revenue Growth	8% to 10%
NEB Revenue Share (Consolidated)	7% to 8%
Industrial Vertical Growth	High single digits
Consolidated Capex	₹1,500 – ₹1,700 crore
NEB Milestones	Full-scale operation of the CQP and e+ Energy Labs
Giga-1 Cell Phase 1 SOP	H1-CY2027
BESS (5 GWh) SOP	Q4-FY27 / end-CY2026
Recycling (ARCS) "battery breaking" ops	Targeted start Q2-FY27

4.48.1 6.1 Associate Model Stress-Test (worked example from the research session)

The NotebookLM session includes a worked example of stress-testing a bottom-up analyst model against management's FY27 guidance — a useful template to reuse each year:

Segment	FY26 Base (₹ Cr)	Growth Assumption	FY27 Forecast (₹ Cr)	% Consolidated Share
Automotive Replacement	6,463.6	12.0%	7,239.2	47.3%
Automotive OEM	2,770.1	9.0%	3,019.4	19.7%
Industrial Vertical	3,771.6	8.0%	4,073.3	26.7%
I. Total LAB	13,005.3	~10.2% (blended)	14,331.9	93.7%
Lithium Battery Packs	647.0	20.0%	776.4	5.1%
Chargers & Power Systems	161.7	15.0%	186.0	1.2%
II. Total NEB	808.7	~19.0% (blended)	962.4	6.3%
Consolidated Total	13,814.0	~10.7%	15,294.3	100%

[ANALYST INTERPRETATION] Verdict from this exercise: LAB-side assumptions (10.2% blended growth) align well with management's 8–10% guidance and are "fundamentally sound." However, a 19% blended NEB growth assumption — which might feel reasonable based on "broader economic activity" — under-delivers materially against management's

implied target: to hit an 8% NEB revenue share, NEB would need to reach roughly ₹1,240 crore, which requires blended NEB growth closer to 50–60% YoY, not ~19%. The gap between a conservative bottom-up NEB estimate and management's implied requirement is the single most important reconciliation question to raise on the next earnings call — i.e., ask management directly what specific growth drivers (BESS SOP timing, charger deliveries, telecom-pack volumes) justify a 50%+ growth assumption when a straightforward extrapolation of current run-rates suggests something closer to 20%.

4.49 7. Competitive Positioning & Distribution Moat

4.49.1 7.1 Market Structure

[FACT] ARE&M operates in a structural duopoly with Exide Industries in India's organised lead-acid battery market — a fact stated directly within the ARE&M research base itself.

[FACT] First-mover claims: first company to manufacture AGM motorcycle batteries in India; first company globally to develop "Punched Grid" battery technology (via the Sovema SpA partnership's Multi-Grid Panel Stamping & Pasting process).

[ANALYST INTERPRETATION] Note: Exide's own research (see the Exide dossier) also describes a "Punch Grid" automation initiative for its 2-wheeler battery lines. This is very likely industry-standard punched/stamped-grid manufacturing technology being adopted by both major Indian players around a similar timeframe, rather than a unique, defensible technology edge belonging to either company alone — worth keeping in mind if either company's investor materials frame it as a differentiator.

4.49.2 7.2 Distribution — The Replacement-Market Moat

[FACT] Retail reach: 100,000+ retail touchpoints, 30,000+ retailers, 3,000+ Amaron franchisees/channel partners. Secondary distribution: 36+ distribution points and 21 branches connecting 4,000+ destinations. Fleet: ~65% company-owned fleet, supporting Just-In-Time (JIT) delivery to OEM customers.

[MGMT COMMENTARY] *The strategic logic, per the research synthesis: a storage battery failure is an "urgent emergency" for the end customer — whether a stranded vehicle owner or a data centre facing a power outage — meaning availability/distribution density, not just product quality, is the primary competitive differentiator in the replacement market.*

[FACT] Evidence of high switching costs / deep customer integration: securing 100% volume share on specific fast-moving 4W OEM platforms, and a "Best Quality" award from a major telecom-tower client (Indus Towers).

4.49.3 7.3 Positioning on the Lithium Transition

[FACT] Management positions ARE&M as pursuing "chemistry-agnostic solutions" — i.e., aiming to retain a given customer relationship (OEM or channel) regardless of whether that customer ultimately prefers lead-acid or lithium-ion, rather than ceding lithium-curious customers entirely to specialised pure-play lithium entrants.

[ANALYST INTERPRETATION] This mirrors, almost exactly, the strategic logic independently identified in the Exide dossier around Exide's own Telecom/E-Rickshaw lithium migration strategy — both major Indian lead-acid incumbents are pursuing the same "defend the customer relationship across the chemistry transition" playbook, rather than treating lithium purely as a threat to be resisted.

4.50 8. Investment Thesis

4.50.1 8.1 Bull Case

- A genuine, distribution-led moat in the high-margin lead-acid replacement/aftermarket business (100,000+ touchpoints, 3,000+ franchisees), which is structurally counter-cyclical to the lower-margin OEM business and provides earnings resilience through auto-sector downturns.
- A commercially live, technologically differentiated, and already-scaling EV-charger business (ARPS) — India's first IP67-localised off-board charger, PMP-compliant, 5,000+ unit order book — that is not merely a future promise but an operating proof point for the "Energy & Mobility" thesis.
- A deliberately "chemistry-agnostic" strategy across both the core Telecom vertical and the export business, aimed at retaining customer relationships through the lead-acid-to-lithium transition rather than ceding them to specialist entrants — already evidenced by 1 GWh+ of telecom lithium-pack installations.
- Vertical integration via ARCS (lead recycling, scaling toward 150,000 MTPA and 25–30% captive lead supply) providing a genuine, if currently modest (0.5–0.6%, targeting 1.0–1.5%), structural margin buffer against LME lead volatility.
- A conservative, historically debt-free balance sheet (gearing 0.00x–0.03x) funding the ₹9,500 crore Giga Corridor primarily through internal accruals, preserving financial flexibility even as capex intensity rises to record levels.
- BESS's claimed high asset-turnover profile (9–10x) offers a potential path to the fastest EBITDA-accretive scaling within the NEB portfolio, if the 5 GWh facility hits its Q4 FY27 SOP target.

4.50.2 8.2 Bear Case

- A demonstrated, specific track record of missing New Energy Business targets — the ₹1,000 crore FY25 NEB revenue guidance was missed by roughly 50% — which should discount the credibility of the current 7–8% FY27 revenue-share target (implying ~50–60% NEB growth) until proven otherwise.
- Consolidated EBITDA margin (10.8% FY26) sits well below management's own stated 14% long-term aspiration, with management itself acknowledging a multi-year path back (first to a 13% "milestone") given a structurally higher lead-price base (~₹2.1 lakh/ton) and an increasing, lower-margin OEM revenue mix.

- ROCE has already declined from 19.2% (FY24) to 16.2% (FY25) as capital has been deployed into long-gestation New Energy assets; with FY27 capex guided even higher (₹1,500–1,700 cr) and Giga-1 commercial revenue not expected before H1-CY2027, further ROCE compression is likely before any recovery.
- Explicit management acknowledgement that ARE&M cannot compete with Chinese cell import prices today, targeting only a "China plus 15–20%" cost position even at scale — with a real risk that continued Chinese cell oversupply/dumping (cells reportedly near \$50/kWh) could keep this gap wider than planned, threatening Phase 1 utilisation economics.
- Firm, binding offtake commitments for the 16 GWh Giga capacity remain unconfirmed — publicly disclosed relationships with Ather Energy and Piaggio are described as MoUs for "long-term product alignment," not confirmed multi-year supply contracts, leaving genuine demand risk for the bulk of the committed capacity.
- Simultaneous execution across four distinct new-business fronts (Li-ion cells, Li-ion packs, chargers/BESS, and recycling) inside four separate subsidiaries places significant strain on management bandwidth and capital-allocation discipline.

4.50.3 8.3 Variant Perception

[ANALYST INTERPRETATION] The market narrative on ARE&M likely centres on either "legacy lead-acid duopoly member facing EV disruption" or "emerging Indian lithium/EV-infrastructure story." The more useful framing, consistent with the research base's own repeated "synthetic bond funding a call option" language, is to explicitly separate valuation of (a) a genuinely high-quality, high-distribution-moat lead-acid replacement business generating real, if margin-pressured, cash flow, from (b) a high-risk, high-optionality, execution-dependent New Energy bet spread across four subsidiaries with a specific and recent history of guidance misses. Whether the current market price properly discounts the New Energy bet's execution risk — given the ₹1,000 cr FY25 miss as a directly comparable precedent — or is instead pricing in an assumption that FY27's 7–8% NEB share target will simply be hit, is the central question this dossier should keep testing every quarter.

4.50.4 8.4 Key Catalysts / Monitorables

- CQP full-scale operation and successful 4W-OEM sample qualification (targeted Q2 FY27) — the single most important near-term leading indicator for Giga-1 offtake viability.
- Any conversion of the Ather Energy / Piaggio MoUs into firm, disclosed multi-year offtake contracts.
- BESS facility actually reaching 5 GWh SOP by the guided Q4 FY27 / end-CY2026 window.

- Whether FY27 NEB revenue actually approaches the ~₹1,240 crore implied by the 7–8% share target, or repeats the pattern of falling meaningfully short.
- Trajectory of the recycling (ARCS) margin accretion as "battery breaking" operations begin (targeted Q2 FY27) — watch for the 0.5–0.6% figure moving toward the aspired 1.0–1.5%.
- Drawdown (or non-drawdown) of the new ₹500 cr Commercial Paper facility — an early signal of whether the group is shifting toward a more leveraged funding posture for the remaining ~₹8,000 crore of Giga Corridor capex.

4.51 9. Open Questions & Items to Reconcile

[OPEN QUESTION] FY26 LAB revenue and its growth rate are quoted inconsistently within the same research session (₹13,005.29 cr with an implied ~5.3% growth in one table, vs. a separately stated ~7.5% blended LAB growth figure elsewhere). Needs reconciliation against the primary FY26 Annual Report segment note.

[OPEN QUESTION] FY26 export revenue is quoted as both ~₹1,404 crore (standalone) and ₹1,684.37 crore / 12.2% share (consolidated) at different points in the chat — likely a standalone-vs-consolidated distinction, but not explicitly reconciled.

[OPEN QUESTION] Consolidated FY26 total revenue growth is quoted as both ~7.5% and ~8.6% at different points — needs a single, reconciled figure.

[OPEN QUESTION] Whether Ather Energy and Piaggio's MoUs with ARACT have progressed to firm, binding, multi-year offtake agreements — critical for justifying the remaining ~₹8,000 crore of Giga Corridor capex.

[OPEN QUESTION] The LFP-vs-NMC chemistry decision for Giga Corridor Phase 2 has not been finalised/disclosed, despite a 25–30 GWh BESS demand projection by FY31 that would logically favour LFP for stationary storage.

[OPEN QUESTION] Exact cash break-even revenue for the standalone pack-assembly business, and the cash-vs-non-cash composition of the ₹135 crore FY26 NEB segment loss.

[OPEN QUESTION] The specific lead price and/or recovery ratio required for the ARCS recycling plant's margin accretion to move from its current 0.5–0.6% toward the aspired 1.0–1.5%.

[OPEN QUESTION] Whether the company will eventually take on long-term project debt for Giga Corridor Phases 2 and 3 (the remaining ~₹8,000 crore) if ROCE remains suppressed in the low-double-digits — management has not addressed this directly.

[OPEN QUESTION] Margin profile of the new Lubricants business (Amaron Hi-Life Lubes) — genuinely accretive, or a distribution-network volume-filler.

[OPEN QUESTION] The precise nature of ARPS's Charge-Point-Operator contracts (e.g., with Yellow Stone EV) — one-time equipment sales versus recurring AMC-type revenue.

[OPEN QUESTION] Potential transfer-pricing dynamics between ARCS and the parent — given ARCS's revenue is almost entirely captive (₹1,424 cr purchased by the parent in FY26), there is a structural risk (not yet evidenced, but worth monitoring) that intercompany pricing could be used to manage/smooth standalone LAB margins during periods of high LME lead prices.

4.52 10. Monitoring Dashboard — Quarterly Checklist

4.52.1 10.1 Core Business

- LAB revenue growth (vs. 8–10% FY27 guidance) and the Automotive/Industrial mix split.
- OEM vs. Replacement growth differential and its impact on blended margins (OEM mix dilution is a known, explicitly acknowledged headwind).
- Material cost as % of Sale of Products (track sequentially — currently 69.7% FY26, up from 68.5% FY25); lead, antimony, tin, sulfuric-acid price trends.
- Telecom LAB revenue share (de-risking checkpoint — currently <5%) vs. Data Center/UPS growth (currently the primary offsetting driver).
- Export performance vs. the "technology-agnostic" strategy, and North American tariff / Middle East geopolitical developments.

4.52.2 10.2 New Energy Business Execution

- CQP progress toward full-scale operation (Q2 FY27 target) and any disclosed 4W-OEM sample-qualification milestones.
- Giga-1 Phase 1 (2 GWh cylindrical) construction/commissioning progress against the H1-CY2027 SOP target.
- BESS facility progress against the Q4 FY27 / end-CY2026 5 GWh SOP target, and the specific ₹280 crore capex drawdown.
- ARACT and ARPS quarterly revenue and segment loss trajectory — watch specifically whether losses continue widening in absolute terms even as revenue grows (as occurred FY25→FY26).
- Any conversion of Ather Energy / Piaggio MoUs into firm offtake contracts.
- Cumulative equity infusion into ARACT against the ₹9,500 crore total Giga Corridor commitment.

4.52.3 10.3 Balance Sheet & Capital Allocation

- Gearing ratio and any drawdown of the new ₹500 cr Commercial Paper / ₹500 cr working-capital facilities.
- ROCE trend (declining from 19.2% FY24 → 16.2% FY25) — watch for further compression or first signs of stabilisation.

- Consolidated capex actual vs. the ₹1,500–1,700 cr FY27 guidance (note the FY26 pattern of guided capex being exceeded).
- Recycling (ARCS) margin accretion — track the move from 0.5–0.6% toward 1.0–1.5% as "battery breaking" operations begin (Q2 FY27 target).

4.52.4 10.4 Guidance-Credibility Tracking

- Maintain and update the hit/miss scorecard in section 5 every quarter — this is the single highest-value recurring exercise for this company given the demonstrated NEB guidance-miss pattern.
- Specifically track whether FY27 NEB revenue approaches the ~₹1,240 crore implied by the 7–8% share target, or repeats a shortfall.

4.53 11. Source Log & Documentation Standards

[FACT] This dossier (v1.0) is built entirely from a single Gemini NotebookLM research session, "Amara Raja Energy & Mobility," covering 16 underlying source documents: Annual Reports FY19 through FY26 (AR19.pdf–AR26.pdf), and Q1–Q4 FY26 Earnings Call transcripts and Investor Presentations (CC Q1FY26.pdf–CC Q4FY26.pdf; PPT Q1FY26.pdf–PPT Q4FY26.pdf). Chat sessions reviewed are dated 21–22 July 2026.

[ANALYST INTERPRETATION] Because this version has not yet been cross-checked against the primary source PDFs directly (unlike the Exide dossier, which incorporated primary annual report/concall text alongside NotebookLM synthesis), every figure in this document should be treated as provisional. The internal inconsistencies flagged in section 9 (FY26 LAB revenue/growth, export revenue, consolidated growth rate) are exactly the kind of errors that primary-source cross-checking would catch — prioritise uploading the underlying PDFs directly in the next research session.

4.53.1 11.1 Notes on Documentation Discipline

- Where the same metric was quoted with different values at different points in the same chat session, both values are recorded here rather than silently resolved, with the discrepancy flagged as an open question.
- Management commentary is recorded with direct quotation wherever the research base captured it verbatim, since exact phrasing (e.g., the shift from an absolute ₹1,000 cr NEB target to a relative 7–8% share target) is itself analytically meaningful.
- This document should be updated once the primary annual reports/concall transcripts/investor presentations are uploaded directly, and again after every subsequent quarterly result. Cross-references to the companion Exide Industries dossier have been included where a genuinely comparable industry-wide pattern was identified (antimony price shock, EPR cost impact, Telecom lithium migration, chemistry-agnostic strategy) — these should be revisited once both dossiers are on a fully primary-source footing, to build a proper peer-comparison section.

This section draws directly on facts and figures already established in Parts VIII and IX; refer back to those sections for full sourcing and tagging detail. Figures below are FY26 (standalone for Exide, consolidated for ARE&M unless stated) as disclosed in the respective dossiers and financial models.

4.54 1. Business Mix

Both companies run the same underlying “dual-pillar” strategic model: a mature, cash-generative lead-acid business (LAB) funding a capital-intensive new-energy lithium-ion pivot. **[ANALYST INTERPRETATION]** Both dossiers independently arrive at a similar mental model — the LAB business as a “synthetic bond” funding a “new-energy call option” — developed separately for each company, which strengthens confidence that this is a genuine industry-structural pattern rather than an artifact of one research session’s framing.

- **Exide:** Organized since FY25 under the functional “One Exide” model (Trade/B2C, Institutional/B2B, International clusters) rather than the older Automotive/Industrial SBU split. Approximately 92% of standalone revenue sits in the “core” business (ex-Telecom/E-rickshaw and Exports), historically split roughly 69:31 between Automotive and Industrial (a ratio flagged in the dossier as stale/contaminated by Solar’s growth and Telecom’s decline — see Part VIII, Section 5 and Part XI below).
- **Amara Raja:** Reports Automotive and Industrial verticals within the LAB segment, plus a distinct New Energy Business (NEB) segment housed across four entities (ARACT for cells/Giga Corridor, ARCS for lead recycling, ARPS for power electronics/chargers, plus the BESS-specific facility). NEB revenue surged 61.4% YoY in FY26 off a small base, while still posting a segment loss (~₹135 crore in FY26, per the dossier).

4.55 2. Market Share and Manufacturing Footprint

Both companies are the only two organized-sector incumbents in India’s lead-acid duopoly, with broadly comparable domestic manufacturing scale in the legacy business. In lithium-ion, both are mid-pack entrants relative to Reliance (largest, most diversified, PLI-anchored) and Tata/Agratas (large, automotive-anchored), but ahead of most smaller pack-assembly-only players in pursuing actual cell-manufacturing capability:

Metric	Exide (EESL)	Amara Raja (ARACT/Giga Corridor)
Ultimate cell capacity target	12 GWh	16 GWh
Phase 1 capacity	6 GWh (3 GWh NMC cylindrical + 3 GWh LFP prismatic)	2 GWh (Giga-1, cylindrical) + separate 5 GWh BESS facility target

Metric	Exide (EESL)	Amara Raja (ARACT/Giga Corridor)
Technology partner	SVOLT Energy (China)	Gotion High-Tech (via GIB EnergyX)
Cumulative equity invested (as of FY26 disclosures)	₹4,802 crore	Smaller cumulative base; ₹500-crore Customer Qualification Plant commissioned in FY26
Total committed capex	~₹6,000 crore	~₹9,500 crore (Giga Corridor total)
Commercialization stage (mid-2026)	Cylindrical customer samples from May 2026; prismatic from June–July 2026; commercial revenue guided from ~Q3 FY27	Giga-1 targeted for H1 CY2027 start of production; BESS facility targeted for 5 GWh by Q4 FY27/end-CY2026

4.56 3. Product Portfolio and Technology Roadmap

Exide's Phase 1 deliberately hedges across both major chemistries (an even 3:3 GWh NMC:LFP split), with its revenue realization sequenced LFP-first (shorter qualification cycles for telecom/BESS/3-wheeler applications) even though the underlying capacity split is 50:50 — a distinction the dossier notes is frequently conflated in market commentary. Amara Raja's Giga-1 begins with NMC cylindrical cells, with LFP capacity concentrated in the separate BESS facility — a sequencing choice not yet as fully chemistry-hedged at the Phase 1 level as Exide's, though both companies are pursuing the same eventual LFP-majority, NMC-minority end-state that the broader Indian market is converging toward (Part VI).

4.57 4. Capital Allocation and Balance Sheet Strength

- **Exide:** Standalone balance sheet has been debt-free for over a decade (AAA/A1+ rated); the new-energy build-out is funded through internal cash generation/equity infusion into EESL rather than standalone debt, with the model assuming further EESL-level debt drawdown against a ₹2,000 crore facility. Exide also holds a strategic stake in HDFC Life, a distinct, non-operating source of embedded value carried in the SOTP valuation (see Part XI).
- **Amara Raja:** The Giga Corridor is being funded through a mix of internal accruals and a rising share of debt — the model assumes ~70%-funded debt for the cumulative Giga Corridor capex by FY31, alongside new ₹500 crore Commercial Paper and ₹500 crore working-capital facilities recently established — a materially more leveraged funding posture for its new-energy build-out than Exide's, and a specific balance-sheet risk flagged for monitoring in both the dossier and the financial model (see Part XI, Section 4).

4.58 5. Financial Performance, Margins, and Returns (FY26)

Metric	Exide (Standalone)	Amara Raja (Consolidated)
Revenue	₹17,269 cr (+4.1% YoY)	₹13,814 cr
EBITDA margin	11.3% (FY26 full-year; Q4 alone 11.7%)	LAB EBITDA of ₹1,714 cr on a consolidated base, with NEB still loss-making at the net-contribution level in FY26
PAT	₹1,111 crore (6.4% margin)	₹896 crore (consolidated, combined LAB+NEB)
ROCE (core business)	Two conflicting disclosed series — 16–18% (ex-HDFC Life/EESL-inclusive denominator) vs. 26–29% (ex-HDFC Life/EESL-exclusive denominator); unresolved in the dossier pending a primary-source ratio-note check	Declining trend, 19.2% (FY24) → 16.2% (FY25), a compression the dossier flags as a monitoring priority
FCFE	Positive and growing across the standalone forecast (model-derived; ~₹1,853 cr by FY30E in the Base case)	Volatile/negative in recent actuals (FY26 actual approximately -₹220 cr per the model) before recovering through the forecast as Giga Corridor capex normalizes

[ANALYST INTERPRETATION] On the figures disclosed to date, Exide’s core lead-acid business displays a meaningfully stronger and more stable return profile (debt-free, high and — on at least one disclosed methodology — rising core ROCE) than Amara Raja’s, whose consolidated returns are being visibly diluted by NEB losses and a more debt-funded capex program. This should not be read as a definitive relative-quality judgment, however, given (a) the unresolved ROCE-methodology discrepancy in Exide’s own disclosures, and (b) the fact that Amara Raja’s dossier is explicitly flagged as not yet cross-checked against primary annual-report sources (unlike Exide’s), so some of the apparent gap may narrow once primary verification occurs.

4.59 6. Growth Opportunities

Both companies share the same core opportunity set — Auto OEM share gains, aftermarket/replacement growth, Solar/UPS/data-centre industrial growth, and the multi-year lithium-ion ramp — but Amara Raja’s NEB structure (four separate entities spanning cells, recycling, power electronics, and BESS) provides more distinct, separately trackable growth vectors (e.g., the BESS facility as a standalone near-term revenue opportunity, ahead of Giga-1 cell revenue) than Exide’s single-subsidiary (EESL) structure, which bundles cell manufacturing and pack/module assembly together.

4.60 7. Risks

Common risks across both companies: commodity-cost volatility (lead, antimony, tin, silver, sulfur for the core business; lithium, nickel, cobalt for new energy); execution risk on

gigafactory yield/utilization ramp-up; reliance on a single foreign technology-transfer partner exposed to Chinese export-control policy; guidance-credibility risk (both dossiers independently flag patterns of management framing results/guidance in a favourable light — see Part VIII Section 6.1 and Part IX’s guidance hit/miss scorecard).

Company-specific risks: Exide’s core-ROCE and inventory-days disclosure inconsistencies represent a near-term data-discipline/model-input risk (Part XI); Amara Raja’s more leveraged capex funding posture and its wider set of unresolved internal revenue/growth-rate inconsistencies (flagged extensively in the dossier’s Open Questions, Part IX Section 9) represent a comparatively higher near-term informational and balance-sheet risk.

4.61 8. Management Quality

Both dossiers independently document management commentary that is directionally informative but occasionally selectively framed (Exide: Q4 YoY margin “expansion” language sitting alongside a full-year margin contraction in the same document; Amara Raja: a demonstrated pattern of NEB revenue/guidance misses tracked via an explicit hit/miss scorecard in the dossier). Neither pattern is characterized as dishonest in either dossier — both are framed as a normal degree of promotional emphasis common to management communication generally — but both are flagged as reasons to weight guidance-based modeling assumptions conservatively rather than take them at face value (a principle applied directly in the Part XI modeling discussion).

4.62 9. Valuation (as of the financial models built to date)

	Exide (Standalone SOTP)	Amara Raja (Consolidated SOTP)
Bear case value/share	₹309.4	₹638.4
Base case value/share	₹341.0	₹759.5
Bull case value/share	₹354.9	₹876.1
CMP referenced in model	—	₹876
Implied upside/(downside), Base case	n/a (CMP not embedded in this model version)	(13.3)%

[OPEN QUESTION] The Exide model as currently built does not carry a live CMP reference in the Executive Summary in the same way the Amara Raja model does; adding one would allow a direct, side-by-side upside/downside comparison consistent with the Amara Raja presentation.

4.63 10. Where One Company Has an Advantage — Summary

- **Exide’s advantage:** A debt-free, higher-disclosed-return core business; a simpler, single-subsidiary new-energy structure; an additional, distinct pool of embedded value from the HDFC Life stake that Amara Raja has no equivalent for; more advanced primary-source verification within its own dossier (reducing near-term data risk).

- **Amara Raja's advantage:** A more diversified new-energy structure (cells, recycling, power electronics, and BESS as four distinct, separately monitorable vectors) that may offer more granular visibility into which specific piece of the new-energy bet is working; an early, already-revenue-generating lead in distributed lithium BESS deployment (telecom-site storage) ahead of its own cell-manufacturing capacity coming online; explicit management guidance for FY27 LAB growth (8–10%) providing a clearer near-term anchor than some of Exide's more qualitative segment commentary.

This comparative section should be revisited and expanded once (a) Amara Raja's dossier completes primary-source cross-checking, and (b) both companies' next quarterly results and updated investor presentations are incorporated. # Part XI — Financial Models

This section documents the modelling work completed to date on both companies — the reasoning behind key assumptions, the debates that shaped them, and explicit judgment calls/limitations — rather than simply restating output numbers. Full model files are the Exide Industries Standalone Model & SOTP workbook and the ARE&M Model v2 workbook; this section is the narrative companion to those files.

4.64 1. Modelling Philosophy (Common to Both Models)

The stated approach in both modelling exercises was explicitly **not** “fill cells in Excel fastest,” but “build the most defensible model” — meaning every material assumption should have a traceable rationale (a primary-source data point, a management statement weighed against its promotional bias, or an explicit, flagged placeholder where no better information exists), and every assumption should be checked for internal consistency against every other assumption (revenue, margins, working capital, capex, depreciation, financing, cash flow, and returns must all logically cohere). Where this consistency check failed, the inconsistency was surfaced and resolved (or explicitly flagged as unresolved) before proceeding — this discipline is documented throughout the sub-sections below.

4.65 2. Exide Industries — Standalone Model & SOTP

4.65.1 2.1 Data Reconciliation (Pre-Modelling Discipline)

Before any growth assumption was set, five cross-source reconciliation problems were identified and explicitly flagged rather than silently resolved:

1. **Core-business ROCE:** two disclosed series with a 10–12 percentage-point gap (16.3–17.6% in the Nov-2025 presentation vs. 26.0–28.8% in the May-2026 presentation), likely reflecting different capital-employed denominators (inclusion vs. exclusion of the HDFC Life stake and EESL CWIP) — flagged as unusable until clarified against a primary-source ratio note.
2. **Inventory days:** a similarly large gap (93 days vs. 73.8 days for what should be a comparable period) likely reflecting different averaging/formula conventions — direction (improving) is trusted; absolute level is not.
3. **FY26 EBITDA margin:** management's Q4 concall narrative emphasized YoY margin expansion (+50bps) while the same document's full-year figure showed a slight full-

year contraction (11.4% → 11.3%) — noted as a selective-framing pattern to weight into confidence on other guidance.

4. **Auto:non-auto revenue mix:** two apparently incompatible splits (a concall-quoted “53:47” auto:non-auto versus an Annual-Report-sourced 69:31 Automotive:Industrial) were reconciled as likely reflecting different taxonomies rather than a genuine contradiction — the AR-sourced 69:31 split was adopted as the more reliable, audited-adjacent anchor.
5. **Full-year revenue growth:** a minor 4.1% vs. 4.39% discrepancy (standalone vs. consolidated, or a restatement difference) — flagged to avoid inadvertently mixing the two figures.

4.65.2 2.2 Revenue Architecture — The Segment-Split Method

With no rupee-denominated segment revenue table available in the annual reports reviewed, the revenue build was constructed as an explicitly-labelled, moderate-confidence approximation rather than left undone:

- Management’s own framing (92% of business from the “core,” 5% Exports, 3% Telecom+E-rickshaw) was adopted as the top-level split.
- Within the 92% core, Solar was isolated using its disclosed absolute rupee figure (having “crossed ₹1,000 crore”), and the remaining Automotive/Industrial split was re-derived rather than mechanically applying the stale 69:31 ratio — because that ratio’s Industrial component still implicitly contained Solar’s and Telecom’s *historical* (larger) weight, which would have inflated Industrial and, by construction, deflated Automotive had it been applied unadjusted. The corrected approach nets out Solar’s and Telecom’s estimated historical embedded weight before back-solving, arriving at an estimated Industrial (ex-Solar, ex-Telecom) share of roughly 22–23% of total revenue and Automotive at roughly 63.5%, explicitly labelled as a placeholder estimate rather than a defensible primary-sourced figure.
- E-rickshaw was flagged as arguably mis-classified when bundled with Telecom (both “declining,” but for different structural reasons — Telecom is stationary-infrastructure lithium substitution; E-rickshaw is mobile-product lithium substitution) — accepted as a conscious, stated simplification given its small (<1%) revenue impact.

FY27 growth-rate assumptions used, and the reasoning behind each:

Bucket	Assumed FY27 growth	Key supporting evidence	Key counter-evidence / caveat
Automotive	15%	Auto OEM +20–25% YoY for two consecutive quarters; aftermarket independently described as “mid-teens”	A meaningful share of the OEM growth reflects one-off Share-of-Business wins on specific vehicle platforms (Kia Seltos facelift, Tata Sierra) rather than a

Bucket	Assumed FY27 growth	Key supporting evidence	Key counter-evidence / caveat
Industrial (ex-Solar, ex- Telecom)	10%	Plausible blended rate across Data Centres, UPS, and other industrial infrastructure	repeatable, compounding growth rate once 100% SOB is reached Data Centre “~20% CAGR” language in investor materials describes <i>market</i> TAM growth, not necessarily Exide’s own segment revenue growth (share is undisclosed)
Telecom + E- rickshaw	(5)%	Deceleration thesis: private- tower lithium migration is largely complete, so the <i>rate</i> of further lead-acid loss should mathematically decelerate as the convertible base shrinks	No comparable deceleration evidence exists specifically for E-rickshaw, whose decline was independently described as “faster than anticipated”
Solar	20%	Arguably conservative rather than aggressive: trailing growth was 25–27% in FY25, GST cut from 12% to 5% is a real, dateable, still-unwound tailwind, and a hard government rooftop-solar target underpins demand	Base-effect consideration: 20% off a >₹1,000cr base is a larger absolute rupee ask than 25% off a smaller base
Exports	Not point- estimated	Management’s own language is the most hedged in the entire concall (“cautiously optimistic,” explicit uncertainty through at least H1 FY27)	Given genuinely bimodal (recovery vs. continued-drag) outcomes and only ~5% revenue weight, a scenario range was judged more honest than a single point estimate

A key internal-consistency check: applying the above rates to the (corrected) FY26 base produced an aggregate FY27 growth of roughly 12–13%+ before even assigning Exports a positive contribution — materially above management’s own “high-single-digit to early-double-digit” (effectively ~7–11%) guidance ceiling. Rather than mechanically forcing the individual rates down to fit management’s guidance (a “backing into a number” failure mode explicitly flagged and avoided), this gap was left as an open, monitorable signal that at least one input — most likely the Automotive rate, given the Share-of-Business caveat above — may need revision as further quarters of data arrive, or that management’s guidance itself may prove conservative (“sandbagged”).

4.65.3 2.3 Cost Structure — Fixed/Variable Classification

Rather than a statistical fit, Other Expenses (Note 32) and Employee Benefits (Note 29) were classified line-by-line (27 total line items) based on the underlying economic driver of each:

- **Variable costs** (9 line items, ₹1,744.25 cr / 10.10% of FY26 revenue): Power & Fuel, Warranty, Freight & Forwarding, Stores & Spares, Battery Charging/Assembly, After-Sales Service, Royalty/Technical Aid, Clearing & Forwarding, Commission — each recomputed every forecast year as a direct percentage of that year's Revenue.
- **Fixed costs** (18 line items, ₹1,590.76 cr / 9.21% of FY26 revenue): Salaries & Wages, Miscellaneous Expenses, PF contributions, Staff Welfare, Publicity & Sales Promotion, Software, Rent, Travel, Repairs & Maintenance, Insurance, Auditors' Remuneration, Rates & Taxes, and several smaller items — grown at a flat 8%/year (revised up from an initial 5%/year during the modelling discussion) compounding from the FY26 base, independent of Revenue growth.
- **Mechanical effect:** because Variable costs move in lockstep with Revenue while Fixed costs grow at a flat rate below the model's assumed Revenue growth in every forecast year, Fixed costs mechanically shrink as a share of Revenue over time — the source of the model's projected EBITDA margin expansion (from ~11.3% toward ~13%+ by the end of the forecast) rather than a flat or runaway margin trajectory.
- **[OPEN QUESTION / LIMITATION]** This classification was cross-checked as stable across FY24–FY26, but that stability occurred during a period of slow revenue growth (3.5–4.1%/year) where fixed-cost inflation (5–8%/year) happened to be similar in magnitude — this says little about whether the classification holds once growth accelerates to the 8–14%/year range now being forecast.

4.65.4 2.4 Lease Liability — A Genuine Model Gap Found and Fixed

Initial model versions held lease liability flat. On inspecting the Ind AS 116 lease note directly, the liability was found to represent a single, well-defined 25-year captive solar Power Purchase Agreement at Exide's factories (not general office/warehouse leases, which are already captured as short-term rent within the Fixed cost bucket above), with a clean, exactly-reconciling roll-forward and a disclosed maturity ladder (₹42.86cr due within 1 year; ₹166.96cr due 1–5 years, averaging ~₹41.74cr/year), implying an internally consistent 8.55% discount rate. This was built as a proper declining amortization schedule (₹375.21cr FY26 → ₹316.28cr FY31) rather than held flat.

A real, previously-undetected error was surfaced and corrected in the process: the model's FCFE build had only ever captured the *interest* portion of lease payments (via reduced PAT) and never the *principal repayment* — a real cash outflow that never touches the P&L under lease accounting. This meant FCFE had been silently overstated by roughly ₹11–14cr/year throughout the forecast. Adding the missing principal-repayment line reduced Base-case value per share modestly (by roughly ₹1.5–2/share) — a small but genuine correction, illustrating the value of checking mechanical completeness (does every

real cash flow appear somewhere in the FCFE build?) rather than only checking whether a schedule “looks right” in isolation.

4.65.5 2.5 Terminal Value Anchor — Extending to FY32E

The Gordon Growth terminal value was initially anchored on FY31, a year that still carried the tail end of the EESL equity-infusion drag (~₹510cr) on FCFE. Because that drag does not persist into perpetuity once the ₹8,000cr new-energy capex program completes, the model was extended by one additional year (FY32E) to provide a cleaner, genuinely steady-state terminal anchor — retaining the ongoing interest cost on the ~₹5,600cr debt balance (which does not disappear) while dropping the one-off equity draw. This single change moved the Base-case value per share meaningfully (Part A’s DCF value rose from roughly ₹3,921cr to ₹6,733cr on this change alone in an earlier model iteration), underscoring how sensitive DCF/SOTP output is to the specific year chosen as the terminal anchor when a forecast period still contains one-off items.

4.65.6 2.6 Sum-of-the-Parts (SOTP) Structure and the EESL Valuation Debate

The SOTP combines three components: (1) the standalone core lead-acid business, valued on a DCF basis; (2) EESL (the lithium-ion new-energy subsidiary), valued at Book Value × an assumed Price-to-Book multiple; and (3) the HDFC Life stake, valued at its audited mark net of a holding-company discount.

A specific methodological debate arose over what P/B multiple to apply to EESL. The case for borrowing Exide’s own consolidated trading multiple (2.2–2.48x) was rejected on two grounds: (a) that multiple reflects the market’s view of the *blended* entity — a mature, high-ROCE, debt-free core business plus whatever option value the market already assigns to EESL plus the HDFC Life stake — and was never earned by EESL on its own, pre-revenue, currently loss-making, single-asset, unproven-technology merits; and (b) using a multiple derived from the current market price to value a component that feeds into an *independent* view of that same market price is circular — it would just reconstruct today’s market cap rather than genuinely test it.

A reverse-engineering cross-check (solving for what the market must implicitly be paying for EESL, given the Core DCF value, the HDFC Life value, and the actual market cap) produced an implied EESL P/B of approximately 0.58x — *below* the model’s own conservative 1.0x baseline assumption, suggesting (with the important caveat that this check is only as good as the Core DCF’s own placeholder 12% discount-rate assumption) that the burden of proof runs toward justifying any premium to book value for EESL, not toward assuming one. Rather than resolve this by picking a single “right” multiple, a sensitivity table across a 0.5x–2.48x range was built (spanning roughly ₹388–₹486/share in an earlier iteration) so the assumption can be examined explicitly rather than defended by fiat.

4.65.7 2.7 Current Model Output (Latest Version on File)

	Bear	Base	Bull
Value per share (₹)	309.4	341.0	354.9

SOTP build-up (selected/Base scenario): Standalone core business DCF ₹20,926 cr + EESL (book value × P/B) ₹5,600 cr + HDFC Life stake (net of holding discount) ₹3,601 cr = Total equity value ₹30,127 cr, or ₹354.4/share on this specific build (the Executive Summary’s headline Base figure of ₹341.0 reflects the scenario-toggle-driven calculation, which may differ slightly from the static build shown here due to which scenario/assumption vintage is currently toggled — reconcile the two before quoting a single number externally).

4.65.8 2.8 Explicitly Documented Model Limitations (carried in the model file itself)

- The Fixed/Variable cost split is a line-item classification, not a statistical fit, and its stability has only been tested during a slow-growth historical period (see 2.3 above).
- Other Income is modelled as 5% interest on the *prior* year’s closing cash balance (deliberately non-circular, since using current-year cash was tested and found to create an unstable, runaway feedback loop) — worth revisiting whether a rising dividend payout ratio or a buyback assumption is more realistic than letting cash accumulate unchecked across the forecast.
- The discount rate (12%) and EESL infusion phasing (₹500cr/₹300cr FY27/FY28) are explicitly flagged as placeholders pending further refinement.
- The Balance Sheet forecast covers only selected line items by scope and will not fully reconcile Assets = Liabilities + Equity.
- The Non-Institutional revenue growth assumption (“mid-teens”) rests on unverified management commentary and carries ~71% of total revenue weight — flagged as Medium confidence, though partially corroborated by the audited Ind AS 115 disaggregation note (Institutional/Non-institutional 24.6%/75.4% FY26), which closely validates the 25%/75% split used to build the revenue module.
- The Export FY28E share-recovery mechanic implies a greater-than-100% single-year increase, which is aggressive; phasing this over 2–3 years should be considered.
- The HDFC Life stake is valued at its audited March-2026 mark; the live market price has since moved (a further ~6% decline observed at one point), and this input should be refreshed periodically rather than treated as static.

4.66 3. Amara Raja Energy & Mobility — Consolidated Model

4.66.1 3.1 Model Structure

Built as a consolidated model combining a bottom-up LAB (lead-acid) forecast with a single-line NEB (new-energy business) forecast, plus a Sum-of-the-Parts valuation combining: (1) the core business on a DCF basis, (2) the BESS facility at Book Value × P/B, and (3) the Giga Cell Corridor at Book Value × P/B, less (4) Giga Corridor debt (modelled at a 70%-funded assumption on cumulative FY31E capex, ~₹4,400 cr).

4.66.2 3.2 Comprehensive Ratios Framework

A full ratio suite was built spanning Historical Actuals (FY24A–FY26A, hardcoded from reported consolidated results since the P&L tab’s detailed cost stack below EBITDA was

only ever built for forecast years) through FY32E, organized into eight sections: Growth (Revenue/EBITDA/PAT), Margins (Gross/EBITDA/PAT), Cash Conversion (CFO/PAT, CFO-excluding-NEB-capex/PAT, FCFE/PAT, FCFE Yield, Dividend Payout), Returns (ROE, ROCE), Leverage & Coverage (Debt/Equity, Net Debt/EBITDA, Interest Coverage), Efficiency (Asset Turnover), and Per-Share & Multiples (EPS, BVPS, EV/EBITDA, trailing and forward P/E and P/B).

A specific definitional judgment call was made and explicitly documented: “CFO (excluding NEB capex) to PAT” was interpreted as CFO net of the LAB business’s own maintenance capex, but *before* the NEB equity-investment drag — i.e., a lens on what cash conversion would look like if the New Energy Business were not consuming capital — rather than the alternative reading (CFO net of *all* capex including NEB). This is flagged as a one-line formula change should the intended meaning differ.

Two build-time errors were caught and corrected before the file was finalized: a Gross Margin FY32E cell that had been accidentally overwritten by its own footnote text (a column-reuse mistake), and an initial attempt to fix it using a row-insertion function that would have silently broken every self-referential formula below the insertion point (since row-insertion shifts cell *content* but does not rewrite formula *text* — a subtle and easy-to-miss class of spreadsheet bug). Both were caught by checking actual formula text rather than only visual output, and the sheet was rebuilt without shifting rows.

4.66.3 3.3 Leverage Trajectory — A Specific Flagged Risk

Interest coverage is modelled to fall from approximately 34x to approximately 5x by FY32E as the Giga Corridor’s debt-funded capex program scales — a large, mechanical consequence of the 70%-debt-funding assumption on a multi-thousand-crore capex program, and a specific figure worth monitoring closely against actual disclosed gearing as the corridor build-out proceeds (see also the dossier’s own open question on whether Amara Raja will take on long-term project debt at scale if ROCE remains suppressed).

4.66.4 3.4 Current Model Output (Latest Version on File)

	Bear	Base	Bull
Value per share (₹)	638.4	759.5	876.1
Upside/(downside) vs. CMP ₹876	(27.1)%	(13.3)%	~0.0%

SOTP build-up (Base scenario): Core business DCF ₹8,754 cr + BESS (book value × P/B) ₹492 cr + Giga Cell Corridor (book value × P/B) ₹9,640 cr – Giga Corridor debt (FY31E cumulative, 70%-funded) –₹4,400 cr = Total equity value ₹14,487 cr, or ₹791.5/share on this specific static build.

[ANALYST INTERPRETATION] Notably, even the model’s *Bull* case lands almost exactly at the current market price (₹876), implying the market is currently pricing in an outcome at or beyond this model’s most optimistic scenario for the Giga Corridor/BESS build-out — a useful, if uncomfortable, sanity check worth holding alongside the equivalent question already raised on the Exide model (Section 2.7’s implied gap to Exide’s own CMP once that reference is added), and consistent with the general pattern, seen in both models, that

these DCF/SOTP frameworks currently imply the market may be paying for more new-energy optionality than either core-DCF-plus-conservative-subsidiary-multiple framework can fully justify.

4.67 4. Cross-Model Observations

- Both models independently surfaced the same category of error — a cash-flow build silently omitting a real, non-P&L cash outflow (lease principal for Exide; implicitly, the broader capex-financing mechanics for Amara Raja) — reinforcing the value of the “does every real cash flow appear somewhere in FCFE” discipline as a standard check to run on any future model update for either company.
- Both models treat their respective new-energy subsidiary’s valuation via Book Value $\times$ P/B rather than a standalone DCF, reflecting the genuinely pre-revenue/early-revenue state of both EESL and ARACT/the BESS facility as of mid-2026 — this is a reasonable simplification today but should be revisited (potentially replaced with a standalone DCF once 2–3 years of commercial revenue and margin data exist) as both new-energy businesses mature.
- Both models currently rely on a placeholder discount rate (Exide: 12%) or an implicit one embedded in the P/B assumptions (Amara Raja) rather than a formally derived WACC — a natural next refinement for both models.
- Neither model has yet incorporated a live, continuously-updated CMP/upside-downside display in a fully consistent way across both companies (Amara Raja’s Executive Summary carries a CMP field; Exide’s does not yet) — worth standardizing for direct side-by-side use. # Part XII — Monitoring Framework

A practical, recurring checklist combining both companies’ individual monitoring dashboards (Parts VIII/12 and IX/10) with sector-level triggers from Parts II–VII. Use this alongside each quarterly results release and earnings call for both companies.

4.68 1. Sector-Level Triggers (Check Every Quarter)

- Movement in India’s operational (not merely announced) domestic lithium-ion cell-manufacturing capacity, and the gap versus the ~180–220+ GWh cumulative announced pipeline.
- LFP vs. NMC chemistry-mix developments across major Indian projects, and any movement on LMFP or sodium-ion commercialization.
- Chinese export-control policy on lithium-ion raw materials, precursor chemicals, or manufacturing technology transfer (a specific hard date — a China restriction delay to November 2026 — is on the calendar per Exide’s dossier).
- Government policy: GST rate changes on batteries/solar; PLI-scheme developments; import-duty changes; any movement on India’s own domestic lithium-mining projects (Jammu & Kashmir, Chhattisgarh, Gujarat).
- Commodity trackers: LME lead, antimony, sulfur, tin, silver, copper (core lead-acid business); lithium carbonate/hydroxide, nickel, cobalt, graphite (new-energy business); INR/USD.
- BESS tender activity from major utility-scale developers (Adani Green, Reliance New Energy, JSW Energy, NTPC Green Energy, Tata Power Renewable) as a leading

indicator of near-term addressable demand for both companies' new-energy capacity.

4.69 2. Exide Industries — Core Business Health

- Segment trend/outlook grid (Auto OEM, 2W/4W Replacement, Industrial Infra, Telecom, Solar, Home UPS/Inverters, Exports), reproduced quarter-over-quarter for direction changes.
- Standalone revenue growth (YoY/QoQ) and domestic-ex-Telecom growth specifically.
- Gross and EBITDA margins, sequential and YoY, alongside specific commodity/price-hike commentary each quarter.
- Price-hike announcements: date, quantum, channel (trade/OEM/both).

4.70 3. Exide Industries — New Energy (EESL) Execution

- Cumulative equity infusion into EESL against the ₹5,600 cr Phase 1 budget, and any Phase 2/Hyundai-Kia capex disclosures.
- Line-by-line commissioning status (L1–L4), MPAs signed, shopfloor headcount.
- Progress along the six-stage roadmap: Facility & utilities → Equipment trials → Process validation → Customer samples → Homologation → Commercial supply.
- Any disclosed utilization% or yield% figures once production begins (targets: >85% utilization, 90% yield).

4.71 4. Exide Industries — Balance Sheet & Capital Allocation

- Standalone debt status (should remain zero); EESL-level debt drawdown against the ₹2,000 cr facility.
- HDFC Life stake fair value and any OCI/MTM movement; any partial-monetisation signal.
- Working-capital metrics (inventory days, receivable/payable turnover) and resolution of the core-ROCE methodology discrepancy flagged in Part VIII/XI.

4.72 5. Amara Raja — Core Business

- LAB revenue growth versus the 8–10% FY27 guidance, and the Automotive/Industrial mix split.
- OEM vs. Replacement growth differential and its blended-margin impact.
- Material cost as % of Sale of Products (currently 69.7% FY26, up from 68.5% FY25); lead, antimony, tin, sulfuric-acid price trends.
- Telecom LAB revenue share (currently <5%) vs. Data Centre/UPS growth as the offsetting driver.

4.73 6. Amara Raja — New Energy Business Execution

- Customer Qualification Plant (CQP) progress toward full-scale operation (Q2 FY27 target) and any disclosed 4W-OEM sample-qualification milestones.
- Giga-1 Phase 1 (2 GWh cylindrical) construction/commissioning progress against the H1-CY2027 start-of-production target.

- BESS facility progress against the Q4 FY27/end-CY2026 5 GWh target, and the specific ₹280 crore capex drawdown.
- ARACT/ARPS quarterly revenue and segment-loss trajectory — specifically whether losses continue widening in absolute terms even as revenue grows.
- Any conversion of the Ather Energy / Piaggio MoUs into firm, binding offtake contracts.

4.74 7. Amara Raja — Balance Sheet & Capital Allocation

- Gearing ratio and any drawdown of the ₹500 cr Commercial Paper / ₹500 cr working-capital facilities.
- ROCE trend (declining from 19.2% FY24 → 16.2% FY25) — watch for further compression or stabilisation.
- Consolidated capex actual versus the ₹1,500–1,700 cr FY27 guidance (note the FY26 pattern of guided capex being exceeded).
- Recycling (ARCS) margin accretion — track the move from 0.5–0.6% toward the aspired 1.0–1.5% as battery-breaking operations begin (Q2 FY27 target).

4.75 8. Guidance-Credibility Tracking (Both Companies)

- Maintain a running hit/miss scorecard for each company's new-energy revenue and margin guidance every quarter — the single highest-value recurring exercise given the demonstrated guidance-miss pattern already documented for Amara Raja's NEB, and the selective-framing pattern already observed in Exide's Q4 FY26 communications.
- Specifically track whether Amara Raja's FY27 NEB revenue approaches the ~₹1,240 crore implied by its 7–8% revenue-share target, or repeats a shortfall.
- Specifically track whether Exide's FY27 core growth lands closer to management's "high-single-digit to early-double-digit" guidance or closer to the ~12–13% implied by the bottom-up segment build in Part XI, Section 2.2 — and, if a gap persists, which specific input (most likely the Automotive rate) needs revision.

4.76 9. Governance & Sustainability (Both Companies)

- Board changes, committee composition, related-party transactions (for Exide, monitor in the context of the broader Rajan Raheja Group structure mapped in Part VII).
- ESG ratings (S&P Global CSA score, EcoVadis) and any environmental/regulatory compliance events at lead-recycling or manufacturing facilities.

4.77 10. Financial-Model Maintenance Checklist

- Re-run the "does every real cash flow appear in FCFE" check on both models after any structural update (the lease-liability principal-repayment omission found in the Exide model, Part XI Section 2.4, is the kind of error this check is designed to catch).
- Refresh the HDFC Life stake mark-to-market value in the Exide model periodically rather than leaving it at the last audited figure.

- Revisit the EESL and ARACT/BESS Book-Value- $\times$ -P/B valuation approach for a possible transition to standalone DCF once 2–3 years of commercial revenue/margin data exist for each.
- Add a live CMP field to the Exide model's Executive Summary to match the Amara Raja model's presentation, enabling direct upside/downside comparison across both companies.
- Revisit the Exide Fixed/Variable cost classification's stability once several quarters of higher (8%+) revenue growth have actually been observed, given the caveat that its historical stability was tested only during a slow-growth period.